Optimal monetary and fiscal policy without fiscal backing *

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Abstract

This paper studies optimal monetary and fiscal policy when the Treasury cannot provide fiscal backing to the central bank. A Ramsey problem characterizes equilibrium adjustments, policy responses, and welfare in response to no fiscal backing. Higher nominal interest rates increase interest payments on reserves and can tighten the central bank's funding constraint, limiting stabilization by policy. For demand shocks, equilibrium allocations and welfare losses are summarized by a single sufficient statistic measuring the shortfall between financing required for full stabilization and the central bank's available capacity. For cost-push shocks, a single sufficient statistic no longer characterizes the equilibrium, as more distant positive shocks impose smaller welfare costs of no fiscal backing even when their discounted present values are identical. Finally, the welfare cost of no fiscal backing declines as taxation becomes more distortionary.

Keywords: optimal policy, central bank, fiscal backing, inflation, reserves
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1 Introduction

The conventional assumption in macroeconomic models of monetary-fiscal policy is that the monetary and fiscal authorities are part of a "government" with a single consolidated budget constraint (Sargent and Wallace, 1981). The starting point of this paper is a departure from the consolidated government budget. Under the Federal Reserve Act, when income exceeds expenses, the Federal Reserve is required to remit net earnings to the Treasury. When expenses exceed income, however, the Treasury does not compensate the central bank for the resulting losses. With roughly $3 trillion of reserve balances outstanding, the central bank has incurred substantial interest expenses as interest rates have risen. From September 2022 to January 2026, accumulated net losses reached approximately $245 billion. Providing fiscal backing for these losses would require approximately 4.3% of annual federal tax revenue. These facts motivate our interest in environments in which the monetary authority is not fiscally backed by fiscal authority, in the sense that transfers from the central bank to the Treasury must be nonnegative.

This paper studies how the lack of fiscal backing for the central bank affects monetary and fiscal policy in an optimal policy framework. An optimal policy framework allows us to characterize which equilibrium variables adjust, how adjustments are allocated across policy instruments, and the resulting welfare cost. We formulate a Ramsey problem in which the government maximizes household welfare subject to implementability conditions. Households consume, supply labor, and hold government bonds and interest-bearing reserves, while firms operate in a standard New Keynesian environment with costly price adjustment. The Treasury finances transfers to the central bank through taxation, while the central bank supplies reserves and pays interest on them. In response to an expansionary demand shock, the planner offsets the shock by raising nominal interest rates, but higher rates increase interest payments on reserves. Without fiscal backing, these losses cannot be offset by Treasury transfers, restricting the set of implementable allocations.

Our first result characterizes how the absence of fiscal backing constrains stabilization by monetary policy. With fiscal backing, the planner can adjust nominal interest rates to fully stabilize output and inflation following expansionary demand shocks. Without fiscal backing, the economy cannot be fully stabilized because higher interest rates increase the central bank's interest payments on reserves beyond what available surpluses can support. More importantly, for demand shocks, we show that the output and inflation are characterized by a single sufficient statistic measuring the shortfall between the resources required for full stabilization and the central bank's available financing capacity.

The absence of fiscal backing affects the inflation-output stabilization problem in ways that depend on the nature and timing of shocks. For demand shocks, only expansionary shocks worsen the central bank's funding position. Moreover, the same present discounted sum implies the same

welfare cost, regardless of shock timing. For cost-push shocks, positive shocks can tighten the funding constraint when inflation stabilization receives greater weight, whereas negative shocks can tighten it when output stabilization is more important. Also, unlike demand shocks, timing also matters: holding the present discounted value fixed, a near-term cost-push shock leads to a greater welfare cost than the same shock occurring further in the future.

We then study the key determinants of the welfare value of fiscal backing. The first question asks whether longer-duration assets make the central bank more vulnerable to the absence of fiscal backing (Bernanke, 2012). A natural concern is that long-duration assets expose the central bank to larger valuation losses when interest rates rise. We find that increasing asset duration reduces the welfare cost of no fiscal backing under reasonable assumptions, because changes in bond prices are increasingly absorbed through the continuation value of the central bank's asset position. In the limiting case of a consol, the absence of fiscal backing never restricts the optimal allocation, and full stabilization remains feasible regardless of the size or time path of the demand shock.

A second question considers a case in which the central bank can issue reserves beyond the amount backed by future fiscal surpluses. In the baseline model, the intertemporal budget constraint limits reserve issuance to available surpluses. In practice, however, the private sector may be willing to hold additional reserves because of their liquidity services (Bailey, 2024). We therefore extend the model by allowing households to derive direct utility from reserve holdings. For a wide range of parameter values, these liquidity services reduce the welfare cost of no fiscal backing by allowing the central bank to raise funds cheaply. However, the liquidity value of reserves does not necessarily mitigate the cost of no fiscal backing. When the coefficient of relative risk aversion is sufficiently low, the welfare loss is larger than that in the benchmark without liquidity services.

A third question is whether fiscal backing remains desirable when supporting the central bank is costly due to tax distortions (Carstens, 2023; Ramsden, 2023). In the baseline model, transfers to the central bank are financed by lump-sum taxation and create no additional tax distortion. We extend the model by allowing fiscal support to be financed through distortionary taxation. As taxation becomes more distortionary, the welfare gain from fiscal backing decreases because financing higher interest payments requires raising costly taxes, reducing the planner's willingness to support the central bank. Importantly, however, this welfare gain remains strictly positive even as tax adjustments become arbitrarily distortionary. In this limit, the Treasury effectively loses its ability to raise tax, yet fiscal backing still strictly improves welfare.

Finally, we develop a richer quantitative model to evaluate the macroeconomic dynamics and welfare consequences of the absence of fiscal backing. We introduce a banking sector in which banks issue deposits and hold central bank reserves, while households derive liquidity services from deposits. The model explicitly incorporates both sides of the central bank's balance sheet, with interest-bearing assets and reserve liabilities.

The quantitative model reveals a key insight of Ramsey problem: the absence of fiscal backing leads to a weaker interest rate response even when a stronger response remains financially feasible under the central bank's budget constraint. Following a shock, the remittance constraint does not bind in every period, leaving room for further monetary tightening. Nevertheless, the planner optimally refrains from exploiting this capacity because additional tightening generates excess reserve liabilities that require further distortionary taxation, resulting in welfare losses that outweigh the benefits of greater macroeconomic stabilization. The limited fiscal backing affects optimal monetary policy not only by restricting feasible policies but also by altering the welfare trade-offs among financially feasible policies.

We quantify the welfare cost of no fiscal backing using impulse responses. Under perfect foresight, a one-time 4% expansionary demand shock with persistent effects generates an additional welfare loss of approximately 1.5 basis points of permanent consumption relative to fiscal backing. Moreover, the lack of fiscal backing improves the welfare through providing more liquidity by printing reservese tough this welfare gain is dominated by the cost of real allocation of excess production.

Related Literature A closely related literature examines how central bank balance sheets affect monetary policy when fiscal support is limited. Studies of central bank solvency highlight the roles of seigniorage, remittance rules, and fiscal support in maintaining price stability (Hall and Reis, 2015; Reis, 2015; Del Negro and Sims, 2015; Benigno, 2020; Benigno and Nisticò, 2020). These issues become particularly relevant when central banks pay interest on reserves and hold long-duration assets, creating financial risks and fiscal consequences (Bassetto and Messer, 2013; Dominguez and Gomis-Porqueras, 2023; Bassetto and Caracciolo, 2021). Such balance-sheet constraints can also limit the implementation of optimal monetary policy (Berriel and Bhattarai, 2009) and affect stabilization policy in liquidity traps (Amador and Bianchi, 2023).

Our paper contributes to this literature in three ways. First, while Berriel and Bhattarai (2009); Del Negro and Sims (2015) highlight how central bank balance-sheet concerns can limit stabilization, we show how these effects depend on shock types, the planner's stabilization priorities, and shock timing. We also find that the absence of fiscal backing leads to a weaker interest rate response even when a stronger response remains financially feasible under the central bank's budget constraint. Second, we introduce distortionary taxation to finance fiscal support, extending the lump-sum benchmark. Although greater tax distortions reduce the welfare value of backing, this value remains strictly positive even when tax adjustments become arbitrarily distortionary, reflecting the benefits of intertemporal transfers between the Treasury and the central bank. Third, we allow reserves to provide liquidity services though treating reserves as purely financial claims provides a useful benchmark in the literature. While liquidity services can ease central bank financing,

they can also increase the welfare cost of no fiscal backing when consumption utility is sufficiently weakly curved, by altering both reserve valuations and the discounting of future financing needs and resources.

A broad literature studies monetary–fiscal interactions through the consolidated government budget constraint, including fiscal constraints on monetary policy (Sargent and Wallace, 1981), monetary–fiscal policy regimes and price stability (Leeper, 1991; Woodford, 2001), and the fiscal origins of inflation (Bianchi and Ilut, 2017; Eusepi and Preston, 2018; Bianchi et al., 2023; Schmidt, 2023; Bassetto and Miller, 2025). Related work studies optimal monetary–fiscal policy in the presence of distortionary taxation (Chari et al., 1991; Benigno and Woodford, 2003; Schmitt-Grohé and Uribe, 2004; Sims, 2025). Our contribution is to depart from the consolidated-budget approach by imposing separate budget constraints on the central bank and the Treasury and restricting fiscal transfers between them.

A related literature examines how central banks should conduct monetary policy when reserves are abundant. This literature studies how the quantity and remuneration of reserves affect monetary policy implementation, financial intermediation, and welfare (Curdia and Woodford, 2011; Kashyap and Stein, 2012; Arce et al., 2020; Canzoneri et al., 2017; Diba and Loisel, 2026). An important insight is that optimal policy arrangements also depend on the fiscal resources available to the central bank (Berentsen et al., 2014, 2018; Hu et al., 2025). Building on these insights, we examine how limited fiscal backing affects Ramsey stabilization when reserves support the provision of liquid bank deposits. We quantify the resulting welfare costs and decompose them into real allocation effects, nominal price adjustment costs, and deposit liquidity services. We find that the welfare cost of no fiscal backing arises primarily from distortions in real allocations rather than direct inflation adjustment costs.

Layout The paper is organized as follows. Section 2 develops the baseline analytical model. Section 3 presents the main results. Section 4 examines the implications of several key determinants of the value of fiscal backing. Section 5 develops a richer quantitative model. Section 6 concludes.

2 Analytical Model

2.1 Households

The representative household chooses consumption, labor supply, and holdings of one-period risk-free nominal reserves and bonds to maximize

$$\max_{\{C_t,N_t,B_t^C,B_t^T\}_{t=0}^{\infty}} \quad E_0 \sum_{t=0}^{\infty} \beta^t \delta_t \left[\log C_t - \frac{1}{1+\nu} N_t^{1+\nu} \right],$$

where C_t is the Dixit–Stiglitz consumption aggregate of differentiated goods,

$$C_t \equiv \left[\int_0^1 c_t(i)^{\frac{\theta-1}{\theta}} di\right]^{\frac{\theta}{\theta-1}}$$

with an elasticity of substitution equal to $\theta > 1$. Also, N_t is the labor supply. The representative households supply all types of labor as

$$N_t = \int_0^1 N_t(i) di.$$

A shock to δ_t represents a demand shock. The budget constraint is

$$P_t C_t + Q_t^C B_t^C + Q_t^T B_t^T + P_t T_t = B_{t-1}^C + B_{t-1}^T + w_t N_t + P_t \Phi_t,$$

where Q^C is the price of reserves, Q^T is the price of bond, B^C is reserve holdings, B^T is the bond holdings, Φ_t is the profit from the goods producers, w_t is wage, and T_t is lump-sum tax. The first-order conditions are

$$\frac{Q_t^C}{C_t} = \beta E_t\left[\frac{\delta_{t+1}}{\delta_t C_{t+1}\pi_{t+1}}\right], \tag{1}$$

$$\frac{Q_t^T}{C_t} = \beta E_t\left[\frac{\delta_{t+1}}{\delta_t C_{t+1}\pi_{t+1}}\right]. \tag{2}$$

$$\frac{w_t}{P_t} = C_t N_t^{\nu} \tag{3}$$

In this section, households directly hold reserves, and reserves and bonds are perfect substitutes. These assumptions are imposed for analytical tractability. In Section 5, we consider a more general setting in which banks issue deposits and hold reserves, and reserves and bonds play distinct roles. The assumption of direct reserve holdings is not critical because reserves support deposit issuance through banks' collateral constraints.

2.2 Firms

There is a continuum of monopolistically competitive firms indexed by $i \in [0,1]$. Each firm faces the Dixit–Stiglitz demand curve

$$y_t(i) = Y_t\left(\frac{p_t(i)}{P_t}\right)^{-\theta}. \tag{4}$$

Production is linear in labor, $y_t(i) = N_t(i)$. Following Rotemberg (1982), firm i chooses $p_t(i)$ to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \Lambda_t \left[p_t(i) y_t(i) - (1-\varsigma)\mu_t^w w_t y_t(i) - \frac{\varphi}{2}\left(\frac{p_t(i)}{p_{t-1}(i)} - 1\right)^2 P_t Y_t\right], \tag{5}$$

where $\Lambda_t \equiv C_t^{-1}/P_t$, φ governs price adjustment costs, μ_t^w is a wage-markup shock with steady-state value $\bar{\mu}^w = 1$, and ς is a constant employment subsidy. The employment subsidy $\varsigma = 1/\theta$ offsets the steady-state monopolistic distortion and is financed by lump-sum taxation.[1] We interpret the wage-markup shock as a wedge introduced by labor intermediaries, whose profits and losses accrue to households. In a symmetric equilibrium, firm optimality implies

$$N_t C_t^{-1} \left[\varphi(\pi_t - 1)\pi_t + (\theta - 1)\left(1 - \mu_t^w N_t^\nu C_t\right)\right] = \beta E_t \left[N_{t+1} C_{t+1}^{-1} \varphi(\pi_{t+1} - 1)\pi_{t+1}\right]. \tag{6}$$

2.3 Treasury and central bank

The Treasury issues bond and collect tax through non-distortionary tax. The Treasury's budget constraint is given by

$$Q_t^T B_t^T + P_t T_t + P_t H_t = B_{t-1}^T, \tag{7}$$

where T_t denotes net lump-sum taxes and H_t denotes real remittances from the central bank to the Treasury, with $H_t > 0$ indicating a transfer to the Treasury.

The central bank issues one-period nominal reserves and holds no assets in the baseline model. Its budget constraint is

$$Q_t^C B_t^C = B_{t-1}^C + P_t H_t, \tag{8}$$

We relax the no-asset assumption in Sections 4.1 and 5.

The key institutional restriction is a lower bound on remittances:

$$H_t \geq \underline{H}, \tag{9}$$

where $\underline{H}$ is an exogenous real bound. This limits transfers from the Treasury to the central bank without imposing a corresponding upper bound on remittances. Such an asymmetry is broadly consistent with institutional arrangements in the United States, Canada, Australia, Germany, the Netherlands, and Switzerland, where central bank losses are absorbed through financial buffers or future earnings rather than automatically covered by government transfers.

2.4 Market Clearing and Equilibrium

Market Clearing In a symmetric equilibrium, the goods market clearing condition requires

$$Y_t = N_t = C_t + \frac{\varphi}{2}(\pi_t - 1)^2 Y_t. \tag{10}$$

The price index is defined as

$$P_t \equiv \left[\int_0^1 p_t(i)^{1-\theta} di\right]^{\frac{1}{1-\theta}}, \qquad \pi_t \equiv \frac{P_t}{P_{t-1}}.$$

[1] Since the subsidy payments and the corresponding taxes offset each other in the aggregate household budget, we omit these flows from the household and Treasury budget constraints.

Competitive equilibrium Given symmetric initial prices $p_{-1}(i) = P_{-1}$, the stochastic processes for the exogenous shocks, the initial liabilities B^C_{-1} and B^T_{-1}, and a government policy plan, a symmetric competitive equilibrium consists of allocations, asset holdings, and prices such that households and firms optimize given prices and government policy, the Treasury and central-bank budget constraints and the remittance constraint are satisfied, and all goods, labor, and asset markets clear.

2.5 Ramsey Problem

At date zero, a benevolent Ramsey planner commits to a complete state-contingent plan for allocations, prices, and policy instruments to maximize household welfare.[2]

Since reserves and Treasury bonds have identical one-period nominal payoffs and provide no liquidity services in the baseline model, the household Euler equations (1) and (2) imply

$$Q^C_t = Q^T_t \equiv Q_t.$$

Given the initial price level P_{-1}, the initial nominal liabilities (B^C_{-1}, B^T_{-1}), and the exogenous shock processes, the Ramsey planner solves

$$\max_{\{C_t, N_t, \pi_t, Q_t, B^C_t, B^T_t, T_t, H_t\}_{t=0}^{\infty}} \quad E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t - \frac{N_t^{1+\nu}}{1+\nu} \right], \tag{11}$$

subject to the household Euler equations (1) and (2), the firm optimality condition (6), the Treasury budget constraint (7), the central bank budget constraint (8), the remittance constraint (9), and the resource constraint (10). These conditions must hold at every date and after every history. The price level is determined recursively by $P_t = \pi_t P_{t-1}$.

We additionally impose separate transversality conditions on Treasury debt and central bank liabilities:

$$\lim_{T \to \infty} E_0 \left[\beta^{T+1} \frac{C_0}{C_{T+1}} \frac{B^j_T}{P_{T+1}} \right] = 0, \qquad j \in \{T, C\}. \tag{12}$$

These conditions require the discounted value of each authority's terminal liabilities, evaluated using household marginal utility, to vanish, ruling out Ponzi financing while allowing debt rollover.

In practice, the Treasury and the central bank make their policy decisions separately. If the two authorities have different objective functions, each may choose its own policy taking the other authority's policy as given, potentially giving rise to strategic interactions. We abstract from these interactions by assuming a common objective of maximizing household welfare, allowing us to isolate the effects of institutional restrictions on resource transfers between the two authorities.

[2] In the linear-quadratic analysis of Sections 3, the relevant steady-state Lagrange multipliers are zero. Thus, inheriting these multipliers from the pre-shock steady state yields the same optimality conditions as commitment at date zero.

2.6 Linear-Quadratic Approximation

To obtain an analytically tractable representation of the Ramsey problem, we approximate the nonlinear economy around its deterministic zero-inflation steady state. We take a second-order approximation to household welfare and first-order approximations to the equilibrium conditions. Appendix A provides the details of the approximation and the definitions of the coefficients appearing below.

Throughout the analytical sections, we study the resulting Ramsey problem under perfect foresight. Conditional expectations can therefore be replaced by subsequent realizations. Hats denote deviations from the deterministic steady state. The shock e_t enters the household Euler equation and represents a demand shock, whereas e_t^c enters the Phillips curve and represents a cost-push shock.

No fiscal backing Under no fiscal backing, the Ramsey planner minimizes the quadratic welfare loss subject to the linearized equilibrium conditions, the separate budget constraints of the Treasury and the central bank, and the lower bound on remittances. The resulting problem is

$$\min_{\{\hat{y}_t,\hat{\pi}_t,\hat{T}_t,\hat{b}_t^T,\hat{b}_t^C,\hat{H}_t,\hat{Q}_t\}_{t=0}^{\infty}} \quad \sum_{t=0}^{\infty}\beta^t\left[\frac{1}{2}q_y\hat{y}_t^2+\frac{1}{2}q_\pi\hat{\pi}_t^2\right], \tag{13}$$

subject to

$$\hat{\pi}_t = \kappa\hat{y}_t + \beta\hat{\pi}_{t+1} + e_t^c, \tag{14}$$

$$\hat{Q}_t = \hat{y}_t - \hat{y}_{t+1} - \hat{\pi}_{t+1} + e_t, \tag{15}$$

$$\beta\left(\hat{b}_t^T + \hat{Q}_t\right) = \hat{b}_{t-1}^T - \hat{\pi}_t - (1-\beta)b_h\left(\hat{H}_t + \hat{T}_t\right), \tag{16}$$

$$\beta\left(\hat{Q}_t + \hat{b}_t^C\right) = \hat{b}_{t-1}^C - \hat{\pi}_t + (1-\beta)a_h\hat{H}_t, \tag{17}$$

$$\hat{H}_t \geq H^*. \tag{18}$$

Hats denote log deviations from steady state, except for taxes and remittances, which are level deviations normalized by steady-state output. Exogenous shocks are e^c and e, and parameters are $q_y, q_\pi, a_h, b_h, \kappa, \beta$ and H^*. The key departure from the standard consolidated-government formulation is the remittance constraint (18).

Fiscal backing Under fiscal backing, the planner minimizes the same quadratic loss (13) subject to (14)–(17), but without the remittance constraint (18). Since lump-sum taxes and remittances can adjust freely, the Treasury and central-bank budget constraints impose no additional restrictions on the welfare-relevant allocation.

We define the gross nominal return as $i_t \equiv Q_t^{-1}$, so that $\hat{i}_t = -\hat{Q}_t$. Throughout the analysis, superscripts B and N denote the fiscal-backing and no-fiscal-backing Ramsey equilibria, respectively, and $\mathscr{W}^B$ and $\mathscr{W}^N$ denote their corresponding welfare losses defined by (13).

2.7 Central Bank Solvency

We clarify several notions of central bank solvency and relate the solvency concept in our model to the existing literature. Under perfect foresight, iterating the nonlinear central bank budget constraint (8) forward and imposing the transversality condition yields

$$B_{t-1}^C = -\sum_{s=t}^{\infty} \left(\prod_{j=t}^{s-1} Q_j^T \right) P_s H_s. \tag{19}$$

This intertemporal budget constraint links changes in central bank liabilities to the present discounted value of future deviations in inflation, reserve prices, and remittances. Central bank faces a constraint that reserves must not exceeds the amount of surplus.

One commonly discussed notion in the literature is accounting equity. For a commercial bank, negative equity is typically associated with insolvency. This criterion is less appropriate for a central bank, which can continue to operate with negative accounting net worth. As Sims (2004) emphasizes, "The bank can be at negative net worth all the time." Our model likewise does not impose a nonnegativity constraint on central bank equity.

Instead, our model assumes that the remittance constraint places a lower bound on equity by restricting remittance. Importantly, this restriction does not require equity to be nonnegative. Equation (19) shows that central bank equity equals $-B_{t-1}^C$, which corresponds to the present discounted value of future remittances.

Another related concept is the "deferred asset" recorded by the Federal Reserve when earnings are insufficient to cover expenses. In such periods, the accumulated shortfall is recorded as a deferred asset rather than a reduction in reported equity. Despite its name, this deferred asset is not a marketable asset; it represents the amount of future net earnings required before remittances resume. In our model, such losses reduce the central bank's economic net worth, as is consistent with the practice.

3 Baseline Analysis

3.1 Demand shocks

We first consider demand shocks, represented by e_t in the household Euler equation, and begin with the fiscal-backing benchmark.

Proposition 1. *Under fiscal backing, the Ramsey equilibrium is*

$$\hat{y}_t = 0, \qquad \hat{\pi}_t = 0, \qquad \hat{i}_t = -e_t.$$

Proof. See Appendix B.2. □

The planner fully stabilizes output and inflation by adjusting the nominal interest rate to offset the demand shock. When this adjustment increases the central bank's interest expenses, the Treasury can cover the resulting shortfall through transfers financed by lump-sum taxes. Because these taxes are non-distortionary, fiscal support does not affect the equilibrium allocation.

We next examine whether full stabilization remains feasible without fiscal backing. This feasibility can be summarized by the funding gap

$$\Delta \equiv \underbrace{\hat{b}^C_{-1} + a_h H^*}_{\text{Balance-sheet and remittance burden}} \qquad \underbrace{-\beta \sum_{t=0}^{\infty} \beta^t e_t}_{\text{financing required for full stabilization}} . \tag{20}$$

The first term captures the financing burden associated with inherited reserve liabilities and the remittance requirement, while the second reflects the financing needs induced by demand shocks. In particular, an expansionary demand shock raises Δ because full stabilization requires a higher nominal interest rate, increasing interest payments on reserves. When $\Delta > 0$, the central bank lacks sufficient financing capacity to implement full stabilization while satisfying the remittance constraint.

In principle, the remittance constraint could bind in some periods and be slack in others. The following proposition, however, shows that whenever the constraint prevents full stabilization, it binds in every period.

Proposition 2. *If $\Delta > 0$, the remittance constraint binds in every period:*

$$\hat{H}_t = H^*, \qquad t \geq 0.$$

If $\Delta < 0$, full stabilization is feasible with the remittance constraint slack in every period.

Proof. See Appendix B.3. □

The intuition is that reserve issuance allows the planner to shift the central bank's financing burden across periods, equalizing the marginal value of financing over time. When $\Delta > 0$, the planner sets remittances at their lower bound in every period to retain as many resources as possible within the central bank. When $\Delta \leq 0$, sufficient financing is available to achieve full stabilization without violating the remittance constraint.

The following proposition characterizes the Ramsey equilibrium when the funding gap is positive and full stabilization is infeasible. In this case, output and inflation both increase on impact, while the nominal interest rate is lower than under fiscal backing. We first consider the benchmark case $q_y = q_\pi$.[3]

Proposition 3. *Suppose that $q_y = q_\pi$. If the funding constraint binds, so that $\Delta > 0$, the Ramsey equilibrium allocation is*

$$\hat{\pi}_0 = (1-r_\kappa)\left[1+\frac{\beta}{\kappa}(1-r_\kappa)\right]\frac{\Delta}{\Omega_\kappa}, \qquad \hat{y}_0 = r_\kappa\left[\kappa+\beta(2-r_\kappa)\right]\frac{\Delta}{\Omega_\kappa},$$
$$\hat{\pi}_t = -(1-r_\kappa)\left(\kappa-1+\frac{1-\beta}{\kappa}\right)r_\kappa^t\frac{\Delta}{\Omega_\kappa}, \qquad \hat{y}_t = -(\kappa^2-\kappa+1-\beta)r_\kappa^{t+1}\frac{\Delta}{\Omega_\kappa}, \qquad t \geq 1.$$

In particular,

$$\hat{\pi}_0 > 0, \qquad \hat{y}_0 > 0.$$

The impact response of the interest-rate variable satisfies

$$\hat{i}_0 < -e_0.$$

Here, $0 < r_\kappa < 1$ and $\Omega_\kappa > 0$, as defined in Appendix B.4.

Proof. See Appendix B.4. □

An expansionary demand shock calls for monetary tightening, but higher nominal interest rates increase the central bank's interest payments on reserves. Without fiscal backing, these higher expenses reduce the central bank's surplus and cannot be offset by additional Treasury transfers. When the funding constraint binds, the interest-rate increase required for full stabilization would leave reserve liabilities insufficiently backed by the present value of available resources. The planner therefore chooses a lower nominal interest rate than under fiscal backing, resulting in higher output and inflation on impact.

The analytical benchmark illustrates this mechanism in a particularly stark setting: the remittance constraint binds in every period, yet the central bank lacks sufficient resources to achieve full stabilization, so the interest rate must adjust. In practice, however, the constraint need not bind in every period. Section 5 shows that no fiscal backing can still affect optimal monetary policy even when the remittance constraint is slack in some periods.

Proposition 3 also shows that the funding gap is a sufficient statistic for the paths of output and inflation, and hence welfare, regardless of the timing of demand shocks.

[3] The relative weights on output and inflation stabilization, q_y and q_π, are determined by structural primitives. We consider equal weights and limiting cases in which one stabilization objective dominates to illustrate the underlying mechanisms. Appendix B.1 provides the corresponding solution for general q_y and q_π.

Corollary 1. *Consider two shock sequences with the same initial conditions. If*

$$\sum_{t=0}^{\infty} \beta^t e_t = \sum_{t=0}^{\infty} \beta^t \tilde{e}_t,$$

then they generate the same funding gap. Consequently, in the binding regime, they induce the same paths of output and inflation and the same welfare loss.

Proof. See Appendix B.5. □

We next examine how the planner's stabilization priorities shape the equilibrium when the funding constraint binds. Figure 1 illustrates two limiting cases: $q_y/q_\pi \to \infty$ (output stabilization) and $q_\pi/q_y \to \infty$ (inflation stabilization).

The two cases generate distinct adjustment patterns. When output stabilization dominates, output and inflation remain constant from $t = 1$ onward, following an initial adjustment. In contrast, when inflation stabilization dominates, adjustments in output and inflation are concentrated in periods $t = 0$ and $t = 1$, with both returning to zero from $t = 2$ onward.

Consider first the output stabilization case. Because output fluctuations are relatively costly, the planner relies on persistent inflation to limit output adjustments. Persistent inflation is from the New Keynesian Phillips curve

$$\hat{y}_t = \frac{1-\beta}{\kappa}\hat{\pi}_t, \qquad t \geq 1.$$

Thus, when β is close to one, persistent inflation can be sustained with relatively small output deviations. Following an expansionary demand shock, monetary tightening increases the central bank's interest payments on reserves. Positive and persistent inflation reduces the real value of outstanding reserve liabilities, helping to offset the additional interest burden.

Consider next the inflation stabilization limit, $q_\pi/q_y \to \infty$. In this case, the planner concentrates inflation adjustments in the first two periods. Combining the Phillips curve with the central bank's intertemporal budget constraint yields

$$\Delta = \frac{\kappa+\beta}{\kappa}\hat{\pi}_0 - \frac{\beta}{\kappa}\hat{\pi}_1.$$

Thus, inflation from $t = 2$ onward provides no additional financing capacity, as the reduction in the real value of reserve liabilities is offset by the associated increase in interest expenses. Since such inflation is costly without relaxing the funding constraint, the optimal allocation in the limit satisfies $\hat{\pi}_t = 0$ and $\hat{y}_t = 0$ for $t \geq 2$.

We next characterize the welfare cost of no fiscal backing. The following proposition shows that this cost is quadratic in the funding gap.

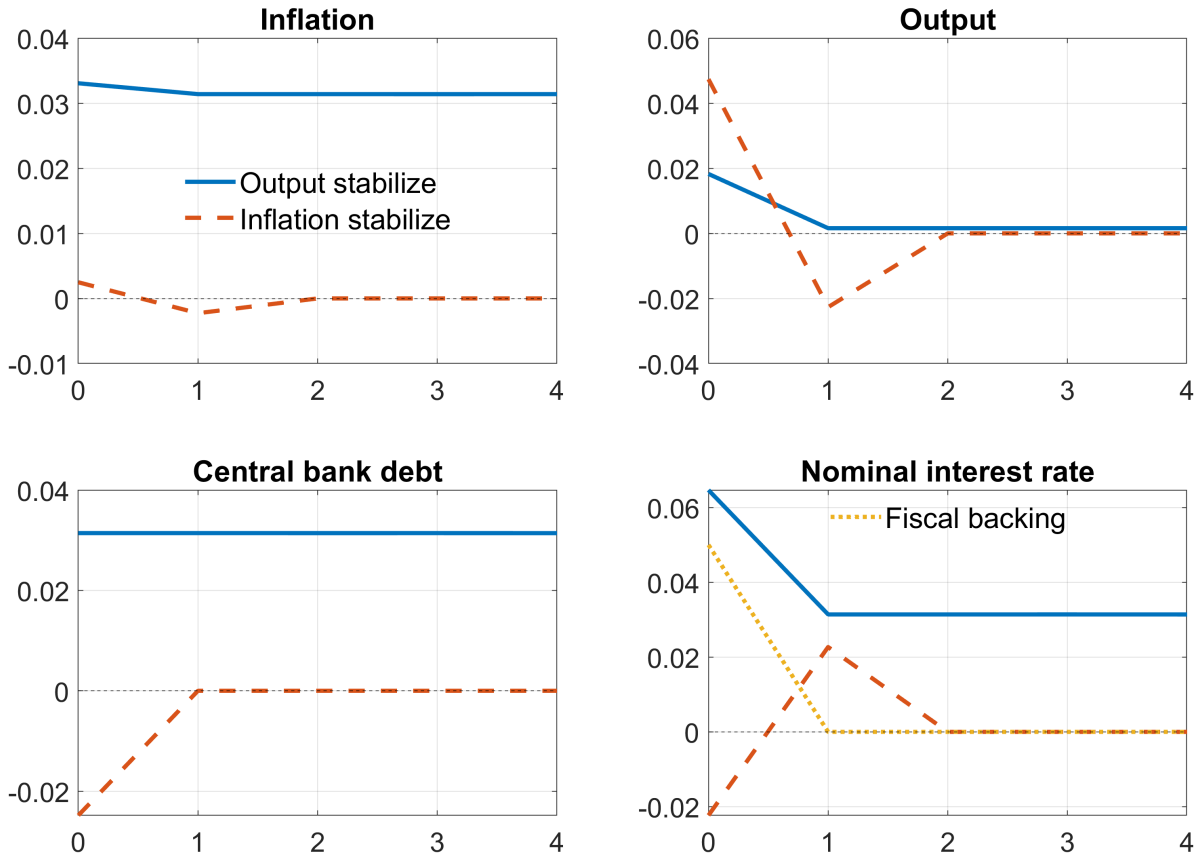


Figure 1: Dynamics under alternative stabilization objectives

Notes: The solid and dashed lines correspond to output stabilization ($q_y/q_\pi \to \infty$) and inflation stabilization ($q_\pi/q_y \to \infty$), respectively. The dotted line in the nominal interest rate panel represents the fiscal-backing benchmark with full stabilization. The figure considers a one-time expansionary demand shock, $e_0 = -0.05$, with $e_t = 0$ for $t \geq 1$. Parameters are $\kappa = 0.1$, $\beta = 0.995$, and $a_h = b_h = \frac{1}{1-\beta}$.

Proposition 4. *Suppose* $\Delta > 0$. *The welfare loss relative to the fiscal-backing benchmark is*

$$\mathscr{W}^N - \mathscr{W}^B = \frac{\Delta^2}{2\Omega},$$

where $\Omega > 0$ *is defined in Appendix B.6.*

Proof. See Appendix B.6. □

3.2 Cost-push shocks

We next consider cost-push shocks, e_t^c, which enter the New Keynesian Phillips curve.

Binding condition We first characterize when cost-push shocks make the remittance constraint binding. As established in Proposition 2, the multiplier on the remittance constraint is constant across dates. Thus, a strictly positive multiplier at any date implies that the constraint binds in

every period. What differs from the demand-shock case is which features of the shock sequence determine whether the constraint binds.

For demand shocks, the funding gap Δ in (20), which depends on the discounted present value of shocks, determines both the binding regime and, conditional on binding, the equilibrium allocation. For cost-push shocks, however, the binding condition depends on two statistics:

$$\Delta^c \equiv \sum_{t=0}^{\infty} \beta^t e_t^c, \qquad \Delta^r \equiv \sum_{t=0}^{\infty} (\beta r)^t e_t^c, \tag{21}$$

where $r \in (0,1)$ is the stable root defined in Appendix C.1. While Δ^c is the standard discounted present value, Δ^r places relatively greater weight on earlier shocks, capturing the role of shock timing in the central bank's financing position. The following proposition provides the necessary and sufficient condition for strict binding.

Proposition 5. *Consider a cost-push shock sequence $\{e_t^c\}_{t=0}^{\infty}$ and suppose $\hat{b}^C_{-1} = 0$. Then the remittance constraint binds strictly in every period if and only if*

$$-a_h H^* + \frac{1-\beta}{\kappa}\Delta^c + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\frac{q_\pi}{q_y}\right] r\Delta^r < 0. \tag{22}$$

Proof. See Appendix C.2. □

Proposition 5 highlights two features of cost-push shocks that distinguish them from demand shocks: how the sign of the shock interacts with the planner's relative stabilization objectives, and how shock timing affects the binding condition. We examine each in turn.

The direction of the shock For demand shocks, Proposition 2 shows that an expansionary shock makes the remittance constraint binding, whereas a recessionary shock does not. This result does not extend to cost-push shocks. Depending on the planner's relative weights on inflation and output stabilization, either a positive or a negative cost-push shock can make the constraint binding. The following proposition characterizes this dependence for a one-time shock.

Proposition 6. *Suppose that $H^* = 0$, $\hat{b}^C_{-1} = 0$, and consider a one-time cost-push shock, $e_0^c \neq 0$ and $e_t^c = 0$ for all $t \geq 1$. Then the remittance constraint binds if and only if*

$$e_0^c > 0 \quad \text{and} \quad \frac{q_\pi}{q_y} \geq \frac{1+\beta+(1-\beta)\sqrt{1+\frac{4\beta}{\kappa}}}{2\beta\kappa},$$
$$e_0^c < 0 \quad \text{and} \quad \frac{q_\pi}{q_y} \leq \frac{1+\beta+(1-\beta)\sqrt{1+\frac{4\beta}{\kappa}}}{2\beta\kappa}.$$

Proof. See Appendix C.3. □

Proposition 6 identifies two distinct mechanisms. For a positive cost-push shock, a planner placing sufficiently large weight on inflation stabilization contracts output to limit inflation. This requires higher nominal interest rates, increasing interest payments on reserves, while the relatively small increase in inflation provides limited relief to the real value of nominal liabilities. Consequently, the remittance constraint binds when q_π/q_y is sufficiently large.

For a negative cost-push shock, the mechanism operates through deflation. When the planner places relatively greater weight on output stabilization, it allows inflation to adjust more while limiting output fluctuations. The resulting deflation raises the real value of the central bank's outstanding nominal liabilities, potentially outweighing the reduction in interest expenses and tightening its funding constraint. Thus, when q_π/q_y is sufficiently small, a negative rather than a positive cost-push shock makes the constraint binding.

The timing of the shock Unlike demand shocks, the timing of cost-push shocks matters even when their discounted present value is held fixed. To isolate this effect, consider one-time positive cost-push shocks that differ only in their arrival date, with their sizes adjusted to preserve the same discounted present value. Proposition 5 implies that delaying a shock cannot turn a nonbinding constraint into a binding one.

Proposition 7. *Suppose that $H^* = 0$, $\hat{b}^C_{-1} = 0$, and $\bar{e}^c > 0$. For each $T \in \{0,1,2,\ldots\}$, consider the one-time cost-push shock*

$$e_t^{(T)} = \begin{cases} \dfrac{\bar{e}^c}{\beta^T}, & t = T, \\ 0, & t \neq T, \end{cases}$$

so that all shock sequences have the same discounted present value. Either the funding constraint is slack for every T, or there exists a finite cutoff $\bar{T} \geq 0$ such that, under $e^{(T)}$, the funding constraint binds for every $T \leq \bar{T}$ and is slack for every $T > \bar{T}$.

Proof. See Appendix C.6. □

The economic intuition follows from the forward iteration of the New Keynesian Phillips curve:

$$\hat{\pi}_0 = \kappa \sum_{t=0}^{T} \beta^t \hat{y}_t + \beta^{T+1} \hat{\pi}_{T+1} + \bar{e}^c,$$

where $\beta^T e_T^{(T)} = \bar{e}^c$. Although the shock's direct contribution to current inflation is identical across arrival dates, a more distant shock allows the planner to adjust output over more periods before the shock arrives, mitigating its effects on inflation and the central bank's funding position.

In summary, the difference between cost-push and demand shocks arises from the inflation-output stabilization trade-off. Demand shocks do not create such a trade-off and can be offset

through nominal interest rate adjustments. Their effects on the optimal allocation operate through the central bank's intertemporal budget constraint, which depends only on their discounted sum. Cost-push shocks instead shift the Phillips curve directly, making their timing and the planner's relative preferences for output and inflation stabilization relevant.

Allocation We next examine how the binding remittance constraint changes the Ramsey allocation. We compare the equilibrium in which the funding constraint binds with the corresponding nonbinding equilibrium.

Proposition 8. *Suppose $H^* = 0$, $\hat{b}^C_{-1} = 0$, and consider a one-time cost-push shock and suppose that the funding constraint binds. Relative to the nonbinding equilibrium, the binding equilibrium features higher inflation and higher output on impact:*

$$\hat{\pi}^N_0 > \hat{\pi}^B_0, \qquad \hat{y}^N_0 > \hat{y}^B_0.$$

Moreover, if $e^c_0 > 0$, then, for all $t \geq 1$,

$$\hat{\pi}^N_t < \hat{\pi}^B_t, \qquad \hat{y}^N_t < \hat{y}^B_t.$$

Proof. See Appendix C.4. □

Figure 2 illustrates the dynamics following a positive one-time cost-push shock. On impact, the binding constraint limits the increase in the nominal interest rate relative to the nonbinding equilibrium, resulting in higher output and inflation. Under commitment, the planner partly compensates by promising a stronger contraction in subsequent periods, generating greater deflation and a larger output decline from $t = 1$ onward. The expectation of lower future inflation helps restrain current inflation through the forward-looking Phillips curve, mitigating the inflation-output trade-off despite the constraint on monetary policy.

Welfare We next examine how the funding constraint affects the composition of welfare losses. For demand shocks, fiscal backing permits full stabilization, so a binding constraint raises both the output and inflation components of welfare loss. For cost-push shocks, however, even the unconstrained Ramsey allocation involves an inflation-output stabilization trade-off. Although the total welfare loss is necessarily weakly higher under a binding constraint, the individual components need not follow the same ordering. The constraint alters the optimal allocation between output and inflation stabilization, potentially reducing one component while increasing the other. The following proposition characterizes this comparison.

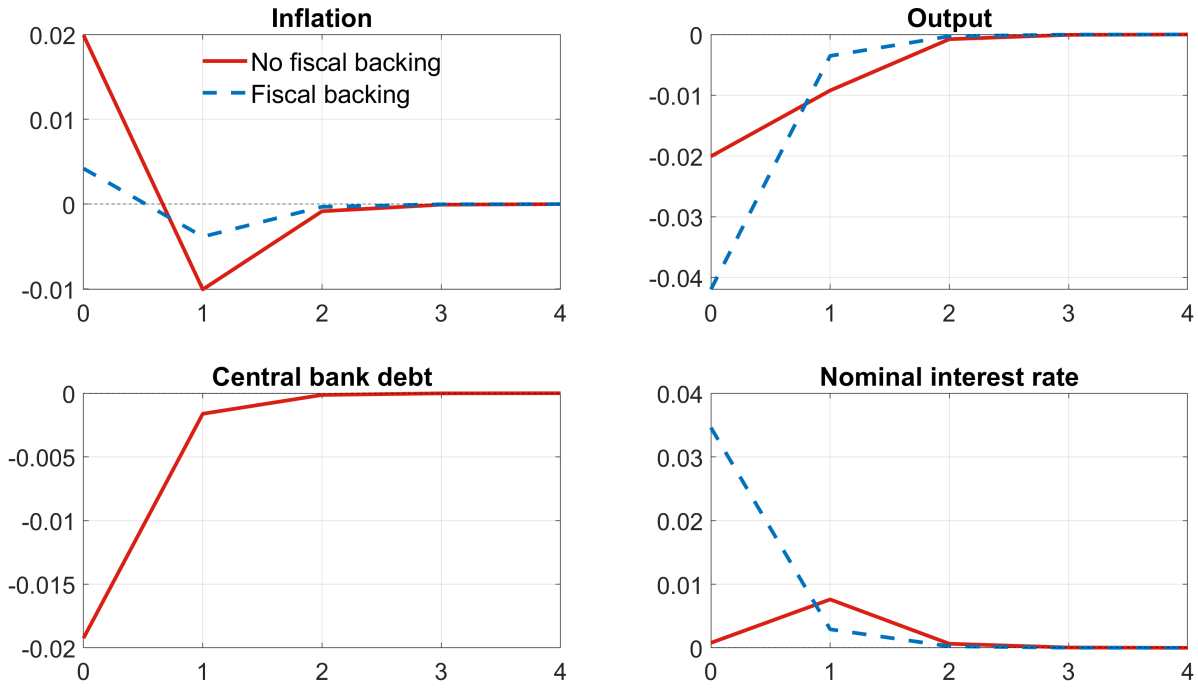


Figure 2: Dynamics following a positive cost-push shock

Notes: The solid and dashed lines correspond to the no-fiscal-backing equilibrium with a binding remittance constraint and the fiscal-backing equilibrium, respectively. The figure considers a one-time positive cost-push shock, $e_0^c = 0.05$, with $e_t^c = 0$ for $t \geq 1$. Parameters are $q_\pi = 10$, $q_y = 1$, $\kappa = 1$, and $\beta = 0.995$. The central-bank debt panel shows only the no-fiscal-backing equilibrium because reserve liabilities are not uniquely determined under fiscal backing.

Proposition 9. *Under the assumptions of Proposition 8, define*

$$\mathscr{W}_\pi^j \equiv \frac{q_\pi}{2}\sum_{t=0}^{\infty}\beta^t(\hat{\pi}_t^j)^2, \qquad \mathscr{W}_y^j \equiv \frac{q_y}{2}\sum_{t=0}^{\infty}\beta^t(\hat{y}_t^j)^2, \qquad j \in \{B,N\},$$

where B and N denote the fiscal-backing and no-fiscal-backing equilibria, respectively. Suppose that the funding constraint binds strictly. If $e_0^c > 0$, then

$$\mathscr{W}_\pi^N > \mathscr{W}_\pi^B, \qquad \mathscr{W}_y^N < \mathscr{W}_y^B.$$

If $e_0^c < 0$, then

$$\mathscr{W}_y^N > \mathscr{W}_y^B,$$

while the ordering of the inflation components is generally ambiguous, as characterized by the necessary and sufficient condition in Appendix C.5.

Proof. See Appendix C.5. □

For a positive cost-push shock, the binding constraint increases inflation losses but reduces output losses. By limiting monetary tightening, it leads to greater inflation variation but a smaller

overall output distortion. For a negative cost-push shock, output losses increase, while the effect on inflation losses depends on the magnitude of the distortion induced by the constraint.

We next examine how the timing of a cost-push shock affects the welfare cost of no fiscal backing, holding its discounted present value fixed.

Proposition 10. *Under the assumptions of Proposition 7, let* $\mathscr{W}^{\mathrm{N}}(T)$ *and* $\mathscr{W}^{\mathrm{B}}(T)$ *denote the date-zero welfare losses under no fiscal backing and fiscal backing for the anticipated cost-push shock* $e^{(T)}$*, respectively. For any* $0 \leq T_1 < T_2$*,*

$$0 \leq \mathscr{W}^{\mathrm{N}}(T_2) - \mathscr{W}^{\mathrm{B}}(T_2) \leq \mathscr{W}^{\mathrm{N}}(T_1) - \mathscr{W}^{\mathrm{B}}(T_1),$$

where the second inequality is strict whenever the remittance constraint binds strictly under $e^{(T_1)}$*. Moreover,* $\mathscr{W}^{\mathrm{B}}(T_1) < \mathscr{W}^{\mathrm{B}}(T_2)$*.*

Proof. See Appendix C.7. □

This proposition states that when the shock is distance, the welfare cost of no fiscal backing decreases. Its intuition is similar to Proposition 7. When the shock is distanced, the planner has a longer time to offset the shock by changing the future output. However, a perhaps surprising result is that, in the fiscal backing, the welfare cost of more distanced shock is larger than that of near future shock. Therefore, a distanced shock is not necessarily a good news.

4 Determinants of the Value of Fiscal Backing

We consider three elements that are key determinant of the value of fiscal backing but are absent in the baseline model.

4.1 Long-Duration Assets

We extend the baseline model to examine how the duration of central bank assets affects its funding position. Central bank assets typically have longer maturities than their liabilities, exposing the central bank to interest-rate risk. When nominal interest rates rise, the market value of long-duration bonds falls, potentially generating substantial valuation losses. As Bernanke (2012) notes, "the Federal Reserve could incur financial losses should interest rates rise to an unexpected extent."

Long-duration bond structure To isolate the role of asset duration, we modify the financial structure of the baseline model while retaining household preferences, firms' price-setting problem, and the remittance constraint. Specifically, we remove one-period reserves, replace one-period Treasury bonds with long-duration bonds, and allow the central bank to hold these bonds

alongside households. Let D_t^T denote total long-duration Treasury debt outstanding, with

$$D_t^T = D_t^{T,H} + D_t^{T,CB},$$

where $D_t^{T,H}$ and $D_t^{T,CB}$ are holdings by households and the central bank, respectively. Following Woodford (2001), a bond issued at t pays nominal coupons of one unit at $t+1$, ρ units at $t+2$, and more generally ρ^j units at $t+j+1$. Thus, $\rho = 0$ corresponds to one-period debt, while larger values of ρ represent longer-duration bonds. Each inherited bond pays a current coupon and retains a continuation value of ρQ_t^T. The Treasury budget constraint is therefore

$$Q_t^T D_t^T + P_t T_t + P_t H_t = (1 + \rho Q_t^T) D_{t-1}^T.$$

Unlike in the baseline model, the central bank holds Treasury bonds as assets. Its budget constraint becomes

$$(1 + \rho Q_t^T) D_{t-1}^{T,CB} = Q_t^T D_t^{T,CB} + P_t H_t.$$

The coupon payments and continuation value of inherited bonds finance new bond purchases and remittances to the Treasury.

Households hold long-duration Treasury bonds instead of one-period nominal claims. Their budget constraint is

$$P_t C_t + Q_t^T D_t^{T,H} + P_t T_t = (1 + \rho Q_t^T) D_{t-1}^{T,H} + w_t N_t + P_t \Phi_t.$$

Given unchanged household preferences, the corresponding Euler equation is

$$\frac{Q_t^T}{C_t} = \beta E_t \left[\frac{\delta_{t+1}(1 + \rho Q_{t+1}^T)}{\delta_t C_{t+1} \pi_{t+1}} \right].$$

The assumption that the central bank holds only assets is for analytical tractability. Section 5 considers the more general case in which the central bank holds long-duration assets while issuing reserves.

Ramsey problem We retain the baseline quadratic welfare objective, the New Keynesian Phillips curve, and the remittance constraint. The changes in the financial structure affect only the household Euler equation and the Treasury and central-bank budget constraints, which are replaced by their long-duration counterparts.

Under the same perfect-foresight convention as in Section 2.6, the no-fiscal-backing planner minimizes (13) over

$$\left\{ \hat{y}_t, \hat{\pi}_t, \hat{Q}_t^T, \hat{d}_t^T, \hat{d}_t^{T,CB}, \hat{T}_t, \hat{H}_t \right\}_{t=0}^{\infty},$$

subject to the unchanged Phillips curve (14), the remittance constraint (18), and

$$\hat{Q}_t^T = \hat{y}_t - \hat{y}_{t+1} - \hat{\pi}_{t+1} + \rho\beta\hat{Q}_{t+1}^T + e_t, \tag{23}$$

$$\hat{d}_{t-1}^T - \hat{\pi}_t - (1-\beta)b_h^D(\hat{H}_t + \hat{T}_t) = \beta\left[\hat{d}_t^T + (1-\rho)\hat{Q}_t^T\right], \tag{24}$$

$$\beta\hat{d}_t^{T,CB} = \hat{d}_{t-1}^{T,CB} - \hat{\pi}_t - \beta(1-\rho)\hat{Q}_t^T - (1-\beta)a_h^D\hat{H}_t. \tag{25}$$

Here, $d_t^T \equiv D_t^T/P_t$ and $d_t^{T,CB} \equiv D_t^{T,CB}/P_t$ denote real bond holdings, while a_h^D and b_h^D are the steady-state coefficients defined in Appendix D.1. In fiscal backing case, the remittanace constraint is absent.

Relative to the baseline model, the Euler equation (23) introduces the continuation value of long-duration bonds through $\rho\beta\hat{Q}_{t+1}^T$. The budget constraints (24) and (25) account for the value of inherited bond holdings. In particular, changes in the current bond price affect both the cost of new bond purchases and the continuation value of inherited bonds, with a net effect proportional to $1-\rho$.

Binding property As in the baseline model, the remittance constraint either binds at every date or permits full stabilization. However, replacing reserve liabilities with long-duration assets changes the condition under which the constraint binds. The following proposition shows that this condition remains characterized by a single statistic, which now depends on asset duration.

Proposition 11. *Suppose* $0 \le \rho < 1$, $\hat{d}_{-1}^{T,CB} = H^* = 0$, *and a bounded deterministic demand-shock sequence* $\{e_t\}_{t=0}^{\infty}$. *Define*

$$\Delta_\rho \equiv \sum_{t=0}^{\infty} \beta^{t+1}(1-\rho^{t+1})e_t.$$

The remittance constraint binds strictly at every date if and only if $\Delta_\rho > 0$. *If* $\Delta_\rho \le 0$, *the optimal allocation achieves full stabilization:*

$$\hat{\pi}_t = \hat{y}_t = 0, \qquad t \ge 0.$$

Proof. See Appendix D.2. □

Unlike in the baseline model, asset duration enters the binding condition through the weights. For nonnegative demand shocks, increasing ρ reduces Δ_ρ, making the remittance constraint less likely to bind. The role of duration becomes particularly transparent in the consol limit, $\rho = 1$.

Proposition 12. *Suppose* $\rho = 1$ *and the initial asset position satisfies* $\hat{d}_{-1}^{T,CB} = a_h^D H^*$. *Then, for any bounded deterministic demand-shock sequence* $\{e_t\}_{t=0}^{\infty}$, *the optimal allocation achieves full*

stabilization. Hence, the remittance constraint does not restrict the optimal allocation. The shock is absorbed entirely through the consol price:

$$\hat{Q}_t^T = \sum_{j=0}^{\infty} \beta^j e_{t+j}.$$

Proof. See Appendix D.3. □

The mechanism is that changes in bond prices affect both the cost of acquiring new bonds and the continuation value of inherited holdings. For a one-period bond, the inherited claim has no continuation value, so only the purchase-price effect remains. With longer-duration bonds, the change in the value of inherited holdings increasingly offsets the purchase-price effect. In the consol limit, this offset is exact: the bond-price terms disappear from the central-bank budget constraint (25), allowing demand shocks to be absorbed entirely through asset prices without affecting output or inflation.

This result distinguishes mark-to-market valuation losses from changes in the central bank's funding position. Although higher interest rates reduce the market value of long-duration assets, the resulting lower bond prices also reduce the cost of maintaining or acquiring these assets. In this asset-only model, the net effect on the funding constraint diminishes with duration and vanishes in the consol limit.

Dynamics and welfare effects of duration Figure 3 illustrates the responses to a one-time 10% demand shock for different asset durations. As ρ increases, the responses of output and inflation become smaller, approaching full stabilization in the consol limit, consistent with Proposition 12. A larger share of the shock is instead absorbed through bond-price adjustments.

Two mechanisms explain this result. First, the resale-value channel discussed above reduces the net effect of bond-price movements on the central bank's funding position, as reflected in the coefficient $1-\rho$ in (25). Second, the intertemporal asset-price channel allows demand shocks to be absorbed through both current and future bond prices, captured by the continuation-value term $\rho\beta\hat{Q}_{t+1}^T$ in (23). Together, these channels reduce the output and inflation adjustments required to accommodate demand shocks as duration increases.

Although the consol limit achieves full stabilization, the welfare loss need not decrease monotonically with duration. Let $\mathscr{W}_\beta(\rho)$ denote the minimized welfare loss for a given $\beta < 1$, and define $\mathscr{W}(\rho) \equiv \lim_{\beta\uparrow 1} \mathscr{W}_\beta(\rho)$. The following proposition characterizes when welfare losses decrease monotonically with duration in the patient limit.

Proposition 13. *Consider a one-time positive demand shock and suppose that the central bank starts from a steady-state funding position, $\hat{d}_{-1}^{T,CB} = a_h^D H^*$. In the patient limit $\beta \uparrow 1$, longer asset*

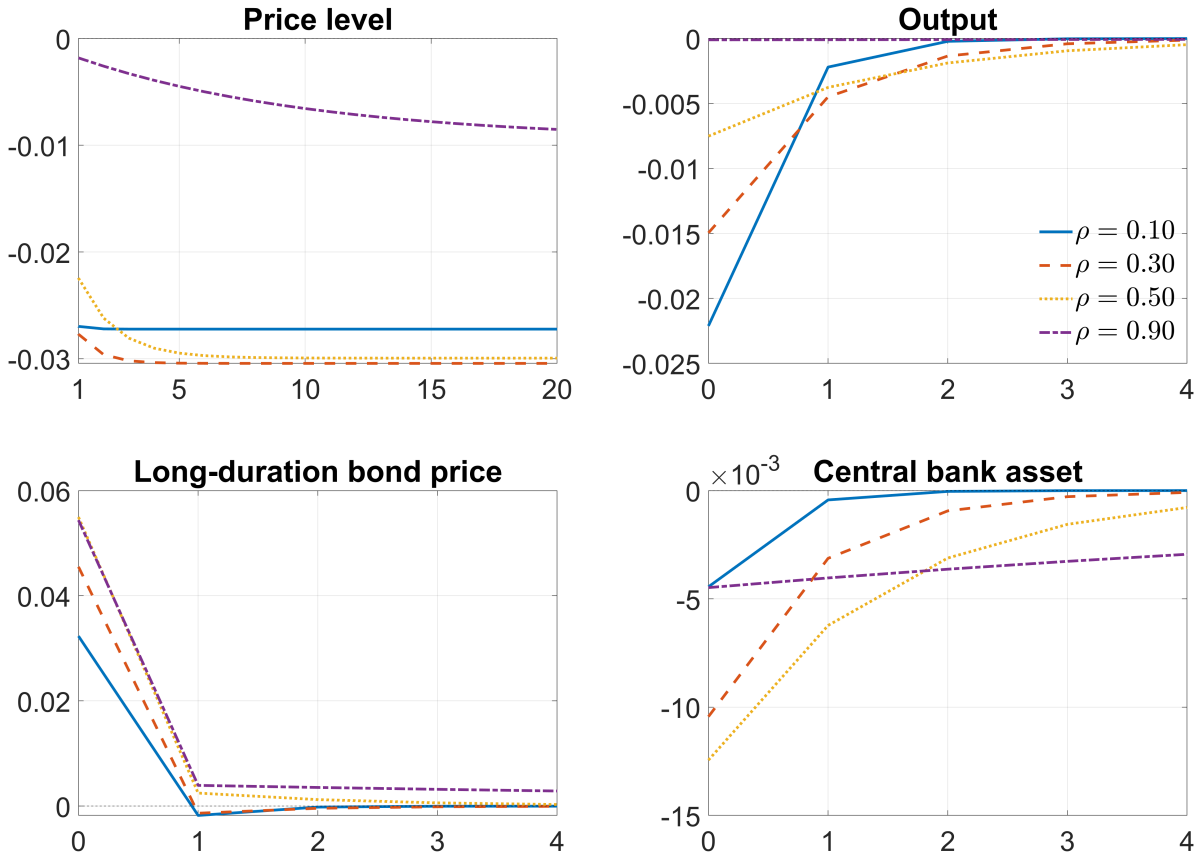


Figure 3: Dynamics with different asset durations

Notes: The figure shows equilibrium responses to a one-time demand shock, $e_0 = 0.10$, with $e_t = 0$ for $t \geq 1$, under different values of the asset-duration parameter ρ. Parameters are $q_\pi = q_y = 1$, $\kappa = 1$, and $\beta = 0.995$.

duration strictly reduces the welfare loss, $\mathscr{W}'(\rho) < 0$ for all $\rho \in (0,1)$, if and only if

$$\kappa \geq 1 \quad or \quad \begin{cases} 0 < \kappa < 1, \\ \dfrac{q_\pi}{q_y} \leq \dfrac{4 - \kappa + \sqrt{\kappa^2 + 8\kappa}}{2\kappa(1-\kappa)}. \end{cases} \tag{26}$$

Proof. See Appendix D.4. □

This condition holds for a wide range of parameters. When $\kappa \geq 1$, condition (26) holds automatically, so longer duration always reduces welfare losses in the patient limit. When $0 < \kappa < 1$, monotonicity depends on the planner's relative weight on inflation stabilization. For example, when $\kappa = 0.1$, the condition reduces to $\frac{q_\pi}{q_y} \leq \frac{80}{3}$. Thus, longer duration improves the central bank's ability to absorb demand shocks through bond-price adjustments though its welfare benefits need not be monotonic.

4.2 Liquidity Value of Reserves

The baseline model imposes an intertemporal backing requirement on central bank liabilities: outstanding reserves must ultimately be supported by the present discounted value of the central bank's future surpluses. The constraint that the central bank faces in the reality may be weaker than the intertemporal backing requirement. Private agents may be willing to hold central bank liabilities even when they are not fully backed by future pecuniary surpluses. For example, reserves provide liquidity services. We therefore extend the baseline model by allowing households to derive utility directly from reserve holdings. The central bank issues reserves that are valued not only for their financial payoff but also for their liquidity services.

Model We extend the baseline household preferences to allow reserves to provide liquidity services and generalize consumption utility to CRRA. To isolate this mechanism, we also eliminate Treasury debt, leaving reserves as the only nominal asset held by households. Firms' price setting, the central bank budget constraint, and the remittance lower bound remain unchanged.

Let $b_t^C \equiv B_t^C/P_t$ denote real reserve holdings. Household preferences become

$$\max_{\{C_t,N_t,B_t^C\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t \delta_t \left[\frac{C_t^{1-\sigma}}{1-\sigma} + \frac{\omega}{1-\eta}(b_t^C)^{1-\eta} - \frac{N_t^{1+\nu}}{1+\nu} \right]. \tag{27}$$

The reserve Euler equation is

$$Q_t^C C_t^{-\sigma} = \beta E_t \left[\frac{\delta_{t+1}}{\delta_t} \frac{C_{t+1}^{-\sigma}}{\Pi_{t+1}} \right] + \omega (b_t^C)^{-\eta}. \tag{28}$$

The additional term captures the marginal liquidity value of reserves, allowing households to hold them at a lower pecuniary return than otherwise identical nonliquid assets.

Without Treasury debt, the Treasury budget constraint reduces to $\hat{T}_t + \hat{H}_t = 0$, while reserve-market clearing equates household reserve holdings with central bank liabilities.

Substituting linearized Euler equation (28) into central bank budget constraint and iterating forward gives

$$\hat{b}_{-1}^C = E_0 \sum_{t=0}^{\infty} \underbrace{\left[\bar{Q}^C \left(1-(1-\bar{r})\eta\right)\right]^t}_{\text{additional discounting}} \left[\hat{\pi}_t + \beta \hat{Q}_t^{nl} - (1-\beta) a_h \hat{H}_t + \underbrace{(\bar{Q}^C - \beta)\sigma \hat{y}_t}_{\text{liquidity valuation}} \right], \tag{29}$$

$$\hat{Q}_t^{nl} = \sigma \hat{y}_t - \sigma E_t \hat{y}_{t+1} - E_t \hat{\pi}_{t+1} + e_t.$$

Here, $\hat{Q}_t^{nl}$ denotes the price of a nonliquid one-period claim. This representation reveals two channels through which reserve liquidity affects the central bank's financing capacity.

The first is a *discounting channel*. Issuing more reserves reduces their marginal liquidity value, lowering their price and raising their return. Higher return leads to more discounting of future

financing needs and resources. When $\eta > 1$, the effective discount factor falls below its baseline value, reducing the weight on both future funding shortfalls and surpluses.

The second is a *liquidity valuation channel*. Through this channel, higher output increases the central bank's surplus. Higher current output raises the value of reserve liquidity relative to consumption, thereby increasing reserve prices. Higher reserve prices allow the central bank to obtain financing from the market at a lower cost, increasing its surplus. The second mechanism states that when the coefficient of relative risk aversion is small, the central bank cannot cheaply raise funds from the market despite the liquidity value.

Together, these channels change the central bank's financing capacity, but their net effect depends on the paths of output and inflation and the timing of financing needs and resources. Consequently, reserve liquidity need not reduce the stabilization loss from the absence of fiscal backing.

Ramsey problem Because household utility now includes reserve liquidity services, its welfare approximation generally contains terms beyond output and inflation stabilization. To isolate how reserve liquidity changes the financing constraint, we nevertheless retain the baseline quadratic stabilization objective (13), without directly penalizing deviations in reserve holdings. Our welfare comparisons therefore refer to this stabilization criterion rather than the full household welfare implied by (27).

The planner minimizes (13) subject to the Phillips curve (14), the modified reserve Euler equation (28), the central bank budget constraint (17), the remittance constraint (18), and the Treasury budget constraint.

Results One might expect that the liquidity value of reserves necessarily reduces the welfare cost of no fiscal backing. Since households value the liquidity services of reserves, the central bank can raise funds on more favorable terms, potentially easing its financing constraint. However, liquidity services also alter the discounting of future financing needs and resources, which can worsen the central bank's funding position. Consequently, whether useful reserves reduce the welfare loss depends on the relative strength of the liquidity valuation and discounting channels.

For the following propositions, we consider a one-time expansionary demand shock with $\hat{b}^C_{-1} = H^* = 0$ and a common funding gap $\Delta > 0$. We restrict attention to bounded equilibrium deviations and hold κ, q_y, and q_π fixed when varying σ or η.

We first establish a sufficient condition under which liquidity services improve welfare. Rather than directly comparing the optimal allocations, the following proposition identifies when a feasible allocation exists with uniformly smaller output and inflation deviations than the no-liquidity optimum.

Proposition 14. *Suppose* $0 < 1-(1-\bar{r})\eta \leq \bar{r} < 1$. *There exists a threshold* $\sigma_c(\eta) \geq 0$ *such that, if and only if* $\sigma > \sigma_c(\eta)$, *some* $m > 1$ *makes the allocation*

$$\hat{y}_t^{RF} = \frac{1}{m}\hat{y}_t^{NL}, \qquad \hat{\pi}_t^{RF} = \frac{1}{m}\hat{\pi}_t^{NL}, \qquad t \geq 0,$$

feasible in the reserve-in-utility model for the same funding gap. Moreover, $\sigma_c(1) = 0$, *and* $\sigma_c(\eta)$ *is strictly increasing in* η, *holding other parameters fixed. Here, NL denotes the no-liquidity optimum and RF a feasible reserve-in-utility allocation. The threshold is defined in Appendix F.2.*

Proof. See Appendix F.2. □

Proposition 14 provides a sufficient condition for welfare improvement. When the condition holds, the reserve-in-utility economy can accommodate the same funding gap with smaller fluctuations in output and inflation, implying a strictly lower welfare loss. However, the condition is not necessary: even when it fails, a different allocation may still improve welfare.

The key mechanism depends on the curvature of consumption utility, governed by σ. Higher output raises the liquidity value of reserves relative to consumption, increasing their price and allowing the central bank to raise funds more cheaply. A larger σ strengthens this financing benefit. Conversely, when σ is small, higher output generates only a limited increase in reserve prices, weakening the central bank's ability to raise additional funds on favorable terms. Meanwhile, when $\eta > 1$, the lower effective discount factor reduces the contribution of future financing resources, potentially offsetting the liquidity valuation benefit.

The next proposition shows that this adverse effect can dominate. Under more restrictive assumptions, it directly characterizes when liquidity services increase the minimized welfare loss relative to the no-liquidity benchmark.

Proposition 15. *Suppose* $q_y = q_\pi > 0$, $\kappa > 0$, $0 < \bar{r} < 1$, *and* $1 < \eta < 1/(1-\bar{r})$. *For any fixed* $\beta \in (0,1)$, *there exists a unique threshold* $\sigma_w \geq 0$ *such that*

$$\mathscr{W}^R > \mathscr{W}^{NL} \quad \text{if and only if} \quad 0 < \sigma < \sigma_w.$$

Here, R denotes the reserve-in-utility Ramsey optimum. Moreover, $\sigma_w > 0$ *whenever* $\kappa \leq 1$, *whereas* $\sigma_w = 0$ *is possible when* $\kappa > 1$. *The threshold is characterized in Appendix F.3.*

Proof. See Appendix F.3. □

Proposition 15 establishes that liquidity services can make the absence of fiscal backing more costly. When the curvature of consumption utility is sufficiently low, the central bank gains little additional financing capacity from higher reserve prices, while the stronger discounting of future resources worsens its funding position. Under the conditions of the proposition, this adverse effect can outweigh the liquidity valuation benefit, generating a strictly larger welfare loss than in the benchmark without liquidity services.

4.3 Distortionary Tax

A practical concern is that fiscal support for the central bank may require distortionary taxation (Carstens, 2023; Ramsden, 2023). This raises the question of whether fiscal backing remains desirable once the fiscal costs of supporting the central bank are taken into account. To examine this issue, we relax the baseline assumption that the Treasury finances fiscal support through nondistortionary lump-sum taxation. Instead, we introduce a distortionary sales tax, whose adjustments affect firms' pricing decisions and enter the New Keynesian Phillips curve. Consequently, fiscal support now affects the inflation–output stabilization trade-off.

Sales tax and fiscal financing Firm i faces a sales tax τ_t, retaining only the fraction $1-\tau_t$ of its revenue. Keeping the baseline production technology and price adjustment costs, the firm chooses its price path to solve

$$\max_{\{p_t(i)\}_{t=0}^{\infty}} \quad E_0 \sum_{t=0}^{\infty} \beta^t \Lambda_t \left[(1-\tau_t) p_t(i) y_t(i) - (1-\varsigma) w_t N_t(i) - \frac{\varphi}{2} \left(\frac{p_t(i)}{p_{t-1}(i)} - 1 \right)^2 P_t Y_t \right], \tag{30}$$

where $\Lambda_t \equiv C_t^{-1}/P_t$, subject to the demand curve (4).

As in the baseline model, firms receive a constant employment subsidy ς, now chosen to offset both the steady-state monopolistic distortion and the sales-tax wedge. The subsidy is financed by a lump-sum tax on households. Since these payments offset each other in the aggregate, we omit them from the household and Treasury budget constraints.

Using the subsidy rule and household labor supply condition, the firm's first-order condition in a symmetric equilibrium becomes

$$N_t C_t^{-1} \left[\varphi(\pi_t - 1)\pi_t + (\theta - 1) \left\{ (1-\tau_t) - (1-\bar{\tau}) N_t^{\nu} C_t \right\} \right] = \beta E_t \left[\frac{\delta_{t+1}}{\delta_t} N_{t+1} C_{t+1}^{-1} \varphi(\pi_{t+1} - 1)\pi_{t+1} \right]. \tag{31}$$

We assume that the Treasury rebates the cyclical component of sales-tax revenue to households through a lump-sum transfer:

$$R_t \equiv \tau_t \left[\int_0^1 p_t(i) y_t(i)\, di - P_t \bar{Y} \right],$$

where $\bar{Y}$ denotes deterministic steady-state output. Because the rebate is lump-sum, it does not affect firms' pricing decisions. In a symmetric equilibrium, net sales-tax revenue is

$$\tau_t P_t Y_t - R_t = \tau_t P_t \bar{Y},$$

and the Treasury budget constraint becomes

$$Q_t^T B_t^T + \tau_t P_t \bar{Y} + P_t H_t = B_{t-1}^T.$$

Thus, tax-rate adjustments affect firms' pricing decisions, while fluctuations in output do not directly affect the Treasury's net tax revenue. This assumption is to highlight the distortion from changes in the tax rate rather than fluctuations in the tax base.

Ramsey problem We retain the baseline quadratic welfare objective (13), household Euler equations, central-bank budget constraint, and remittance constraint. Replacing lump-sum taxation with a distortionary sales tax modifies only the Phillips curve and the Treasury budget constraint.

The Phillips curve becomes

$$\hat{\pi}_t = \kappa \hat{y}_t + \gamma \hat{\tau}_t + \beta \hat{\pi}_{t+1}, \tag{32}$$

where γ measures the effect of tax-rate adjustments on firms' pricing decisions. Its derivation is provided in Appendix E.3. The linearized Treasury budget constraint is

$$\hat{b}^T_{t-1} - \hat{\pi}_t - (1-\beta) b_h (\hat{H}_t + \hat{\tau}_t) = \beta (\hat{b}^T_t + \hat{Q}^T_t).$$

Unlike a conventional sales tax, output fluctuations do not directly enter the Treasury budget constraint because the lump-sum rebate offsets cyclical changes in the tax base.

The unchanged household Euler equations (15), central-bank budget constraint (17), and remittance constraint (18).

The key difference from the baseline model is that fiscal financing now affects the inflation–output trade-off. Under fiscal backing, monetary tightening may require additional Treasury transfers to cover higher interest payments on reserves. Financing these transfers through the sales tax affects firms' pricing decisions and, through (32), the equilibrium paths of output and inflation.

The parameter γ governs the strength of this distortion. When $\gamma = 0$, tax adjustments do not affect firms' pricing decisions, and the Ramsey problem reduces to the nondistortionary-tax benchmark. As γ increases, fiscal support becomes more costly in terms of output and inflation stabilization, potentially reducing the welfare value of fiscal backing.[4]

Results We next examine how the degree of tax distortion affects the Ramsey allocation and the welfare value of fiscal backing. Figure 4 illustrates the equilibrium responses to a one-time expansionary demand shock for different values of γ. Under fiscal backing, transfers from the Treasury to the central bank are unrestricted, but financing these transfers through distortionary taxation affects firms' marginal product, making full stabilization by raising nominal rate suboptimal.

The welfare-relevant no-fiscal-backing allocation is invariant to γ. This is because the no-fiscal-backing equilibrium requires the present discounted value of tax-rate adjustments to be zero.

[4]The parameter γ is determined by structural primitives. Our comparative statics vary γ while holding κ, q_y, and q_π fixed to isolate the effect of the tax wedge on firms' pricing decisions. We therefore interpret this exercise as a comparative static of the linear-quadratic Ramsey problem rather than a change in a single structural primitive.

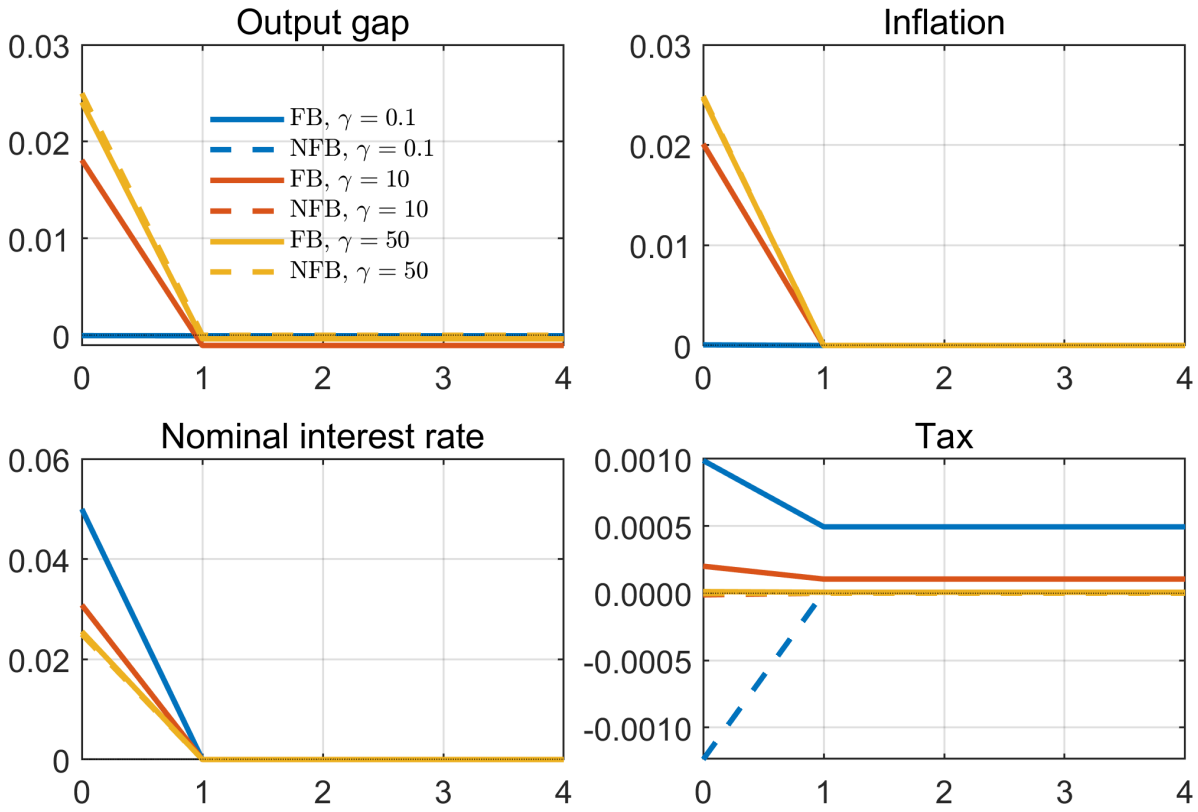


Figure 4: Dynamics under different degrees of tax distortion

Notes: The figure shows equilibrium responses to a one-time expansionary demand shock, $e_0 = -5\%$, with $e_t = 0$ for $t \geq 1$, under different degrees of tax distortion, $\gamma \in \{0.1, 10, 50\}$. Solid lines correspond to fiscal backing and dashed lines to no fiscal backing. Parameters are $q_\pi = q_y = 1$, $\kappa = 1$, and $\beta = 0.995$. Under no fiscal backing, the responses of output, inflation, and the nominal interest rate coincide across values of γ.

In contrast, the fiscal-backing allocation depends on γ. As γ increases, financing transfers to the central bank becomes more distortionary, raising the fiscal-backing welfare loss toward its no-fiscal-backing counterpart.

Figure 4 shows that greater tax distortions reduce the stabilization benefits of fiscal backing, while leaving the no-fiscal-backing allocation unchanged. The following proposition establishes that the welfare value of fiscal backing declines with the degree of tax distortion but remains strictly positive even in the limit.

Proposition 16. *Suppose that the economy is initialized at its deterministic steady state,* $\hat{b}^T_{-1} = \hat{b}^C_{-1} = 0$*, and that* $H^* = 0$*. Consider a one-time expansionary demand shock,* $e_0 < 0$ *and* $e_t = 0$ *for all* $t \geq 1$*. For* $\gamma > 0$*, the welfare value of fiscal backing is strictly decreasing in the degree of tax distortion:*

$$\frac{d}{d\gamma}\left[\mathscr{W}^N - \mathscr{W}^B(\gamma)\right] < 0.$$

Nevertheless, its limiting value remains strictly positive:

$$\lim_{\gamma\to\infty}\left[\mathscr{W}^{N}-\mathscr{W}^{B}(\gamma)\right]>0.$$

Proof. See Appendix E.2. □

The first result reflects the increasing cost of financing central-bank support through distortionary taxation. When γ is small, the Treasury can provide fiscal support with relatively limited effects on the inflation–output trade-off, allowing the planner greater flexibility in adjusting the nominal interest rate. As γ increases, tax-rate adjustments have stronger effects on firms' pricing decisions, making fiscal support more costly. Consequently, the fiscal-backing welfare loss rises toward its no-fiscal-backing counterpart.

The second result is perhaps unexpected: fiscal backing retains a strictly positive welfare value even when raising additional tax revenue becomes prohibitively costly. The reason is that fiscal backing provides not only access to tax revenue but also a mechanism for intertemporal transfers between the Treasury and the central bank. When the central bank faces a temporary funding shortfall, the Treasury can finance a transfer by adjusting its own balance sheet and subsequently recover these resources through future central-bank remittances. Fiscal backing thus acts as an internal credit line between the two authorities, allowing temporary financing needs to be accommodated without relying on permanently higher distortionary taxation. This intertemporal financing channel remains valuable even as γ approaches infinity.

Appendix E.5 allows tax revenue to depend on actual output, $\tau_t P_t Y_t$. Although the welfare value of fiscal backing need not decrease monotonically with tax distortions, it does so over a broad parameter region and remains strictly positive as $\gamma\to\infty$.

5 Quantitative Model and Analysis

This section extends the baseline model to a more realistic setting. We retain the baseline structure of Section 2, including nominal rigidities, separate Treasury and central-bank budget constraints, and a one-sided remittance constraint, while combining the mechanisms studied separately in Section 4. In particular, we introduce financial intermediation, incorporate both assets and liabilities on the central bank's balance sheet, and allow for distortionary fiscal adjustment. By calibrating the model and solving the full nonlinear Ramsey problem, we quantify how limited fiscal backing affects optimal policy responses, adjustments in public and private balance sheets, and the resulting welfare costs.

Households As in the baseline model, households choose consumption, labor supply, and nominal asset holdings. The main difference is that households hold one-period bank deposits instead

of central bank reserves directly, and long-duration Treasury bonds. We also introduce liquidity services from deposits and generalize consumption utility to CRRA preferences. Household preferences are

$$\max_{\{C_t,N_t,D_t,B_t^{T,H}\}_{t=0}^{\infty}} \quad E_0\sum_{t=0}^{\infty}\beta^t\delta_t\left[\frac{C_t^{1-\sigma}}{1-\sigma}+\frac{\omega_D}{1-\eta_D}\left(\frac{D_t}{P_t}\right)^{1-\eta_D}-\chi_N\frac{N_t^{1+\nu}}{1+\nu}\right],$$

where δ_t is the preference shock generating fluctuations in aggregate demand, as in the baseline model. The household budget constraint is

$$P_tC_t+Q_t^DD_t+Q_t^TB_t^{T,H}=D_{t-1}+(1+\rho Q_t^T)B_{t-1}^{T,H}+w_tN_t+P_t\Phi_t+P_t\Phi_t^B+\Phi_t^A.$$

Here, D_t denotes one-period nominal deposits, $B_t^{T,H}$ denotes household holdings of long-duration Treasury bonds, and Φ_t, Φ_t^B, and Φ_t^A denote profits from goods-producing firms, bank profits, and cash flows from other bank assets, respectively. The long-duration bond has the same coupon structure as in Section 4.1.

Let $d_t\equiv D_t/P_t$ denote real deposits. Household optimality implies

$$C_t^{-\sigma}Q_t^D=\beta E_t\left[\frac{\delta_{t+1}C_{t+1}^{-\sigma}}{\delta_t\pi_{t+1}}\right]+\omega_Dd_t^{-\eta_D}, \tag{33}$$

$$C_t^{-\sigma}Q_t^T=\beta E_t\left[\frac{\delta_{t+1}C_{t+1}^{-\sigma}(1+\rho Q_{t+1}^T)}{\delta_t\pi_{t+1}}\right]. \tag{34}$$

Relative to the baseline Euler equations, (33) incorporates the marginal liquidity value of deposits, while (34) accounts for the continuation value of long-duration bonds. Thus, deposits command a liquidity premium relative to otherwise comparable nonliquid claims.

To maintain the demand-shock convention of the analytical model, normalize $\bar{\delta}=1$ and define

$$e_t^n\equiv E_t\widehat{\delta}_{t+1}-\widehat{\delta}_t. \tag{35}$$

As in the baseline model, $e_t^n<0$ represents an expansionary demand disturbance, which raises the nominal interest rate required to stabilize demand.

Banks Unlike the baseline model, central bank reserves are held by banks rather than households. Following the collateral-based framework of Piazzesi et al. (2019), competitive banks issue one-period deposits D_t and invest in one-period central bank reserves B_t^C and other assets A_t, each paying one nominal unit in the following period. Banks maximize discounted cash flows:

$$\max_{\{B_t^C,A_t,D_t\}_{t=0}^{\infty}} \quad E_0\sum_{t=0}^{\infty}\beta^t\Lambda_t\left[B_{t-1}^C+A_{t-1}-D_{t-1}-Q_t^CB_t^C-Q_t^AA_t+Q_t^DD_t\right], \qquad \Lambda_t\equiv\frac{\delta_tC_t^{-\sigma}}{P_t},$$

subject to the collateral constraint

$$Q_t^D D_t \leq \ell \left(Q_t^C B_t^C + \rho_A Q_t^A A_t \right), \qquad 0 < \ell \leq 1, \quad 0 < \rho_A < 1. \tag{36}$$

Here, ℓ governs the collateralizable share of bank assets, while ρ_A captures the lower collateral value of other assets relative to reserves.

Let $\xi_t \geq 0$ denote the multiplier on (36), and define the price of a one-period claim without liquidity or collateral services as

$$Q_t^S \equiv \beta E_t \left[\frac{\Lambda_{t+1}}{\Lambda_t} \right].$$

The bank's first-order conditions are

$$Q_t^C (1 - \ell \xi_t) = Q_t^S, \qquad Q_t^A (1 - \ell \rho_A \xi_t) = Q_t^S, \qquad Q_t^D (1 - \xi_t) = Q_t^S.$$

Because deposits provide liquidity services, the household Euler equation (33) implies $Q_t^D > Q_t^S$. Hence, $\xi_t > 0$, and the collateral constraint binds.

Other bank assets are supplied in a fixed real quantity by a household-owned entity, $\frac{A_t}{P_t} = \bar{a}$, whose net cash flows are rebated to households. This banking structure makes reserve demand endogenous, unlike in the baseline model: reserves relax banks' collateral constraint and thereby support the issuance of liquid deposits.

Firms We retain the baseline production technology, monopolistic competition, Rotemberg price adjustment costs, and employment subsidy, while introducing the distortionary sales tax τ_t as in Section 4.3. Unlike that extension, we do not rebate cyclical fluctuations in the tax base. Consequently, tax revenue is $\tau_t P_t Y_t$ rather than $\tau_t P_t \bar{Y}$, so fluctuations in output directly affect the Treasury's revenue. The nonlinear firm optimality condition is provided in Appendix G.1.

Treasury and central bank Relative to the baseline model, the Treasury issues long-duration bonds, levies distortionary sales tax τ_t, and finances constant government spending G. Its budget constraint is

$$Q_t^T B_t^T + \tau_t P_t Y_t + P_t H_t = (1 + \rho Q_t^T) B_{t-1}^T + P_t G, \tag{37}$$

where $B_t^T = B_t^{T,H} + B_t^{T,C}$ denotes total Treasury debt held by households and the central bank. As in the baseline model, $H_t > 0$ represents remittances from the central bank to the Treasury.

Unlike the baseline model, the central bank holds long-duration Treasury bonds as assets while continuing to issue one-period reserves as liabilities. Its budget constraint becomes

$$Q_t^C B_t^C + (1 + \rho Q_t^T) B_{t-1}^{T,C} = B_{t-1}^C + Q_t^T B_t^{T,C} + P_t H_t.$$

Thus, reserve issuance and the coupon and continuation value of inherited bonds finance maturing reserves, new bond holdings, and remittances.

To isolate the effects of asset-price movements from active portfolio adjustments, we hold the central bank's real bond holdings constant throughout the transition:

$$\frac{B_t^{T,C}}{P_t} = \bar{b}^{T,C}.$$

This assumption allows changes in bond prices to affect the market value of central-bank assets without introducing portfolio choice as an additional policy instrument. The baseline one-sided remittance constraint (9) remains unchanged. The resource constraint becomes

$$C_t + G + \frac{\varphi}{2}(\pi_t - 1)^2 Y_t = Y_t = N_t.$$

Ramsey problem A benevolent Ramsey planner jointly chooses monetary and fiscal policies under commitment. The quantitative model uses the nonlinear household welfare objective and the nonlinear equilibrium conditions. Under perfect foresight, the planner chooses the entire transition path subject to these conditions and, under no fiscal backing, the remittance constraint. The economy starts from the no-shock Ramsey steady state, including inherited state variables and commitment multipliers. For each deterministic shock path, we solve for the transition dynamics over a long horizon with an endogenous stationary continuation. Appendix G.1 provides the full formulation.

We solve the problem using continuation methods and verify equilibrium feasibility, Ramsey first-order and complementarity conditions, local second-order optimality conditions, and convergence with respect to the solution horizon. The resulting allocations are verified local solutions to the nonlinear Ramsey problem; global optimality is not established.

Calibration Table 1 summarizes the baseline calibration. Most preference and price-setting parameters are set to standard values. Details are discussed in Appendix G.2. A key parameter is the curvature of deposit liquidity, η_D. We set $\eta_D = 30$, corresponding to a deposit-demand semi-elasticity of approximately 3.3% for a one-percentage-point change in the deposit rate. Thus, deposit demand is relatively insensitive to changes in its pecuniary return. The utility weight on deposit liquidity, ω_D, appears large in levels, but this magnitude matches the liquidity value of deposit, $\bar{Q}^D - \bar{Q}^S$, at 1 percentage point.

Perfect-foresight dynamics We examine the model's dynamics under perfect foresight. Figure 5 reports impulse responses to a persistent expansionary demand shock. We set $e_0^n = -0.015$, with $e_t^n = \rho_e^t e_0^n$ and $\rho_e = 0.9$. Agents fully anticipate the deterministic shock path from period zero onward.

Table 1: Calibration

Parameter	Interpretation	Value	Calibration target	Target value
β	Discount factor	0.995	Annual real interest rate	2%
σ	Relative risk aversion	1	Risk aversion	1
ν	Inverse Frisch elasticity	2	Frisch elasticity	0.5
θ	Elasticity of substitution	6	Steady-state gross markup	1.2
χ_N	Weight on labor disutility	1.25	Reference real wage	1
φ	Rotemberg adjustment cost	60	NKPC slope, κ	0.215
η_D	Deposit liquidity curvature	30	Deposit semi-elasticity	3.3% decline/pp
ω_D	Weight on deposit liquidity	3.36×10^6	Deposit liquidity premium	1 pp
ℓ	Bank collateral constraint	0.90	Collateralizable share	90%
ρ_A	Other-asset weight	0.842	Collateral value vs. reserves	84.2%
ρ	Treasury coupon decay	0.9675	Bond duration	6.7 years
$\bar{b}^{T,C}$	CB Treasury holdings	0.0343	CB asset/liability ratio	1.00
$\bar{a}$	Supply of other bank assets	1.544	Annual net deposit rate	1%
G	Government spending	0.167	Gov. debt / annual GDP	100%
$\bar{H}$	Steady-state remittance	0.00209	Remittances / annual GDP	0.25%
$\underline{\mathrm{H}}$	Lower bound on remittance	0	Lower bound on remittance	0
ς	Employment subsidy	0.156	Steady-state tax rate	20%
ρ_e	Demand shock persistence	0.90	AR(1) persistence	0.90

Notes: The parameters G, $\bar{b}^{T,C}$, $\bar{H}$, $\bar{a}$, and ς are jointly calibrated to the five steady-state moments reported in the last two columns. The row-wise placement indicates the economically most direct identifying moment and does not imply that the numerical calibration is performed equation by equation. The target of other bank assets equal to 2.4 times quarterly output is the calibration rationale underlying the choice of ρ_A in the previous calibration; it is not imposed as a separate steady-state target in the current joint calibration."CB" denotes central bank.

The figure highlights how fiscal backing changes the way in which the planner accommodates the shock. Under fiscal backing, the planner raises the reserve rate sharply on impact. The resulting increase in the central bank's interest burden is financed by transfers from the Treasury. Output and inflation are not fully stabilized because full stabilization would require a sufficiently high interest rate, and the resulting interest payments would need to be financed through distortionary taxation. Since such tax distortions are costly, the planner optimally allows output and inflation to deviate from their steady-state levels.

Importantly, reserves exhibit relatively smooth dynamics, even though the high interest rate on reserves could potentially increase the quantity of reserves. This is because fluctuations in reserves are transmitted to deposits through the collateral constraint (36). Since households dislike fluctuations in deposit holdings, the planner smooths reserve holdings by relying on transfers from the Treasury.

Without fiscal backing, the planner raises the reserve rate by less than under fiscal backing. Al-

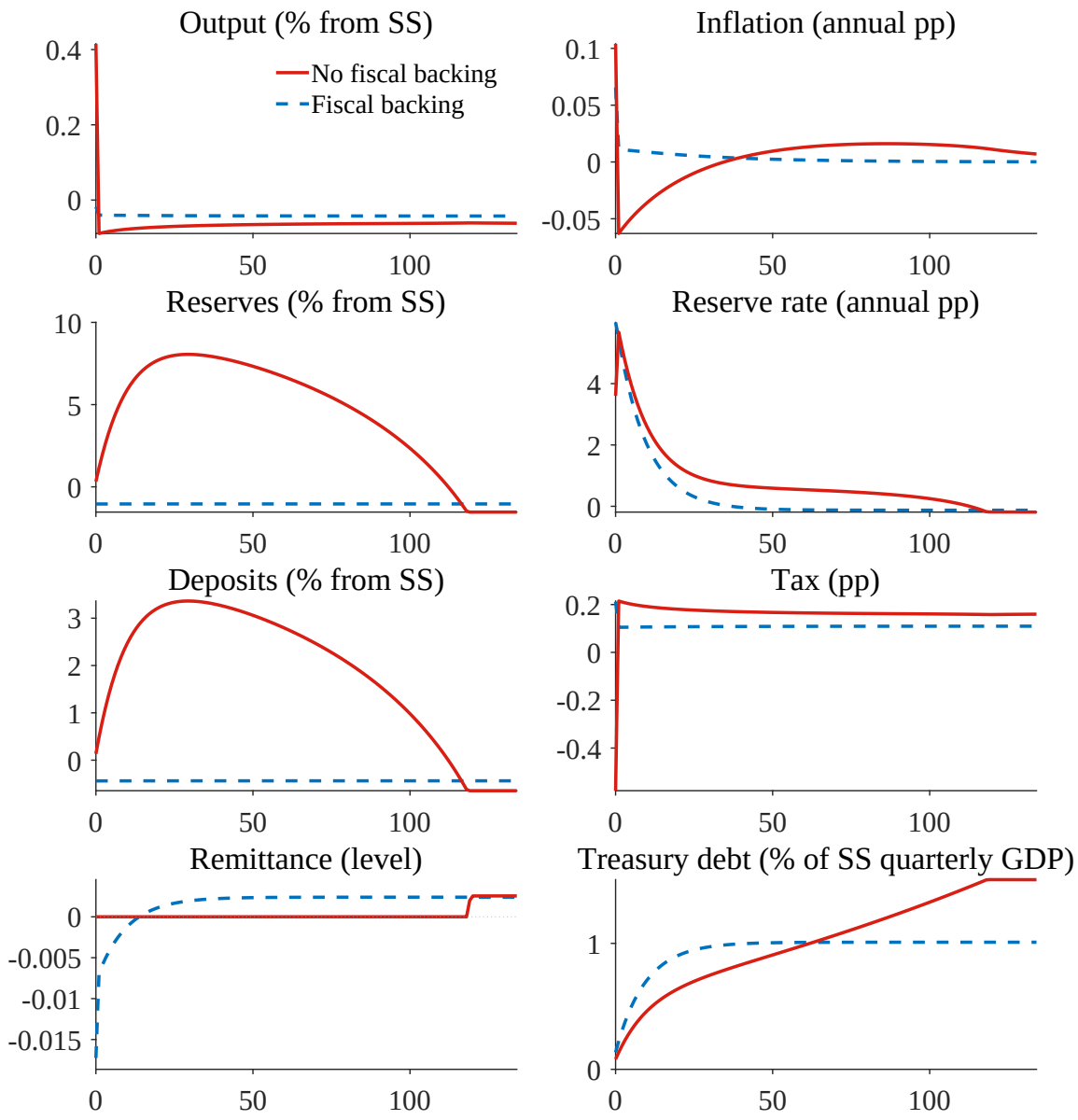


Figure 5: Impulse responses to an expansionary demand shock under fiscal backing and no fiscal backing

Notes: The shock is $e_0^n = -1.5\%$ and follows $e_t^n = \rho_e^t e_0^n$, with $\rho_e = 0.9$. Output, reserves, and deposits are reported as percentage deviations from the corresponding no-shock Ramsey equilibrium. Inflation and the reserve rate are reported as annualized percentage-point deviations, the tax rate in percentage points, and Treasury debt as a percentage of steady-state quarterly GDP. Remittances are reported in levels. Both regimes are solved from the same no-shock Ramsey steady state under perfect foresight.

though the interest rate dynamics resemble those in the baseline model, the underlying mechanism is more subtle: the remittance constraint does not bind in every period. After some periods, the constraint becomes slack. This implies that an interest rate path as high as that under fiscal backing

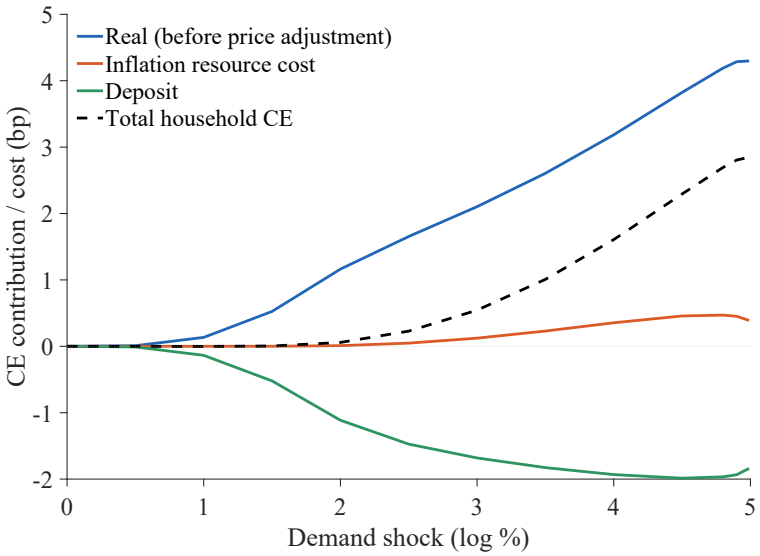


Figure 6: Welfare cost of no fiscal backing and its decomposition

Notes: The horizontal axis measures the expansionary demand shock, $s = -100e_0^n$, in log percentage points. The dashed line shows the welfare cost of no fiscal backing relative to fiscal backing, in basis points of permanent consumption. Solid lines show contributions from the real allocation before price-adjustment costs, inflation resource costs, and deposit services. These contributions use a common conversion factor and sum exactly to the total. Positive (negative) values indicate welfare costs (gains). Calculations use exact nonlinear household utility and the consumption-equivalent transformation, including the stationary continuation.

can be feasible under the central bank's budget constraint, as the central bank can receive greater remittances in the distant future. Despite this financial capacity, the planner optimally chooses a lower nominal interest rate without fiscal backing. Why?

If the planner chooses a higher interest rate without fiscal backing, the quantity of reserves increases. These additional reserve liabilities generate further interest payments, which must be financed through distortionary taxation. Since such financing is costly, the planner optimally chooses a lower nominal interest rate.

Welfare Figure 6 shows how the welfare cost of no fiscal backing varies with the magnitude of the expansionary demand shock and decomposes it into three components. For each shock size, we solve the nonlinear Ramsey problem separately under fiscal backing and no fiscal backing and compare discounted household utility. We express the welfare difference as a permanent-consumption-equivalent cost in basis points, using the exact nonlinear utility and consumption-equivalent transformation. The reported measure excludes the inherited-commitment term in the Ramsey objective.[5] When the 5% of demand shock hits, the welfare loss due to no fiscal backing is about 3 basis points of permanent-consumption-equivalent.

We decompose the welfare difference into real allocation, inflation resource costs, and deposit

[5]Details are provided in Appendix G.3.

services. The real allocation component measures welfare differences from consumption before price-adjustment costs, $Y_t - G$, net of labor disutility. The inflation component captures the additional resources absorbed by price adjustment, while the deposit component captures differences in liquidity services from deposit holdings. The three components are converted using a common factor so that their signed contributions sum exactly to the total consumption-equivalent cost.

The real allocation component is the largest positive contribution and rises with the size of the shock, while inflation resource costs make a smaller positive contribution. By contrast, the deposit component is negative, indicating that higher deposit services under no fiscal backing partially offset these losses. For small shocks, this offset is substantial, but as the shock becomes larger, the real allocation loss dominates and the net welfare cost rises. These results show that the welfare consequences of limited fiscal backing operate not only through conventional macroeconomic stabilization but also through the liquidity services provided by the banking system.

6 Conclusion

This paper studies optimal monetary and fiscal policy when central banks lack unrestricted fiscal backing. Separate Treasury and central bank budgets constrain monetary stabilization by limiting resources available to finance interest payments. For demand shocks, a funding gap determines the equilibrium allocation and welfare loss when the constraint binds. For cost-push shocks, its effects also depend on shock timing and stabilization priorities. Asset duration, the liquidity value of central bank liabilities, and distortionary taxation further shape the welfare value of fiscal backing.

In a nonlinear model with banks, deposits, reserves, long-duration Treasury securities, and distortionary taxation, the absence of fiscal backing weakens monetary tightening and amplifies adjustments in output, inflation, bank balance sheets, and fiscal variables. Central bank losses thus matter not because negative accounting net worth is inherently problematic, but because limited fiscal backing restricts the set of stabilization policies that can be financed and implemented.

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Appendix

A Perturbation of the main model

Define real bond holdings and the normalized level deviations of taxes and remittances as

$$b_t^j \equiv \frac{B_t^j}{P_t}, \qquad \widehat{T}_t \equiv \frac{T_t - \bar{T}}{\bar{Y}}, \qquad \widehat{H}_t \equiv \frac{H_t - \bar{H}}{\bar{Y}},$$

where $j \in \{T, C\}$. For b_t^j, Q_t^T, and inflation, hats denote log deviations, whereas $\widehat{T}_t$ and $\widehat{H}_t$ denote normalized level deviations. We linearize around $\bar{\Pi} = 1$ and use $\bar{Q}^T = \beta$.

A.1 Treasury budget constraint

Dividing the nominal Treasury budget constraint

$$Q_t^T B_t^T + P_t T_t + P_t H_t = B_{t-1}^T$$

by P_t yields

$$Q_t^T b_t^T + T_t + H_t = \frac{b_{t-1}^T}{\Pi_t}.$$

For $j \in \{T, C\}$, the first-order approximations are

$$Q_t^T b_t^j \simeq \bar{Q}^T \bar{b}^j (1 + \widehat{Q}_t^T + \widehat{b}_t^j), \qquad \frac{b_{t-1}^j}{\Pi_t} \simeq \bar{b}^j (1 + \widehat{b}_{t-1}^j - \widehat{\pi}_t).$$

Using these approximations, the steady-state budget constraint, and $\bar{Q}^T = \beta$, we obtain

$$\widehat{b}_{t-1}^T - \widehat{\pi}_t - \frac{\bar{Y}}{\bar{b}^T}(\widehat{T}_t + \widehat{H}_t) = \beta(\widehat{b}_t^T + \widehat{Q}_t^T).$$

Define

$$b_h \equiv \frac{\bar{Y}}{(1-\beta)\bar{b}^T}.$$

A.2 Central bank budget constraint

Dividing the nominal central bank budget constraint

$$Q_t^T B_t^C = B_{t-1}^C + P_t H_t$$

by P_t yields

$$Q_t^T b_t^C = \frac{b_{t-1}^C}{\Pi_t} + H_t.$$

Using the same first-order approximations and the steady-state condition

$$\bar{Q}^T \bar{b}^C = \bar{b}^C + \bar{H},$$

we obtain

$$\widehat{b}^C_{t-1} - \widehat{\pi}_t + \frac{\bar{Y}}{\bar{b}^C}\widehat{H}_t = \beta(\widehat{b}^C_t + \widehat{Q}^T_t).$$

Define

$$a_h \equiv \frac{\bar{Y}}{(1-\beta)\bar{b}^C}.$$

A.3 Remittance constraint

Let $\underline{H}$ denote the lower bound on remittances and define

$$H^* \equiv \frac{\underline{H} - \bar{H}}{\bar{Y}}.$$

Then $H_t \geq \underline{H}$ can be written as

$$\widehat{H}_t \geq H^*. \tag{38}$$

A.4 Household Euler equation and the demand shock

Household optimality, with $Q^C_t = Q^T_t \equiv Q_t$, implies

$$C_t^{-1} Q_t = \beta E_t \left[\frac{\delta_{t+1}}{\delta_t} \frac{C_{t+1}^{-1}}{\pi_{t+1}} \right]. \tag{39}$$

We log-linearize around the deterministic zero-inflation steady state with $\bar{\delta} = \bar{\pi} = 1$ and $\bar{Q} = \beta$. Defining

$$\hat{\delta}_t \equiv \log \delta_t, \qquad e_t \equiv E_t \hat{\delta}_{t+1} - \hat{\delta}_t, \tag{40}$$

the first-order approximation of (39) becomes

$$\hat{Q}_t = \hat{c}_t - E_t \hat{c}_{t+1} - E_t \hat{\pi}_{t+1} + e_t. \tag{41}$$

The resource constraint is

$$Y_t = N_t = C_t + \frac{\varphi}{2}(\pi_t - 1)^2 Y_t.$$

Since price-adjustment costs are second order around $\bar{\pi} = 1$, its first-order approximation implies

$$\hat{c}_t = \hat{y}_t. \tag{42}$$

Substituting this into (41) yields

$$\hat{Q}_t = \hat{y}_t - E_t \hat{y}_{t+1} - E_t \hat{\pi}_{t+1} + e_t. \tag{43}$$

Throughout the analytical sections, we impose perfect foresight conditional on the shock sequence, so expectations equal subsequent realizations. The Euler equation therefore reduces to

$$\hat{Q}_t = \hat{y}_t - \hat{y}_{t+1} - \hat{\pi}_{t+1} + e_t, \tag{44}$$

which is used in the linear-quadratic Ramsey problem.

A.5 Linearization of firm optimality

In a symmetric equilibrium, firm optimality implies

$$N_t C_t^{-1} \left[\varphi(\pi_t - 1)\pi_t + (\theta - 1)\left(1 - \mu_t^w N_t^\nu C_t\right)\right] = \beta E_t \left[N_{t+1} C_{t+1}^{-1} \varphi(\pi_{t+1} - 1)\pi_{t+1}\right]. \tag{45}$$

We log-linearize around the deterministic zero-inflation steady state with $\bar{\pi} = \bar{\mu}^w = 1$, where (45) implies

$$\bar{N}^\nu \bar{C} = 1. \tag{46}$$

Define log deviations as

$$\hat{\pi}_t \equiv \log \pi_t, \quad \hat{\mu}_t^w \equiv \log \mu_t^w, \quad \hat{n}_t \equiv \log(N_t/\bar{N}), \quad \hat{c}_t \equiv \log(C_t/\bar{C}).$$

Since both the bracketed term on the left-hand side of (45) and $(\pi_{t+1} - 1)\pi_{t+1}$ on the right-hand side vanish in steady state, fluctuations in their respective multiplicative terms $N_t C_t^{-1}$ and $N_{t+1} C_{t+1}^{-1}$ do not enter at first order. Using

$$(\pi_t - 1)\pi_t \simeq \hat{\pi}_t, \qquad \mu_t^w N_t^\nu C_t \simeq 1 + \hat{\mu}_t^w + \nu \hat{n}_t + \hat{c}_t,$$

the first-order approximation becomes

$$\varphi \hat{\pi}_t - (\theta - 1)(\hat{\mu}_t^w + \nu \hat{n}_t + \hat{c}_t) = \beta \varphi E_t \hat{\pi}_{t+1}. \tag{47}$$

The symmetric-equilibrium resource constraint,

$$Y_t = N_t = C_t + \frac{\varphi}{2}(\pi_t - 1)^2 Y_t,$$

implies $\hat{c}_t = \hat{n}_t = \hat{y}_t$ at first order, since price-adjustment costs are second order around $\bar{\pi} = 1$. Substituting these relations into (47) yields the New Keynesian Phillips curve

$$\hat{\pi}_t = \kappa \hat{y}_t + \beta E_t \hat{\pi}_{t+1} + e_t^c, \tag{48}$$

where

$$\kappa \equiv \frac{(\theta - 1)(1 + \nu)}{\varphi}, \qquad e_t^c \equiv \frac{\theta - 1}{\varphi} \hat{\mu}_t^w. \tag{49}$$

Thus, a positive wage-markup shock raises firms' marginal costs and enters the Phillips curve as a positive cost-push shock.

A.6 Second-order approximation of household welfare

Household welfare is

$$E_0 \sum_{t=0}^{\infty} \beta^t \delta_t \left[\log C_t - \frac{N_t^{1+\nu}}{1+\nu} \right]. \tag{50}$$

We approximate welfare around the deterministic zero-inflation steady state. Normalize $\bar{\delta} = 1$. Since the employment subsidy eliminates the steady-state monopolistic distortion, the steady state is efficient. The resource constraint implies $\bar{C} = \bar{N} = \bar{Y}$, while household intratemporal optimality implies $\bar{N}^{\nu}\bar{C} = 1$. Hence,

$$\bar{C} = \bar{N} = \bar{Y} = 1. \tag{51}$$

Define log deviations by

$$\hat{c}_t \equiv \log C_t, \qquad \hat{n}_t \equiv \log N_t, \qquad \hat{y}_t \equiv \log Y_t, \qquad \hat{\pi}_t \equiv \log \pi_t. \tag{52}$$

A second-order expansion of period utility gives

$$\delta_t \left[\log C_t - \frac{N_t^{1+\nu}}{1+\nu} \right] \simeq \text{terms independent of policy} + \hat{c}_t - \hat{n}_t - \frac{1+\nu}{2}\hat{n}_t^2. \tag{53}$$

The demand shock δ_t does not generate an additional policy-dependent term at second order. In particular, the first-order welfare variation is $\hat{c}_t - \hat{n}_t$, which is zero by the first-order resource constraint. Hence, products between the demand shock and endogenous first-order deviations are of third order.

To eliminate consumption, consider the resource constraint,

$$Y_t = N_t = C_t + \frac{\varphi}{2}(\pi_t - 1)^2 Y_t. \tag{54}$$

Using $N_t = Y_t$ and expanding to second order around $\bar{\pi} = 1$ gives

$$\hat{y}_t + \frac{1}{2}\hat{y}_t^2 = \hat{c}_t + \frac{1}{2}\hat{c}_t^2 + \frac{\varphi}{2}\hat{\pi}_t^2. \tag{55}$$

Its first-order approximation is

$$\hat{c}_t = \hat{y}_t = \hat{n}_t. \tag{56}$$

Since $\hat{y}_t - \hat{c}_t$ is therefore already second order, $\hat{y}_t^2 - \hat{c}_t^2$ is of third order. Equation (55) thus implies

$$\hat{c}_t - \hat{n}_t = -\frac{\varphi}{2}\hat{\pi}_t^2 + o(2). \tag{57}$$

Substituting (57) and $\hat{n}_t = \hat{y}_t + o(1)$ into (53), the policy-dependent component of household welfare is

$$-\frac{1}{2}\left[(1+\nu)\hat{y}_t^2 + \varphi\hat{\pi}_t^2\right]. \tag{58}$$

Therefore, up to an additive term independent of policy and a positive normalization, maximizing the second-order approximation to household welfare is equivalent to minimizing

$$E_0\sum_{t=0}^{\infty}\beta^t\left[\frac{1}{2}q_y\hat{y}_t^2 + \frac{1}{2}q_\pi\hat{\pi}_t^2\right], \tag{59}$$

where

$$q_y \equiv 1+\nu, \qquad q_\pi \equiv \varphi. \tag{60}$$

This is the quadratic objective used.

B Appendix for demand shock

B.1 General closed-form solution

We derive the equilibrium allocation for general parameters

$$q_y > 0, \qquad q_\pi > 0, \qquad \kappa > 0, \qquad 0 < \beta < 1,$$

conditional on the remittance constraint binding at every date, $\hat{H}_t = H^*$ for $t \geq 0$.

Because lump-sum taxes and Treasury debt are unrestricted, the Treasury budget constraint does not restrict the welfare-relevant allocation. To characterize the remittance KKT condition, let λ_{3t}, λ_{4t}, and $\nu_t \geq 0$ denote the multipliers on the Treasury budget constraint, the central-bank budget constraint, and $\hat{H}_t \geq H^*$, respectively. The first-order conditions with respect to $\hat{T}_t$, $\hat{b}_t^T$, $\hat{b}_t^C$, and $\hat{H}_t$ are

$$\begin{aligned}
-(1-\beta)b_h\lambda_{3t} &= 0,\\
-\beta\lambda_{3t} + \beta\lambda_{3,t+1} &= 0,\\
-\beta\lambda_{4t} + \beta\lambda_{4,t+1} &= 0,\\
-(1-\beta)b_h\lambda_{3t} + (1-\beta)a_h\lambda_{4t} - \nu_t &= 0.
\end{aligned}$$

These imply $\lambda_{3t} = 0$ and $\lambda_{4t} = \lambda_{4,t+1} \equiv \lambda_4$, so

$$\nu_t = (1-\beta)a_h\lambda_4. \tag{61}$$

Eliminating the one-period asset price using the household Euler equation, iterating the central-bank budget constraint forward, and imposing the transversality condition and $\hat{H}_t = H^*$ yield

$$\hat{\pi}_0 + \beta \hat{y}_0 - (1-\beta) \sum_{t=1}^{\infty} \beta^t \hat{y}_t = \Delta, \tag{62}$$

where

$$\Delta \equiv \hat{b}^C_{-1} + a_h H^* - \beta \sum_{t=0}^{\infty} \beta^t e_t$$

is the funding gap.

The binding problem therefore reduces to

$$\min_{\{\hat{y}_t, \hat{\pi}_t\}_{t=0}^{\infty}} \quad \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left(q_y \hat{y}_t^2 + q_\pi \hat{\pi}_t^2 \right)$$

subject to the New Keynesian Phillips curve,

$$\hat{\pi}_t = \kappa \hat{y}_t + \beta \hat{\pi}_{t+1}, \qquad t \geq 0,$$

and (62). Let λ_{1t} denote the multiplier on the Phillips curve and λ_4 the multiplier on the funding constraint. The Lagrangian is

$$\begin{aligned} \mathscr{L} = & \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left(q_y \hat{y}_t^2 + q_\pi \hat{\pi}_t^2 \right) + \sum_{t=0}^{\infty} \beta^t \lambda_{1t} \left(\hat{\pi}_t - \kappa \hat{y}_t - \beta \hat{\pi}_{t+1} \right) \\ & + \lambda_4 \left[\Delta - \hat{\pi}_0 - \beta \hat{y}_0 + (1-\beta) \sum_{t=1}^{\infty} \beta^t \hat{y}_t \right]. \end{aligned}$$

The first-order conditions at $t = 0$ are

$$q_y \hat{y}_0 - \kappa \lambda_{10} - \beta \lambda_4 = 0, \tag{63}$$

$$q_\pi \hat{\pi}_0 + \lambda_{10} - \lambda_4 = 0, \tag{64}$$

and, for $t \geq 1$,

$$q_y \hat{y}_t - \kappa \lambda_{1t} + (1-\beta) \lambda_4 = 0, \tag{65}$$

$$q_\pi \hat{\pi}_t + \lambda_{1t} - \lambda_{1,t-1} = 0. \tag{66}$$

Let $\hat{p}_t$ denote the log price level, with $\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$ and $\hat{p}_{-1} = 0$. The inflation first-order conditions imply

$$\lambda_{1t} = \lambda_4 - q_\pi \hat{p}_t, \qquad t \geq 0. \tag{67}$$

Substituting into the output first-order conditions yields

$$\hat{y}_0 = -\frac{\kappa q_\pi}{q_y} \hat{p}_0 + \frac{\kappa + \beta}{q_y} \lambda_4, \tag{68}$$

$$\hat{y}_t = -\frac{\kappa q_\pi}{q_y} \hat{p}_t + \frac{\kappa + \beta - 1}{q_y} \lambda_4, \qquad t \geq 1. \tag{69}$$

Writing the Phillips curve as

$$\hat{p}_t - \hat{p}_{t-1} = \kappa \hat{y}_t + \beta(\hat{p}_{t+1} - \hat{p}_t)$$

and substituting (69) gives, for $t \geq 1$,

$$\beta \hat{p}_{t+1} - \left(1 + \beta + \frac{\kappa^2 q_\pi}{q_y}\right) \hat{p}_t + \hat{p}_{t-1} + \frac{\kappa(\kappa + \beta - 1)}{q_y} \lambda_4 = 0. \tag{70}$$

The stable root of the homogeneous equation is

$$r \equiv \frac{1 + \beta + \frac{\kappa^2 q_\pi}{q_y} - \sqrt{\left(1 + \beta + \frac{\kappa^2 q_\pi}{q_y}\right)^2 - 4\beta}}{2\beta}. \tag{71}$$

It satisfies

$$\beta r^2 - \left(1 + \beta + \frac{\kappa^2 q_\pi}{q_y}\right) r + 1 = 0, \tag{72}$$

or equivalently,

$$(1 - r)(1 - \beta r) = \frac{\kappa^2 q_\pi}{q_y} r. \tag{73}$$

Since the characteristic polynomial satisfies

$$\mathscr{P}(0) = 1 > 0, \qquad \mathscr{P}(1) = -\frac{\kappa^2 q_\pi}{q_y} < 0,$$

we have $0 < r < 1$.

Define

$$\chi \equiv \frac{\frac{\kappa^2 q_\pi}{q_y} - \kappa + 1 - \beta}{q_y[1 + \beta(1 - r)] + \kappa^2 q_\pi}. \tag{74}$$

The bounded price-level solution is

$$\hat{p}_t = \left[\frac{\kappa + \beta - 1}{\kappa q_\pi} + \frac{q_y \chi}{\kappa q_\pi} r^t\right] \lambda_4, \qquad t \geq 0. \tag{75}$$

Consequently, inflation is

$$\hat{\pi}_0 = \frac{\kappa + \beta - 1 + q_y \chi}{\kappa q_\pi} \lambda_4, \tag{76}$$

$$\hat{\pi}_t = -\frac{q_y(1 - r)}{\kappa q_\pi} \chi r^{t-1} \lambda_4, \qquad t \geq 1, \tag{77}$$

and output is

$$\hat{y}_0 = \left(\frac{1}{q_y} - \chi\right) \lambda_4, \tag{78}$$

$$\hat{y}_t = -\chi r^t \lambda_4, \qquad t \geq 1. \tag{79}$$

The discounted sum of future output is therefore

$$\sum_{t=1}^{\infty} \beta^t \hat{y}_t = -\frac{\beta r}{1-\beta r}\chi\lambda_4. \tag{80}$$

Define

$$\Omega \equiv \frac{\kappa+\beta-1+q_y\chi}{\kappa q_\pi} + \beta\left(\frac{1}{q_y}-\chi\right) + \frac{\beta(1-\beta)r}{1-\beta r}\chi. \tag{81}$$

Substituting (76), (78), and (80) into the funding constraint gives

$$\Delta = \Omega\lambda_4, \qquad \lambda_4 = \frac{\Delta}{\Omega}.$$

Hence, the general closed-form allocation is

$$\hat{\pi}_0 = \frac{\kappa+\beta-1+q_y\chi}{\kappa q_\pi}\frac{\Delta}{\Omega}, \tag{82}$$

$$\hat{\pi}_t = -\frac{q_y(1-r)}{\kappa q_\pi}\chi r^{t-1}\frac{\Delta}{\Omega}, \qquad t \geq 1, \tag{83}$$

$$\hat{y}_0 = \left(\frac{1}{q_y}-\chi\right)\frac{\Delta}{\Omega}, \tag{84}$$

$$\hat{y}_t = -\chi r^t \frac{\Delta}{\Omega}, \qquad t \geq 1. \tag{85}$$

Thus, conditional on binding, output and inflation depend on the demand-shock sequence only through Δ.

Finally, the forward central-bank budget constraint implies

$$\hat{b}_t^C = -a_h H^* + \hat{\pi}_{t+1} + \beta\hat{y}_{t+1} - (1-\beta)\sum_{j=t+2}^{\infty}\beta^{j-t-1}\hat{y}_j + \beta\sum_{j=t+1}^{\infty}\beta^{j-t-1}e_j.$$

Defining

$$\Gamma \equiv \frac{1-r}{1-\beta r}\left[\frac{q_y}{\kappa q_\pi}(1-\beta r)+\beta r\right], \tag{86}$$

and substituting the closed-form allocation yields

$$\hat{b}_t^C = -a_h H^* + \beta\sum_{j=t+1}^{\infty}\beta^{j-t-1}e_j - \Gamma\chi r^t\frac{\Delta}{\Omega}, \qquad t \geq 0. \tag{87}$$

For a one-time shock with $e_t = 0$ for $t \geq 1$, this simplifies to

$$\hat{b}_t^C = -a_h H^* - \Gamma\chi r^t\frac{\Delta}{\Omega}, \qquad t \geq 0. \tag{88}$$

B.2 Proof of Proposition 1

Proof. The objective is nonnegative, so it is sufficient to show that full stabilization is feasible. Consider

$$\hat{y}_t = \hat{\pi}_t = 0, \qquad t \geq 0.$$

This allocation satisfies the New Keynesian Phillips curve. Substituting it into the household Euler equation and the central-bank budget constraint gives

$$\beta \left(e_t + \hat{b}_t^C \right) = \hat{b}_{t-1}^C + (1-\beta) a_h \hat{H}_t.$$

Set

$$\hat{b}_t^C = \hat{b}_t^T = 0, \qquad t \geq 0.$$

The central-bank budget constraint is then satisfied by choosing

$$\hat{H}_t = \frac{\beta e_t - \hat{b}_{t-1}^C}{(1-\beta) a_h}.$$

Under fiscal backing, remittances are unrestricted, so this choice is feasible. Substituting full stabilization into the Treasury budget constraint, lump-sum taxes can then be chosen as

$$\hat{T}_t = \frac{\hat{b}_{t-1}^T - \beta e_t - (1-\beta) b_h \hat{H}_t}{(1-\beta) b_h}.$$

Hence, full stabilization is feasible. Since it yields a welfare loss of zero, it is optimal. □

B.3 Proof of Proposition 2

Proof. Let ν_t denote the multiplier on the remittance constraint $\hat{H}_t \geq H^*$. Equation (61) implies

$$\nu_t = (1-\beta) a_h \lambda_4,$$

so the multiplier is constant across dates. If $\nu_t > 0$ at any date, it is strictly positive at every date, and complementary slackness requires $\hat{H}_t = H^*$ for all $t \geq 0$.

First, suppose $\Delta \leq 0$. Under full stabilization, $\hat{y}_t = \hat{\pi}_t = 0$ for all $t \geq 0$. Substituting the household Euler equation into the central-bank budget constraint and iterating forward yields

$$\hat{b}_{-1}^C = \beta \sum_{t=0}^{\infty} \beta^t e_t - (1-\beta) a_h \sum_{t=0}^{\infty} \beta^t \hat{H}_t.$$

Choose

$$\hat{H}_t = H^* - \frac{\Delta}{a_h}, \qquad t \geq 0.$$

This satisfies $\hat{H}_t \geq H^*$ because $\Delta \leq 0$, and the intertemporal budget constraint by the definition

$$\Delta = \hat{b}^C_{-1} + a_h H^* - \beta \sum_{t=0}^{\infty} \beta^t e_t.$$

Central-bank liabilities can then be determined recursively from the flow budget constraint, while lump-sum taxes adjust to satisfy the Treasury budget constraint. Thus, full stabilization is feasible and achieves zero welfare loss. If $\Delta < 0$, the constructed remittance path is strictly above its lower bound at every date. If $\Delta = 0$, full stabilization is feasible with $\hat{H}_t = H^*$, but the constraint has zero shadow value.

Next, suppose $\Delta > 0$. Under full stabilization, the intertemporal central-bank budget constraint and $\hat{H}_t \geq H^*$ imply

$$\hat{b}^C_{-1} = \beta \sum_{t=0}^{\infty} \beta^t e_t - (1-\beta) a_h \sum_{t=0}^{\infty} \beta^t \hat{H}_t \leq \beta \sum_{t=0}^{\infty} \beta^t e_t - a_h H^*,$$

which requires $\Delta \leq 0$, a contradiction. Full stabilization is therefore infeasible. Since it is the unconstrained optimum, the remittance restriction must have a strictly positive shadow value at the constrained optimum. Equation (61) then implies $\nu_t > 0$ for every $t \geq 0$, and complementary slackness yields

$$\hat{H}_t = H^*, \qquad t \geq 0.$$

□

B.4 Proof of Proposition 3

Proof. Let $q_y = q_\pi \equiv q > 0$. Although the common weight q does not affect the Ramsey allocation, we retain it to demonstrate its cancellation from the general solution in Appendix B.1.

The stable root in (71) becomes

$$r_\kappa \equiv \frac{1+\beta+\kappa^2 - \sqrt{(1+\beta+\kappa^2)^2 - 4\beta}}{2\beta}, \tag{89}$$

with $0 < r_\kappa < 1$. It satisfies

$$(1-r_\kappa)(1-\beta r_\kappa) = \kappa^2 r_\kappa, \tag{90}$$

which implies

$$1 + \beta(1-r_\kappa) + \kappa^2 = \frac{1}{r_\kappa}.$$

Define

$$\chi_\kappa \equiv (\kappa^2 - \kappa + 1 - \beta) r_\kappa.$$

Equation (74) then gives $\chi = \chi_\kappa/q$. The root identity also yields

$$\frac{\kappa+\beta-1+\chi_\kappa}{\kappa} = (1-r_\kappa)\left[1+\frac{\beta}{\kappa}(1-r_\kappa)\right],$$
$$1-\chi_\kappa = r_\kappa[\kappa+\beta(2-r_\kappa)].$$

Define

$$\Omega_\kappa \equiv (1-r_\kappa)\left[1+\frac{\beta}{\kappa}(1-r_\kappa)\right]+\beta r_\kappa[\kappa+\beta(2-r_\kappa)]+\frac{\beta(1-\beta)(\kappa^2-\kappa+1-\beta)r_\kappa^2}{1-\beta r_\kappa}. \tag{91}$$

Substituting $q_y = q_\pi = q$ and $\chi = \chi_\kappa/q$ into (81) gives

$$\Omega = \frac{\Omega_\kappa}{q}, \qquad \lambda_4 = q\frac{\Delta}{\Omega_\kappa}.$$

Equation (61) and $\nu_t \geq 0$ imply $\lambda_4 \geq 0$. Since $\Delta > 0$, we have $\lambda_4 > 0$ and hence $\Omega_\kappa > 0$.

Substituting these expressions into (76)–(79), the common weight q cancels, yielding

$$\hat{\pi}_0 = (1-r_\kappa)\left[1+\frac{\beta}{\kappa}(1-r_\kappa)\right]\frac{\Delta}{\Omega_\kappa}, \qquad \hat{y}_0 = r_\kappa[\kappa+\beta(2-r_\kappa)]\frac{\Delta}{\Omega_\kappa},$$
$$\hat{\pi}_t = -(1-r_\kappa)\left(\kappa-1+\frac{1-\beta}{\kappa}\right)r_\kappa^t\frac{\Delta}{\Omega_\kappa}, \qquad \hat{y}_t = -(\kappa^2-\kappa+1-\beta)r_\kappa^{t+1}\frac{\Delta}{\Omega_\kappa}, \qquad t \geq 1.$$

Since $0 < r_\kappa < 1$, $\Delta > 0$, and $\Omega_\kappa > 0$, both impact coefficients are strictly positive:

$$\hat{\pi}_0 > 0, \qquad \hat{y}_0 > 0.$$

Finally, the household Euler equation gives

$$\hat{Q}_0 - e_0 = \hat{y}_0 - \hat{y}_1 - \hat{\pi}_1.$$

Substituting the closed-form allocation and using (90) yields

$$\hat{Q}_0 - e_0 = r_\kappa\left[1+(1-r_\kappa)\left(2\kappa-1+\beta(2-r_\kappa)+\frac{1-\beta}{\kappa}\right)\right]\frac{\Delta}{\Omega_\kappa}.$$

The expression in parentheses is strictly positive because

$$2\kappa-1+\beta(2-r_\kappa)+\frac{1-\beta}{\kappa} > 2\kappa-1+\beta+\frac{1-\beta}{\kappa} > 0.$$

The last inequality holds immediately for $\kappa \geq 1$; for $0 < \kappa < 1$, it follows from $\beta+(1-\beta)/\kappa > 1$. Therefore, $\hat{Q}_0 > e_0$, or equivalently,

$$\hat{i}_0 < -e_0,$$

since $\hat{i}_0 = -\hat{Q}_0$. □

B.5 Proof of Corollary 1

Proof. Let Δ and $\tilde{\Delta}$ denote the funding gaps associated with $\{e_t\}_{t=0}^{\infty}$ and $\{\tilde{e}_t\}_{t=0}^{\infty}$, respectively. Since the two economies have the same initial central-bank liability and the same remittance lower bound,

$$\Delta - \tilde{\Delta} = -\beta \sum_{t=0}^{\infty} \beta^t \left(e_t - \tilde{e}_t\right) = 0.$$

Hence, $\Delta = \tilde{\Delta}$. Conditional on the remittance constraint binding, the general solution in Appendix B.1 depends on the demand-shock sequence only through the funding gap Δ. The coefficients r, χ, and Ω depend only on β, κ, q_y, and q_π. Therefore,

$$\hat{y}_t = \tilde{y}_t, \qquad \hat{\pi}_t = \tilde{\pi}_t, \qquad t \geq 0.$$

The two economies consequently generate the same welfare loss. □

B.6 Proof of Proposition 4

Proof. Conditional on the remittance constraint binding, Appendix B.1 reduces the Ramsey problem to

$$\min_{\{\hat{y}_t, \hat{\pi}_t\}_{t=0}^{\infty}} \quad \mathscr{W}^N \equiv \frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left(q_y \hat{y}_t^2 + q_\pi \hat{\pi}_t^2\right)$$

subject to

$$\hat{\pi}_t - \kappa \hat{y}_t - \beta \hat{\pi}_{t+1} = 0, \qquad t \geq 0,$$

and the intertemporal funding constraint

$$\mathscr{B} \equiv \hat{\pi}_0 + \beta \hat{y}_0 - (1-\beta) \sum_{t=1}^{\infty} \beta^t \hat{y}_t = \Delta.$$

Using the sign convention in Appendix B.1, the Lagrangian is

$$\mathscr{L} = \mathscr{W}^N + \sum_{t=0}^{\infty} \beta^t \mu_t \left(\hat{\pi}_t - \kappa \hat{y}_t - \beta \hat{\pi}_{t+1}\right) + \lambda_4 (\Delta - \mathscr{B}),$$

where λ_4 is the multiplier on the funding constraint.

The objective is homogeneous of degree two in output and inflation, whereas the constraints are homogeneous of degree one. Multiplying each first-order condition by its corresponding choice variable and summing across dates, the Phillips-curve terms vanish by feasibility, yielding

$$2\mathscr{W}^N - \lambda_4 \mathscr{B} = 0.$$

Since $\mathscr{B} = \Delta$ and Appendix B.1 establishes $\Delta = \Omega\lambda_4$, we obtain

$$\mathscr{W}^N = \frac{1}{2}\lambda_4\Delta = \frac{\Delta^2}{2\Omega}.$$

For $\Delta > 0$, the binding allocation is nonzero. Since $q_y, q_\pi > 0$, the welfare loss is strictly positive, implying $\Omega > 0$.

Under fiscal backing, Proposition 1 establishes full stabilization, so $\mathscr{W}^B = 0$. Therefore,

$$\mathscr{W}^N - \mathscr{W}^B = \frac{\Delta^2}{2\Omega}.$$

Since Ω depends only on structural parameters, the welfare cost of no fiscal backing depends on the demand-shock sequence only through Δ. □

C Appendix for cost-push shock

C.1 Closed-form solution under an arbitrary cost-push shock

Consider an arbitrary bounded deterministic cost-push shock sequence $\{e_t^c\}_{t=0}^{\infty}$, with $\sup_{t\geq 0}|e_t^c| < \infty$. We set demand shocks to zero and assume that the remittance constraint binds at every date, $\hat{H}_t = H^*$ for $t \geq 0$.

Because lump-sum taxes and Treasury debt are unrestricted, the Treasury budget constraint does not restrict the welfare-relevant allocation. Eliminating the one-period asset price and iterating the central-bank budget constraint forward yields

$$\hat{\pi}_0 + \beta\hat{y}_0 - (1-\beta)\sum_{t=1}^{\infty}\beta^t\hat{y}_t = \hat{b}^C_{-1} + a_h H^*.$$

Conditional on binding, the reduced Ramsey problem is therefore

$$\min_{\{\hat{y}_t,\hat{\pi}_t\}_{t=0}^{\infty}} \quad \frac{1}{2}\sum_{t=0}^{\infty}\beta^t\left(q_y\hat{y}_t^2 + q_\pi\hat{\pi}_t^2\right)$$

subject to the intertemporal funding constraint above and

$$\hat{\pi}_t = \kappa\hat{y}_t + \beta\hat{\pi}_{t+1} + e_t^c, \qquad t \geq 0.$$

Let λ_{1t} and λ_4 denote the multipliers on the Phillips curve and the funding constraint, respectively. The first-order conditions at $t = 0$ are

$$q_y\hat{y}_0 - \kappa\lambda_{10} - \beta\lambda_4 = 0,$$
$$q_\pi\hat{\pi}_0 + \lambda_{10} - \lambda_4 = 0,$$

and, for $t \geq 1$,

$$q_y \hat{y}_t - \kappa \lambda_{1t} + (1-\beta)\lambda_4 = 0,$$
$$q_\pi \hat{\pi}_t + \lambda_{1t} - \lambda_{1,t-1} = 0.$$

As shown in Appendix B.1, the multiplier on the remittance constraint satisfies

$$\nu_t = (1-\beta) a_h \lambda_4,$$

so strict binding requires $\lambda_4 > 0$.

Define the log price level by $\hat{\pi}_t = \hat{p}_t - \hat{p}_{t-1}$, with $\hat{p}_{-1} = 0$. The inflation first-order conditions imply

$$\lambda_{1t} = \lambda_4 - q_\pi \hat{p}_t, \qquad t \geq 0.$$

Substituting into the output first-order conditions gives

$$\hat{y}_0 = -\frac{\kappa q_\pi}{q_y}\hat{p}_0 + \frac{\kappa+\beta}{q_y}\lambda_4, \tag{92}$$
$$\hat{y}_t = -\frac{\kappa q_\pi}{q_y}\hat{p}_t + \frac{\kappa+\beta-1}{q_y}\lambda_4, \qquad t \geq 1. \tag{93}$$

Define the stable root

$$r \equiv \frac{1+\beta+\kappa^2\frac{q_\pi}{q_y} - \sqrt{\left(1+\beta+\kappa^2\frac{q_\pi}{q_y}\right)^2 - 4\beta}}{2\beta} \in (0,1),$$

which satisfies

$$\beta r^2 - \left(1+\beta+\kappa^2\frac{q_\pi}{q_y}\right) r + 1 = 0,$$

and let

$$s \equiv \frac{\kappa(\kappa+\beta-1)r}{q_y(1-\beta r)}.$$

For later use, define the shock statistics

$$\Delta^c \equiv \sum_{t=0}^{\infty} \beta^t e_t^c, \qquad \Delta_t^r \equiv \sum_{j=0}^{\infty} (\beta r)^j e_{t+j}^c, \qquad \Delta^r \equiv \Delta_0^r. \tag{94}$$

Here, Δ^c is the usual discounted present value, while Δ^r places relatively greater weight on earlier shocks. The continuation statistics $\{\Delta_t^r\}_{t\geq 1}$ capture the shock timing relevant for subsequent equilibrium dynamics.

For $t \geq 1$, substituting (93) into the Phillips curve yields

$$\beta \hat{p}_{t+1} - \left(1+\beta+\kappa^2\frac{q_\pi}{q_y}\right)\hat{p}_t + \hat{p}_{t-1} + \frac{\kappa(\kappa+\beta-1)}{q_y}\lambda_4 + e_t^c = 0.$$

The unique bounded solution satisfies

$$\hat{p}_t = r\hat{p}_{t-1} + s\lambda_4 + r\Delta_t^r, \qquad t \geq 1.$$

The period-zero Phillips curve determines

$$\hat{p}_0 = r\left[\frac{\kappa(\kappa+\beta)}{q_y} + \beta s\right]\lambda_4 + r\Delta^r. \tag{95}$$

Iterating the recursion gives

$$\hat{p}_t = \left\{r^{t+1}\left[\frac{\kappa(\kappa+\beta)}{q_y} + \beta s\right] + \frac{1-r^t}{1-r}s\right\}\lambda_4 + r^{t+1}\Delta^r + r\sum_{k=1}^{t} r^{t-k}\Delta_k^r, \qquad t \geq 1. \tag{96}$$

Substituting the allocation into the intertemporal funding constraint yields

$$\hat{b}_{-1}^C + a_h H^* = \Omega\lambda_4 + \frac{1-\beta}{\kappa}\Delta^c + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\frac{q_\pi}{q_y}\right] r\Delta^r,$$

where $\Omega > 0$ is defined in Appendix B.1. Thus,

$$\lambda_4 = \frac{\hat{b}_{-1}^C + a_h H^* - \frac{1-\beta}{\kappa}\Delta^c - \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\frac{q_\pi}{q_y}\right] r\Delta^r}{\Omega}. \tag{97}$$

Since $\Omega > 0$, strict binding is equivalent to

$$\hat{b}_{-1}^C + a_h H^* - \frac{1-\beta}{\kappa}\Delta^c - \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\frac{q_\pi}{q_y}\right] r\Delta^r > 0.$$

Under $\hat{b}_{-1}^C = 0$, this coincides with the binding condition in Proposition 5.

The impact allocation is

$$\begin{aligned}
\hat{\pi}_0 &= r\left[\frac{\kappa(\kappa+\beta)}{q_y} + \beta s\right]\lambda_4 + r\Delta^r, \\
\hat{y}_0 &= \left\{\frac{\kappa+\beta}{q_y} - \frac{\kappa q_\pi}{q_y} r\left[\frac{\kappa(\kappa+\beta)}{q_y} + \beta s\right]\right\}\lambda_4 - \frac{\kappa q_\pi}{q_y} r\Delta^r.
\end{aligned}$$

Since λ_4 depends on the shock sequence only through (Δ^c, Δ^r), these two statistics fully characterize the impact allocation.

For $t \geq 1$, the subsequent allocation follows from

$$\begin{aligned}
\hat{\pi}_t &= \hat{p}_t - \hat{p}_{t-1}, \\
\hat{y}_t &= -\frac{\kappa q_\pi}{q_y}\hat{p}_t + \frac{\kappa+\beta-1}{q_y}\lambda_4,
\end{aligned}$$

where $\hat{p}_t$ is given by (96). Central-bank liabilities are recovered from

$$\hat{b}^C_t = -a_h H^* + \hat{\pi}_{t+1} + \beta \hat{y}_{t+1} - (1-\beta) \sum_{j=t+2}^{\infty} \beta^{j-t-1} \hat{y}_j, \qquad t \geq 0.$$

Thus, the multiplier, binding condition, and impact allocation depend on the cost-push shock sequence only through (Δ^c, Δ^r). However, the subsequent allocation also depends on $\{\Delta^r_t\}_{t\geq 1}$. Unlike demand shocks, therefore, cost-push shocks cannot be summarized by a single discounted present value, and even the two initial statistics do not generally determine the entire equilibrium path.

C.2 Proof of Proposition 5

Proof. From Appendix C.1,

$$\lambda_4 = \frac{a_h H^* - \dfrac{1-\beta}{\kappa}\Delta^c - \left[\dfrac{\kappa+\beta-1}{\kappa} - \kappa\dfrac{q_\pi}{q_y}\right] r\Delta^r}{\Omega},$$

where $\Omega > 0$. Since the multiplier on the remittance constraint is $\nu_t = (1-\beta)a_h\lambda_4$, the constraint binds strictly at every date if and only if $\lambda_4 > 0$. Therefore,

$$\lambda_4 > 0 \quad \Longleftrightarrow \quad -a_h H^* + \frac{1-\beta}{\kappa}\Delta^c + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\frac{q_\pi}{q_y}\right] r\Delta^r < 0,$$

which is the stated condition. □

C.3 Proof of Proposition 6

Proof. Define $\omega \equiv q_\pi/q_y$. For a one-time cost-push shock, $e^c_0 \neq 0$ and $e^c_t = 0$ for $t \geq 1$, the shock statistics satisfy $\Delta^c = \Delta^r = e^c_0$. With $H^* = \hat{b}^C_{-1} = 0$, Appendix C.1 gives

$$\lambda_4 = -\frac{e^c_0}{\Omega}\Phi(\omega),$$

where $\Omega > 0$ and

$$\Phi(\omega) \equiv \frac{1-\beta}{\kappa} + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\omega\right] r.$$

The stable root satisfies

$$\beta r^2 - (1+\beta+\kappa^2\omega)r + 1 = 0,$$

or equivalently,

$$\kappa^2\omega r = (1-r)(1-\beta r).$$

Using this identity,

$$\kappa\Phi(\omega) = 1-\beta+(\kappa+\beta-1)r-\kappa^2\omega r = \kappa r-\beta(1-r)^2.$$

This expression is strictly increasing in r and vanishes at the unique $\bar{r}\in(0,1)$ satisfying

$$\kappa\bar{r} = \beta(1-\bar{r})^2,$$

which gives

$$\bar{r} = \frac{\sqrt{1+\frac{4\beta}{\kappa}}-1}{\sqrt{1+\frac{4\beta}{\kappa}}+1}.$$

Since r is strictly decreasing in ω, there is a unique threshold $\bar{\omega}$ satisfying $r(\bar{\omega})=\bar{r}$, such that

$$\Phi(\omega)\begin{cases} >0, & \omega<\bar{\omega},\\ =0, & \omega=\bar{\omega},\\ <0, & \omega>\bar{\omega}. \end{cases}$$

The stable-root identity yields

$$\bar{\omega} = \frac{(1-\bar{r})(1-\beta\bar{r})}{\kappa^2\bar{r}} = \frac{1+\beta+(1-\beta)\sqrt{1+\frac{4\beta}{\kappa}}}{2\beta\kappa}.$$

Because the remittance multiplier is proportional to λ_4, the condition $\lambda_4\geq 0$ is equivalent to

$$\omega\geq\bar{\omega} \quad \text{if } e_0^c>0, \qquad \omega\leq\bar{\omega} \quad \text{if } e_0^c<0.$$

At $\omega=\bar{\omega}$, the multiplier is zero. Thus, the weak inequalities characterize the boundary-inclusive binding condition, while strict binding requires strict inequalities. □

C.4 Proof of Proposition 8

Proof. Define $\omega\equiv q_\pi/q_y$. For a one-time cost-push shock, $e_0^c\neq 0$ and $e_t^c=0$ for $t\geq 1$, the nonbinding allocation is

$$\begin{aligned} \hat{\pi}_0^B &= re_0^c,\\ \hat{\pi}_t^B &= -(1-r)r^t e_0^c, \qquad t\geq 1,\\ \hat{y}_t^B &= -\kappa\omega r^{t+1}e_0^c, \qquad t\geq 0, \end{aligned}$$

where $r\in(0,1)$ is the stable root satisfying

$$\beta r^2-(1+\beta+\kappa^2\omega)r+1=0.$$

This implies

$$\kappa^2\omega r = (1-r)(1-\beta r), \qquad 1+\kappa^2\omega-\beta r = \frac{1-\beta r}{r}.$$

Let $\lambda_4 > 0$ denote the funding-constraint multiplier in the strictly binding equilibrium. Subtracting the nonbinding allocation from the binding solution in Appendix C.1 and using the root identities yields

$$\hat{\pi}_0^N - \hat{\pi}_0^B = \frac{\kappa r[\kappa+\beta(1-r)]}{q_y(1-\beta r)}\lambda_4, \quad \hat{y}_0^N - \hat{y}_0^B = \frac{r[\kappa+\beta(2-r)]}{q_y}\lambda_4.$$

Both expressions are strictly positive, implying

$$\hat{\pi}_0^N > \hat{\pi}_0^B, \qquad \hat{y}_0^N > \hat{y}_0^B.$$

For $t \geq 1$, the differences are

$$\hat{\pi}_t^N - \hat{\pi}_t^B = \frac{\kappa r^{t+1}}{q_y(1-\beta r)}(\kappa+\beta-1-\kappa^2\omega)\lambda_4, \quad \hat{y}_t^N - \hat{y}_t^B = \frac{r^{t+1}}{q_y}(\kappa+\beta-1-\kappa^2\omega)\lambda_4.$$

Suppose $e_0^c > 0$. Under $H^* = \hat{b}_{-1}^C = 0$, Appendix C.3 establishes that strict binding requires $\omega > \bar{\omega}$, where $\bar{\omega}$ is the threshold at which $\lambda_4 = 0$. At $\omega = \bar{\omega}$, the threshold condition and the stable-root identity imply

$$r(\kappa^2\bar{\omega} - \kappa - \beta + 1) = 1 - \beta > 0,$$

so

$$\kappa+\beta-1-\kappa^2\bar{\omega} < 0.$$

Since this expression is strictly decreasing in ω, strict binding implies

$$\kappa+\beta-1-\kappa^2\omega < 0.$$

Consequently, for every $t \geq 1$,

$$\hat{\pi}_t^N < \hat{\pi}_t^B, \qquad \hat{y}_t^N < \hat{y}_t^B.$$

□

C.5 Proof of Proposition 9

Proof. Define

$$\omega \equiv \frac{q_\pi}{q_y}, \qquad \ell \equiv \frac{\lambda_4}{q_y} > 0,$$

where the inequality follows from strict binding. For a one-time cost-push shock, $e_0^c \neq 0$ and $e_t^c = 0$ for $t \geq 1$, Appendix C.4 gives the fiscal-backing allocation

$$\hat{\pi}_0^B = re_0^c, \qquad \hat{\pi}_t^B = -(1-r)r^t e_0^c, \quad t \geq 1,$$
$$\hat{y}_t^B = -\kappa\omega r^{t+1}e_0^c, \quad t \geq 0,$$

and the differences between the no-fiscal-backing and fiscal-backing allocations:

$$\begin{aligned}
\hat{\pi}_0^N - \hat{\pi}_0^B &= \frac{\kappa r[\kappa + \beta(1-r)]}{1-\beta r}\ell, \\
\hat{\pi}_t^N - \hat{\pi}_t^B &= \frac{\kappa(\kappa + \beta - 1 - \kappa^2\omega)}{1-\beta r} r^{t+1}\ell, \qquad t \geq 1, \\
\hat{y}_0^N - \hat{y}_0^B &= r[\kappa + \beta(2-r)]\ell, \\
\hat{y}_t^N - \hat{y}_t^B &= (\kappa + \beta - 1 - \kappa^2\omega) r^{t+1}\ell, \qquad t \geq 1.
\end{aligned}$$

Recall the stable-root identity

$$\kappa^2\omega r = (1-r)(1-\beta r).$$

Define the positive coefficients

$$\begin{aligned}
\mathscr{C} &\equiv \frac{\kappa r^2[\kappa + 2\beta(1-r)]}{1-\beta r^2} > 0, \\
\mathscr{V}_\pi &\equiv \left[\frac{\kappa r[\kappa + \beta(1-r)]}{1-\beta r}\right]^2 + \frac{\beta\kappa^2(\kappa + \beta - 1 - \kappa^2\omega)^2 r^4}{(1-\beta r)^2(1-\beta r^2)} > 0, \\
\mathscr{V}_y &\equiv r^2[\kappa + \beta(2-r)]^2 + \frac{\beta(\kappa + \beta - 1 - \kappa^2\omega)^2 r^4}{1-\beta r^2} > 0.
\end{aligned}$$

Substituting the allocations and summing the geometric series yields

$$\begin{aligned}
\sum_{t=0}^{\infty} \beta^t \hat{\pi}_t^B(\hat{\pi}_t^N - \hat{\pi}_t^B) &= e_0^c \ell \mathscr{C}, \\
\sum_{t=0}^{\infty} \beta^t (\hat{\pi}_t^N - \hat{\pi}_t^B)^2 &= \ell^2 \mathscr{V}_\pi, \\
\sum_{t=0}^{\infty} \beta^t \hat{y}_t^B(\hat{y}_t^N - \hat{y}_t^B) &= -\omega e_0^c \ell \mathscr{C}, \\
\sum_{t=0}^{\infty} \beta^t (\hat{y}_t^N - \hat{y}_t^B)^2 &= \ell^2 \mathscr{V}_y.
\end{aligned}$$

Using

$$\frac{1}{2}\left[(x_t^N)^2 - (x_t^B)^2\right] = x_t^B(x_t^N - x_t^B) + \frac{1}{2}(x_t^N - x_t^B)^2,$$

the welfare-component differences become

$$\mathscr{W}_\pi^N - \mathscr{W}_\pi^B = q_\pi \left[e_0^c \ell \mathscr{C} + \frac{1}{2}\ell^2 \mathscr{V}_\pi\right], \tag{98}$$

$$\mathscr{W}_y^N - \mathscr{W}_y^B = q_y \left[-\omega e_0^c \ell \mathscr{C} + \frac{1}{2}\ell^2 \mathscr{V}_y\right]. \tag{99}$$

Since $q_\pi = q_y\omega$, the cross terms cancel, giving

$$(\mathscr{W}_\pi^N + \mathscr{W}_y^N) - (\mathscr{W}_\pi^B + \mathscr{W}_y^B) = \frac{q_y}{2}\ell^2(\omega\mathscr{V}_\pi + \mathscr{V}_y) > 0.$$

Under the assumptions of Proposition 8, Appendix C.1 gives

$$\lambda_4 = -\frac{e_0^c}{\Omega}\left\{\frac{1-\beta}{\kappa} + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\omega\right] r\right\}.$$

Using the stable-root identity and

$$q_y\Omega = \omega\mathscr{V}_\pi + \mathscr{V}_y,$$

we obtain

$$\ell = -\frac{re_0^c}{\omega\mathscr{V}_\pi + \mathscr{V}_y}\left[1 - \kappa\omega\frac{\beta(1-r)}{1-\beta r}\right].$$

First, suppose $e_0^c > 0$. Proposition 6 implies that strict binding requires

$$1 - \kappa\omega\frac{\beta(1-r)}{1-\beta r} < 0,$$

so $\ell > 0$. Equation (98) immediately implies

$$\mathscr{W}_\pi^N > \mathscr{W}_\pi^B.$$

Moreover, substituting the expression for ℓ and using the stable-root identity yields

$$0 < \frac{\ell}{e_0^c} < \frac{2\omega\mathscr{C}}{\mathscr{V}_y}.$$

Equation (99) therefore gives

$$\mathscr{W}_y^N < \mathscr{W}_y^B.$$

Next, suppose $e_0^c < 0$. Since $\ell > 0$ under strict binding, (99) gives

$$\mathscr{W}_y^N - \mathscr{W}_y^B = q_y\left[\omega|e_0^c|\ell\mathscr{C} + \frac{1}{2}\ell^2\mathscr{V}_y\right] > 0.$$

Thus, $\mathscr{W}_y^N > \mathscr{W}_y^B$. For inflation, (98) becomes

$$\mathscr{W}_\pi^N - \mathscr{W}_\pi^B = q_\pi\left[-|e_0^c|\ell\mathscr{C} + \frac{1}{2}\ell^2\mathscr{V}_\pi\right],$$

implying

$$\mathscr{W}_\pi^N \gtreqless \mathscr{W}_\pi^B \quad\Longleftrightarrow\quad \frac{\lambda_4}{q_y|e_0^c|} \gtreqless \frac{2\mathscr{C}}{\mathscr{V}_\pi}.$$

□

C.6 Proof of Proposition 7

Proof. For each $T \in \{0,1,2,\ldots\}$, consider the one-time cost-push shock

$$e_t^{c,(T)} = \begin{cases} \dfrac{\bar{e}^c}{\beta^T}, & t = T, \\ 0, & t \neq T, \end{cases} \qquad \bar{e}^c > 0.$$

The two shock statistics defined in Appendix C.1 are

$$\Delta^{c,(T)} = \sum_{t=0}^{\infty} \beta^t e_t^{c,(T)} = \bar{e}^c, \qquad \Delta^{r,(T)} = \sum_{t=0}^{\infty} (\beta r)^t e_t^{c,(T)} = r^T \bar{e}^c.$$

Thus, delaying the shock leaves its usual discounted present value unchanged, while reducing its early-weighted present value by the factor r^T. Under $H^* = 0$ and $\hat{b}^C_{-1} = 0$, Appendix C.1 gives

$$\lambda_4 = -\frac{\bar{e}^c}{\Omega} \left\{ \frac{1-\beta}{\kappa} + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa \frac{q_\pi}{q_y} \right] r^{T+1} \right\},$$

where $\Omega > 0$. Define

$$g_T \equiv \frac{1-\beta}{\kappa} + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa \frac{q_\pi}{q_y} \right] r^{T+1}.$$

Because $\bar{e}^c > 0$, the remittance constraint binds strictly if and only if

$$g_T < 0.$$

Since $(1-\beta)/\kappa > 0$, strict binding is possible only if

$$\frac{\kappa+\beta-1}{\kappa} - \kappa \frac{q_\pi}{q_y} < 0.$$

Under this condition,

$$g_{T+1} - g_T = \left[\frac{\kappa+\beta-1}{\kappa} - \kappa \frac{q_\pi}{q_y} \right] r^{T+1}(r-1) > 0,$$

because $0 < r < 1$. Hence, g_T is strictly increasing in T. Moreover,

$$\lim_{T\to\infty} g_T = \frac{1-\beta}{\kappa} > 0.$$

Therefore, if $g_0 \geq 0$, then $g_T > 0$ for every $T \geq 1$, so the funding constraint never binds strictly. If $g_0 < 0$, there exists a finite $\bar{T} \geq 0$ such that

$$g_T < 0, \qquad T = 0,\ldots,\bar{T},$$

and

$$g_T \geq 0, \qquad T > \bar{T}.$$

Thus, the set of arrival dates for which the remittance constraint binds strictly is either empty or an initial finite sequence $\{0,1,\ldots,\bar{T}\}$. □

C.7 Proof of Proposition 10

Proof. Let g_T denote the statistic used in Appendix C.6:

$$g_T \equiv \frac{1-\beta}{\kappa} + \left[\frac{\kappa+\beta-1}{\kappa} - \kappa\frac{q_\pi}{q_y}\right] r^{T+1}.$$

Under the assumptions of the proposition, $\Delta^{c,(T)} = \bar{e}^c$ and $\Delta^{r,(T)} = r^T \bar{e}^c$, and the remittance constraint binds strictly if and only if $g_T < 0$.

Define the funding-capacity functional

$$\mathscr{F}(\hat{y}, \hat{\pi}) \equiv \hat{\pi}_0 + \beta\hat{y}_0 - (1-\beta)\sum_{t=1}^{\infty} \beta^t \hat{y}_t.$$

Since demand shocks are absent and $\hat{b}^C_{-1} = H^* = 0$, the intertemporal central-bank budget constraint implies

$$\mathscr{F}(\hat{y}, \hat{\pi}) = (1-\beta)a_h \sum_{t=0}^{\infty} \beta^t \hat{H}_t.$$

Thus, the additional implementability restriction under no fiscal backing is $\mathscr{F}(\hat{y}, \hat{\pi}) \geq 0$. Under fiscal backing, remittances are unrestricted and this inequality is absent.

For a fixed arrival date T and any scalar f, consider the auxiliary equality-constrained problem

$$\mathscr{J}_T(f) \equiv \min_{\{\hat{y}_t, \hat{\pi}_t\}_{t=0}^{\infty}} \frac{1}{2}\sum_{t=0}^{\infty} \beta^t \left(q_y \hat{y}_t^2 + q_\pi \hat{\pi}_t^2\right)$$

subject to

$$\begin{aligned} \hat{\pi}_t &= \kappa\hat{y}_t + \beta\hat{\pi}_{t+1} + e_t^{(T)}, \qquad t \geq 0, \\ \mathscr{F}(\hat{y}, \hat{\pi}) &= f. \end{aligned}$$

Let $\lambda_4(T, f)$ be the unrestricted multiplier on $f - \mathscr{F}(\hat{y}, \hat{\pi})$, using the sign convention of Appendix C.1. The closed-form solution in that appendix implies

$$f = \Omega\lambda_4(T, f) + \bar{e}^c g_T,$$

where $\Omega > 0$ depends only on $(\beta, \kappa, q_y, q_\pi)$ and is independent of T and f. By the envelope theorem,

$$\mathscr{J}_T'(f) = \lambda_4(T, f) = \frac{f - \bar{e}^c g_T}{\Omega}.$$

Integrating from $\bar{e}^c g_T$ to f gives

$$\mathscr{J}_T(f) = \mathscr{J}_T(\bar{e}^c g_T) + \frac{(f - \bar{e}^c g_T)^2}{2\Omega}. \tag{100}$$

Under fiscal backing, the planner can choose any $f \in \mathbb{R}$, so the optimum satisfies $f = \bar{e}^c g_T$ and

$$\mathscr{W}^{\mathrm{FB}}(T) = \mathscr{J}_T(\bar{e}^c g_T).$$

Under no fiscal backing, the planner instead minimizes $\mathscr{J}_T(f)$ over $f \geq 0$. Equation (100) therefore implies

$$f^{\mathrm{NFB}}(T) = \max\{\bar{e}^c g_T, 0\}.$$

Since $\bar{e}^c > 0$, the welfare cost of no fiscal backing is consequently

$$\mathscr{V}(T) = \frac{(\bar{e}^c)^2}{2\Omega}\left[\max\{-g_T, 0\}\right]^2. \tag{101}$$

It remains to establish monotonicity. If $g_0 \geq 0$, Appendix C.6 implies $g_T \geq 0$ for every T, so $\mathscr{V}(T) = 0$ for every arrival date. Suppose instead that $g_0 < 0$. Then

$$\frac{\kappa + \beta - 1}{\kappa} - \kappa \frac{q_\pi}{q_y} < 0,$$

and hence

$$g_{T+1} - g_T = \left[\frac{\kappa + \beta - 1}{\kappa} - \kappa \frac{q_\pi}{q_y}\right] r^{T+1}(r-1) > 0.$$

Therefore, $\max\{-g_T, 0\}$ is weakly decreasing in T, with a strict decrease between any $T_1 < T_2$ whenever $g_{T_1} < 0$. Equation (101) establishes the stated welfare inequalities. Finally,

$$\lim_{T\to\infty} g_T = \frac{1-\beta}{\kappa} > 0,$$

so $g_T \geq 0$, and therefore $\mathscr{V}(T) = 0$, for all sufficiently large T.

We finally establish the monotonicity of the welfare loss under fiscal backing. Define the price-level deviation $\hat{p}^{\mathrm{B}}_t \equiv \sum_{s=0}^{t} \hat{\pi}^{\mathrm{B}}_s$, with $\hat{p}^{\mathrm{B}}_{-1} = 0$. The first-order conditions of the fiscal-backing Ramsey problem imply

$$\hat{y}^{\mathrm{B}}_t = -\kappa \frac{q_\pi}{q_y} \hat{p}^{\mathrm{B}}_t.$$

Substituting this condition into the Phillips curve yields

$$\beta \hat{p}^{\mathrm{B}}_{t+1} - \left(1 + \beta + \kappa^2 \frac{q_\pi}{q_y}\right) \hat{p}^{\mathrm{B}}_t + \hat{p}^{\mathrm{B}}_{t-1} = -e^{(T)}_t.$$

Using the initial condition and boundedness, the solution at the shock date is

$$\hat{p}^{\mathrm{B}}_T = \frac{r\bar{e}^c}{\beta^T} \sum_{j=0}^{T} (\beta r^2)^j,$$

where $r \in (0,1)$ is the stable root defined in Appendix C.1. Multiplying the Phillips curve by $\beta^t q_\pi \hat{p}_t^{\mathrm{B}}$ and summing over time, the optimality condition above implies

$$2\mathscr{W}^{\mathrm{B}}(T) = q_\pi \sum_{t=0}^{\infty} \beta^t e_t^{(T)} \hat{p}_t^{\mathrm{B}} = q_\pi \bar{e}^c \hat{p}_T^{\mathrm{B}}.$$

Therefore,

$$\mathscr{W}^{\mathrm{B}}(T) = \frac{q_\pi r (\bar{e}^c)^2}{2} \beta^{-T} \sum_{j=0}^{T} (\beta r^2)^j.$$

Since $0 < \beta < 1$, $0 < r < 1$, and $\bar{e}^c > 0$, both β^{-T} and $\sum_{j=0}^{T} (\beta r^2)^j$ are strictly increasing in T. Consequently,

$$\mathscr{W}^{\mathrm{B}}(T_1) < \mathscr{W}^{\mathrm{B}}(T_2) \qquad \text{for all } 0 \le T_1 < T_2,$$

which completes the proof. □

D Appendix for Section 4.1

D.1 Closed-form solution for $\rho < 1$

The coefficients in the budget constraints are defined as

$$a_h^D \equiv \frac{(1-\beta\rho)\bar{Y}}{(1-\beta)\bar{d}^{T,CB}}, \qquad b_h^D \equiv \frac{(1-\beta\rho)\bar{Y}}{(1-\beta)\bar{d}^{T}}.$$

We derive the closed-form solution for $0 \le \rho < 1$, $0 < \beta < 1$, and general $q_y, q_\pi, \kappa > 0$, conditional on the remittance constraint binding at every date:

$$\hat{H}_t = H^*, \qquad t \ge 0.$$

We consider a bounded deterministic demand-shock sequence $\{e_t\}_{t=0}^{\infty}$ and impose the linearized transversality and no-bubble conditions

$$\lim_{T\to\infty} \beta^{T+1} \hat{d}_T^{T,CB} = 0,$$
$$\lim_{J\to\infty} (\rho\beta)^{J+1} \hat{Q}_{t+J+1}^{T} = 0, \qquad t \ge 0.$$

Because lump-sum taxes and Treasury debt are unrestricted, the Treasury budget constraint does not restrict the welfare-relevant allocation and can be omitted from the reduced problem.

Define the bond-price path that implements full stabilization,

$$\hat{Q}_t^{T,0} \equiv \sum_{j=0}^{\infty} (\rho\beta)^j e_{t+j},$$

and let $\widetilde{Q}_t^T \equiv \hat{Q}_t^T - \hat{Q}_t^{T,0}$. The long-duration Euler equation then becomes

$$\widetilde{Q}_t^T = \hat{y}_t - \hat{y}_{t+1} - \hat{\pi}_{t+1} + \rho\beta\widetilde{Q}_{t+1}^T.$$

Under binding, the central-bank budget constraint is

$$\beta\hat{d}_t^{T,CB} = \hat{d}_{t-1}^{T,CB} - \hat{\pi}_t - \beta(1-\rho)\hat{Q}_t^T - (1-\beta)a_h^D H^*.$$

Iterating this constraint forward and imposing the transversality condition gives

$$\hat{d}_{-1}^{T,CB} = \sum_{t=0}^{\infty} \beta^t \left[\hat{\pi}_t + \beta(1-\rho)\hat{Q}_t^T + (1-\beta)a_h^D H^*\right].$$

Define the generalized funding gap

$$\mathscr{G}_\rho \equiv a_h^D H^* - \hat{d}_{-1}^{T,CB} + \beta(1-\rho)\sum_{t=0}^{\infty}\beta^t\hat{Q}_t^{T,0} = a_h^D H^* - \hat{d}_{-1}^{T,CB} + \sum_{t=0}^{\infty}\beta^{t+1}(1-\rho^{t+1})e_t.$$

The second equality follows from

$$\beta(1-\rho)\sum_{t=0}^{\infty}\beta^t\sum_{j=0}^{\infty}(\rho\beta)^j e_{t+j} = \sum_{t=0}^{\infty}\beta^{t+1}(1-\rho^{t+1})e_t.$$

The binding central-bank budget constraint can therefore be written as

$$-\sum_{t=0}^{\infty}\beta^t\left[\hat{\pi}_t + \beta(1-\rho)\widetilde{Q}_t^T\right] = \mathscr{G}_\rho.$$

Under the initial conditions $\hat{d}_{-1}^{T,CB} = H^* = 0$, $\mathscr{G}_\rho$ coincides with the statistic Δ_ρ defined in Proposition 11. The reduced binding problem is thus

$$\min_{\{\hat{y}_t,\hat{\pi}_t,\widetilde{Q}_t^T\}_{t=0}^{\infty}} \quad \frac{1}{2}\sum_{t=0}^{\infty}\beta^t\left(q_y\hat{y}_t^2 + q_\pi\hat{\pi}_t^2\right),$$

subject to

$$\begin{aligned}
\hat{\pi}_t &= \kappa\hat{y}_t + \beta\hat{\pi}_{t+1},\\
\widetilde{Q}_t^T &= \hat{y}_t - \hat{y}_{t+1} - \hat{\pi}_{t+1} + \rho\beta\widetilde{Q}_{t+1}^T,\\
&-\sum_{t=0}^{\infty}\beta^t\left[\hat{\pi}_t + \beta(1-\rho)\widetilde{Q}_t^T\right] = \mathscr{G}_\rho.
\end{aligned}$$

Let λ_{1t}, λ_{2t}, and λ_ρ denote the multipliers on these three restrictions. We use the sign convention under which the multiplier ν_t on the original remittance constraint satisfies

$$\nu_t = (1-\beta)a_h^D\lambda_\rho.$$

Hence, strict binding requires $\lambda_\rho > 0$. The first-order conditions are

$$\begin{aligned} q_y \hat{y}_t - \kappa\lambda_{1t} - \lambda_{2t} + \beta^{-1}\lambda_{2,t-1} &= 0, \\ q_\pi \hat{\pi}_t + \lambda_{1t} - \lambda_{1,t-1} + \beta^{-1}\lambda_{2,t-1} + \lambda_\rho &= 0, \\ \lambda_{2t} - \rho\lambda_{2,t-1} + \beta(1-\rho)\lambda_\rho &= 0, \end{aligned}$$

with

$$\lambda_{1,-1} = \lambda_{2,-1} = 0.$$

The last condition gives

$$\lambda_{2t} = -\beta\lambda_\rho\left(1-\rho^{t+1}\right).$$

Substituting this expression into the output first-order condition yields

$$\lambda_{1t} = \frac{q_y}{\kappa}\hat{y}_t - \frac{\lambda_\rho}{\kappa}\left[(1-\beta) + (\beta\rho - 1)\rho^t\right].$$

Using $\hat{y}_t = \frac{\hat{\pi}_t - \beta\hat{\pi}_{t+1}}{\kappa}$, the inflation first-order condition implies, for $t \geq 1$,

$$\beta\hat{\pi}_{t+1} - \left(1 + \beta + \frac{\kappa^2 q_\pi}{q_y}\right)\hat{\pi}_t + \hat{\pi}_{t-1} = -\frac{\kappa^2}{q_y}\lambda_\rho\left[\frac{(1-\beta\rho)(1-\rho)}{\kappa} - \rho\right]\rho^{t-1}.$$

Define

$$\begin{aligned} A &\equiv 1 + \beta + \frac{\kappa^2 q_\pi}{q_y}, \\ m &\equiv \frac{A - \sqrt{A^2 - 4\beta}}{2\beta}, \qquad 0 < m < 1, \\ \chi(z) &\equiv \frac{(1-\beta z)(1-z)}{\kappa} - z. \end{aligned}$$

The stable root m satisfies

$$\beta m^2 - Am + 1 = 0.$$

For $\rho \neq m$, define

$$a_\rho \equiv \frac{\frac{\kappa^2}{q_y}\chi(\rho)}{\beta\rho^2 - A\rho + 1}, \qquad c_\rho \equiv \frac{1 + \frac{\beta(1-\rho)}{\kappa} - a_\rho\left[q_\pi + \frac{q_y}{\kappa^2}(1-\beta\rho)\right]}{q_\pi + \frac{q_y}{\kappa^2}(1-\beta m)}.$$

The unique bounded inflation path is

$$\hat{\pi}_t = -\lambda_\rho\left(a_\rho\rho^t + c_\rho m^t\right), \qquad t \geq 0.$$

At the isolated resonance point $\rho = m$, the expression is understood by continuity; equivalently, the particular solution contains a term proportional to tm^t.

Output follows from the Phillips curve:

$$\hat{y}_t = -\frac{\lambda_\rho}{\kappa}\left[a_\rho(1-\beta\rho)\rho^t + c_\rho(1-\beta m)m^t\right], \qquad t \geq 0.$$

The endogenous component of the long-duration bond price is

$$\widetilde{Q}_t^T = -\lambda_\rho\left[\frac{a_\rho\chi(\rho)\rho^t}{1-\beta\rho^2} + \frac{c_\rho\chi(m)m^t}{1-\beta\rho m}\right].$$

Hence,

$$\hat{Q}_t^T = \sum_{j=0}^{\infty}(\rho\beta)^j e_{t+j} - \lambda_\rho\left[\frac{a_\rho\chi(\rho)\rho^t}{1-\beta\rho^2} + \frac{c_\rho\chi(m)m^t}{1-\beta\rho m}\right].$$

To determine λ_ρ, define the discounted loss generated by the unit-multiplier allocation:

$$\begin{aligned} D_\rho \equiv\, & q_\pi\left[\frac{a_\rho^2}{1-\beta\rho^2} + \frac{2a_\rho c_\rho}{1-\beta\rho m} + \frac{c_\rho^2}{1-\beta m^2}\right] \\ & + \frac{q_y}{\kappa^2}\left[\frac{a_\rho^2(1-\beta\rho)^2}{1-\beta\rho^2} + \frac{2a_\rho c_\rho(1-\beta\rho)(1-\beta m)}{1-\beta\rho m} + \frac{c_\rho^2(1-\beta m)^2}{1-\beta m^2}\right]. \end{aligned}$$

At $\rho = m$, D_ρ is understood as its continuous extension. Since $q_y, q_\pi > 0$ and the unit-multiplier allocation is nonzero,

$$D_\rho > 0.$$

The objective is homogeneous of degree two in the choice variables, while the equilibrium restrictions entering the reduced problem are linear in those variables. Multiplying the first-order conditions by the corresponding choice variables and summing over time, with the terms associated with the Phillips curve and the long-duration Euler equation vanishing by feasibility, gives

$$2W = \lambda_\rho \mathscr{G}_\rho.$$

On the other hand, the closed-form allocation above implies

$$2W = \lambda_\rho^2 D_\rho.$$

Consequently,

$$\mathscr{G}_\rho = D_\rho\lambda_\rho, \qquad \lambda_\rho = \frac{\mathscr{G}_\rho}{D_\rho}.$$

Thus, the binding candidate satisfies the strict KKT condition if and only if $\mathscr{G}_\rho > 0$. For a one-time demand shock,

$$e_0 = \varepsilon, \qquad e_t = 0, \quad t \geq 1,$$

the generalized funding gap becomes

$$\mathscr{G}_\rho = a_h^D H^* - \hat{d}_{-1}^{T,CB} + \beta(1-\rho)\varepsilon.$$

In particular, when $\hat{d}_{-1}^{T,CB} = H^* = 0$,

$$\lambda_\rho = \frac{\beta(1-\rho)\varepsilon}{D_\rho}.$$

Finally, the central-bank asset position can be recovered from

$$\hat{d}_{t-1}^{T,CB} = \sum_{s=t}^{\infty} \beta^{s-t} \left[\hat{\pi}_s + \beta(1-\rho)\hat{Q}_s^T + (1-\beta)a_h^D H^* \right].$$

Given these paths, the Treasury budget constraint can be satisfied residually through lump-sum taxes and Treasury debt. Hence, the fiscal implementation need not be uniquely specified to determine the welfare-relevant allocation.

D.2 Proof of Proposition 11

Proof. Under the initial conditions

$$\hat{d}_{-1}^{T,CB} = H^* = 0,$$

the generalized funding gap defined in Appendix D.1 becomes

$$\mathscr{G}_\rho = \sum_{t=0}^{\infty} \beta^{t+1} \left(1 - \rho^{t+1}\right) e_t = \Delta_\rho.$$

Appendix D.1 gives

$$\lambda_\rho = \frac{\Delta_\rho}{D_\rho}, \qquad D_\rho > 0,$$

and the multiplier on the original remittance constraint satisfies

$$\nu_t = (1-\beta)a_h^D \lambda_\rho, \qquad t \geq 0.$$

Since $a_h^D > 0$, if $\Delta_\rho > 0$, then $\lambda_\rho > 0$ and hence $\nu_t > 0$ for every $t \geq 0$. Complementary slackness therefore implies

$$\hat{H}_t = H^* = 0, \qquad t \geq 0.$$

Thus, the remittance constraint binds strictly at every date. Now suppose $\Delta_\rho \leq 0$. Consider full stabilization,

$$\hat{y}_t = \hat{\pi}_t = 0, \qquad t \geq 0.$$

The long-duration Euler equation then implies

$$\hat{Q}_t^T = \sum_{j=0}^{\infty} (\rho\beta)^j e_{t+j}.$$

Iterating the central-bank budget constraint forward and imposing the transversality condition gives

$$0 = \Delta_\rho + (1-\beta) a_h^D \sum_{t=0}^{\infty} \beta^t \hat{H}_t.$$

Choose

$$\hat{H}_t = -\frac{\Delta_\rho}{a_h^D}, \qquad t \geq 0.$$

Because $\Delta_\rho \leq 0$ and $a_h^D > 0$, this choice satisfies

$$\hat{H}_t \geq 0 = H^*.$$

Moreover,

$$(1-\beta) a_h^D \sum_{t=0}^{\infty} \beta^t \hat{H}_t = a_h^D \hat{H}_t = -\Delta_\rho,$$

so the intertemporal central-bank budget constraint is satisfied. The corresponding central-bank asset path can be generated recursively from the flow budget constraint, and the Treasury budget constraint can be satisfied residually through lump-sum taxes and Treasury debt. Hence, full stabilization is feasible. Since the quadratic welfare loss is nonnegative and full stabilization yields zero loss, it is optimal. If $\Delta_\rho < 0$, the construction gives

$$\hat{H}_t = -\frac{\Delta_\rho}{a_h^D} > 0 = H^*,$$

for every $t \geq 0$, so the remittance constraint is strictly slack throughout. If $\Delta_\rho = 0$, full stabilization is feasible with

$$\hat{H}_t = H^* = 0, \qquad t \geq 0,$$

but the remittance constraint has zero shadow value. □

D.3 Proof of Proposition 12

Proof. When $\rho = 1$, the long-duration asset-price equation becomes

$$\hat{Q}_t^T = \hat{y}_t - \hat{y}_{t+1} - \hat{\pi}_{t+1} + \beta \hat{Q}_{t+1}^T + e_t,$$

while the bond-price term disappears from the central-bank budget constraint:

$$\beta \hat{d}_t^{T,CB} = \hat{d}_{t-1}^{T,CB} - \hat{\pi}_t - (1-\beta) a_h^D \hat{H}_t.$$

Consider the allocation

$$\hat{y}_t = \hat{\pi}_t = 0, \qquad \hat{H}_t = H^*, \qquad \hat{d}_t^{T,CB} = a_h^D H^*, \qquad t \geq 0.$$

Given the initial condition $\hat{d}_{-1}^{T,CB} = a_h^D H^*$, this allocation satisfies the central-bank budget constraint and the Phillips curve. Under this allocation, the asset-price equation reduces to

$$\hat{Q}_t^T = \beta \hat{Q}_{t+1}^T + e_t.$$

Imposing the no-bubble condition gives

$$\hat{Q}_t^T = \sum_{j=0}^{\infty} \beta^j e_{t+j}.$$

Because lump-sum taxes and Treasury debt are unrestricted, the Treasury budget constraint can be satisfied residually. The allocation generates zero welfare loss:

$$\frac{1}{2} \sum_{t=0}^{\infty} \beta^t \left(q_y \hat{y}_t^2 + q_\pi \hat{\pi}_t^2 \right) = 0.$$

Since $q_y, q_\pi > 0$, the objective is nonnegative for every feasible allocation. Hence, this allocation is optimal, and the welfare-relevant allocation is uniquely characterized by

$$\hat{y}_t = \hat{\pi}_t = 0, \qquad t \geq 0.$$

Although the constructed implementation sets $\hat{H}_t = H^*$, the remittance constraint has zero shadow value because relaxing it cannot improve on the zero-loss allocation. Thus, the remittance constraint does not restrict the optimal allocation. □

D.4 Proof of Proposition 13

Proof. For each $\beta \in (0,1)$, let $\mathscr{W}_\beta(\rho)$ denote the minimized welfare loss. We take the patient limit $\beta \uparrow 1$, holding κ, q_y, q_π, and ρ fixed. The limiting stable root $m \in (0,1)$ satisfies

$$(1-m)^2 = \kappa^2 \frac{q_\pi}{q_y} m.$$

For $0 \le \rho < 1$, consider the one-time shock and initial condition

$$e_0 = \varepsilon > 0, \qquad e_t = 0 \quad (t \ge 1), \qquad \hat{d}_{-1}^{T,CB} = a_h^D H^*.$$

The generalized funding gap defined in Appendix D.1 becomes

$$\mathscr{G}_\rho = a_h^D H^* - \hat{d}_{-1}^{T,CB} + \beta(1-\rho)\varepsilon = \beta(1-\rho)\varepsilon > 0.$$

Thus, the remittance constraint binds at every date, and Appendix D.1 gives

$$\mathscr{W}_\beta(\rho) = \frac{\mathscr{G}_\rho^2}{2D_\rho} = \frac{\beta^2(1-\rho)^2\varepsilon^2}{2D_\rho}.$$

Define

$$\bar{D}_\rho \equiv \lim_{\beta\uparrow 1} D_\rho, \qquad K(\rho) \equiv \frac{\bar{D}_\rho}{(1-\rho)^2}, \qquad \mathscr{W}(\rho) \equiv \lim_{\beta\uparrow 1} \mathscr{W}_\beta(\rho).$$

Taking the patient limit yields

$$\mathscr{W}(\rho) = \frac{\varepsilon^2}{2K(\rho)}.$$

Using the limiting stable-root condition to simplify $\bar{D}_\rho$ gives

$$K(\rho) = \frac{m}{q_y}\Bigg\{ \frac{2}{(1+m)^2(1+\rho)} + \frac{2m}{(1+m)^2(1-m\rho)} + \frac{m(1-m)}{(1+m)(1-m\rho)^2} + \frac{2\kappa}{(1-\rho^2)(1-m\rho)^2} + \frac{\kappa^2}{(1-\rho)^2(1-m\rho)^2}\left[\frac{1}{1-\rho^2} + \frac{m}{1-m}\right]\Bigg\}.$$

This expression holds for $0 \le \rho < 1$, including $\rho = m$ by continuity. The first three terms are strictly convex in ρ, while the remaining terms are products of positive, nondecreasing convex functions. Hence,

$$K''(\rho) > 0, \qquad 0 \le \rho < 1.$$

Since

$$\mathscr{W}'(\rho) = -\frac{\varepsilon^2}{2K(\rho)^2}K'(\rho),$$

and K' is strictly increasing, $\mathscr{W}'(\rho) < 0$ for every $\rho \in (0,1)$ if and only if $K'(0) \ge 0$. Direct differentiation gives

$$K'(0) = \frac{2m(\kappa+1-m)}{q_y(1-m)}\left[\kappa(1+m) - (1-m)^2\right].$$

Since the prefactor is strictly positive, monotonicity is equivalent to

$$\kappa(1+m) \geq (1-m)^2.$$

This condition holds automatically when $\kappa \geq 1$. For $0 < \kappa < 1$, it is equivalent to

$$m \geq \bar{m}(\kappa) \equiv \frac{2+\kappa-\sqrt{\kappa^2+8\kappa}}{2}.$$

The limiting stable-root condition implies

$$\frac{q_\pi}{q_y} = \frac{(1-m)^2}{\kappa^2 m},$$

whose right-hand side is strictly decreasing in $m \in (0,1)$. Therefore,

$$m \geq \bar{m}(\kappa) \quad \Longleftrightarrow \quad 2\kappa(1-\kappa)q_\pi \leq \left(4-\kappa+\sqrt{\kappa^2+8\kappa}\right) q_y.$$

This establishes condition (26) for all $\kappa > 0$. □

E Appendix for Section 4.3

E.1 Closed-form equilibrium

Consider a one-time expansionary demand shock,

$$e_0 < 0, \qquad e_t = 0, \quad t \geq 1,$$

with zero government purchases, $G_t \equiv 0$, as in Section 4.3. We assume $\gamma > 0$, allow unrestricted tax-rate adjustments, and initialize the economy at its deterministic steady state:

$$\hat{b}^T_{-1} = \hat{b}^C_{-1} = 0, \qquad H^* = 0.$$

Under the tax-rebate scheme in Section 4.3, net sales-tax revenue is $\tau_t P_t \bar{Y}$. The linearized Treasury budget constraint is therefore

$$\hat{b}^T_{t-1} - \hat{\pi}_t - (1-\beta) b_h (\hat{H}_t + \hat{\tau}_t) = \beta(\hat{b}^T_t + \hat{Q}_t).$$

Multiplying the central-bank and Treasury budget constraints by β^t, summing over time, and imposing the transversality conditions yields

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) = (1-\beta) a_h \sum_{t=0}^{\infty} \beta^t \hat{H}_t,$$

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) + (1-\beta) b_h \sum_{t=0}^{\infty} \beta^t (\hat{H}_t + \hat{\tau}_t) = 0.$$

Eliminating remittances gives

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) + \frac{(1-\beta)a_h b_h}{a_h + b_h} \sum_{t=0}^{\infty} \beta^t \hat{\tau}_t = 0.$$

Define

$$\chi_\gamma \equiv \frac{(1-\beta)a_h b_h}{(a_h + b_h)\gamma}. \tag{102}$$

Summing the Phillips curve,

$$\hat{\pi}_t = \kappa \hat{y}_t + \gamma \hat{\tau}_t + \beta \hat{\pi}_{t+1},$$

yields

$$\gamma \sum_{t=0}^{\infty} \beta^t \hat{\tau}_t = \hat{\pi}_0 - \kappa \sum_{t=0}^{\infty} \beta^t \hat{y}_t.$$

Substituting into the consolidated present-value budget constraint gives

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) + \chi_\gamma \left[\hat{\pi}_0 - \kappa \sum_{t=0}^{\infty} \beta^t \hat{y}_t\right] = 0.$$

The household Euler equation and the one-time demand shock imply

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) = \hat{y}_0 + \hat{\pi}_0 - (1-\beta) \sum_{t=0}^{\infty} \beta^t \hat{y}_t + \beta e_0.$$

Combining these expressions yields the fiscal-backing implementability condition:

$$\hat{y}_0 + (1+\chi_\gamma)\hat{\pi}_0 - (1 - \beta + \kappa\chi_\gamma) \sum_{t=0}^{\infty} \beta^t \hat{y}_t = -\beta e_0.$$

Define

$$S_y \equiv \sum_{t=1}^{\infty} \beta^t \hat{y}_t.$$

Conditional on $(\hat{y}_0, \hat{\pi}_0, S_y)$, welfare is minimized by

$$\hat{\pi}_t = 0, \qquad \hat{y}_t = \frac{1-\beta}{\beta} S_y, \qquad t \geq 1.$$

The fiscal-backing problem therefore reduces to

$$\min_{\hat{y}_0, \hat{\pi}_0, S_y} \quad \frac{q_y}{2}\hat{y}_0^2 + \frac{q_\pi}{2}\hat{\pi}_0^2 + \frac{q_y}{2}\frac{1-\beta}{\beta}S_y^2$$

subject to

$$(\beta - \kappa\chi_\gamma)\hat{y}_0 + (1+\chi_\gamma)\hat{\pi}_0 - (1-\beta+\kappa\chi_\gamma)S_y = -\beta e_0.$$

Define

$$D_\gamma \equiv \frac{(\beta-\kappa\chi_\gamma)^2}{q_y} + \frac{(1+\chi_\gamma)^2}{q_\pi} + \frac{\beta(1-\beta+\kappa\chi_\gamma)^2}{(1-\beta)q_y} = \frac{\beta}{q_y} + \frac{(1+\chi_\gamma)^2}{q_\pi} + \frac{\kappa^2\chi_\gamma^2}{(1-\beta)q_y}. \tag{103}$$

The fiscal-backing allocation is

$$\begin{aligned}
\hat{y}_0^B &= -\frac{\beta e_0(\beta-\kappa\chi_\gamma)}{q_y D_\gamma}, \\
\hat{\pi}_0^B &= -\frac{\beta e_0(1+\chi_\gamma)}{q_\pi D_\gamma}, \\
\hat{y}_t^B &= \frac{\beta e_0(1-\beta+\kappa\chi_\gamma)}{q_y D_\gamma}, \qquad t \geq 1, \\
\hat{\pi}_t^B &= 0, \qquad t \geq 1.
\end{aligned}$$

The corresponding asset-price responses are

$$\begin{aligned}
\hat{Q}_0^B &= e_0\left(1 - \frac{\beta}{q_y D_\gamma}\right), \\
\hat{Q}_t^B &= 0, \qquad t \geq 1.
\end{aligned}$$

The tax path follows from the Phillips curve:

$$\hat{\tau}_t^B = \frac{\hat{\pi}_t^B - \kappa\hat{y}_t^B - \beta\hat{\pi}_{t+1}^B}{\gamma}.$$

The fiscal-backing welfare loss is

$$W^B(\gamma) = \frac{\beta^2 e_0^2}{2D_\gamma}. \tag{104}$$

We next characterize the no-fiscal-backing equilibrium. Under the fiscal-backing allocation,

$$\hat{\pi}_0^B - \kappa\sum_{t=0}^{\infty}\beta^t\hat{y}_t^B = -\frac{\beta e_0}{D_\gamma}\left[\frac{1+\chi_\gamma}{q_\pi} + \frac{\kappa^2\chi_\gamma}{(1-\beta)q_y}\right] > 0.$$

The consolidated present-value budget constraint therefore implies

$$\sum_{t=0}^{\infty}\beta^t\left(\beta\hat{Q}_t^B + \hat{\pi}_t^B\right) < 0.$$

Under no fiscal backing, however, the central-bank budget constraint and $\hat{H}_t \geq 0$ require

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) \geq 0.$$

Thus, the fiscal-backing allocation is infeasible under no fiscal backing. Since the no-fiscal-backing problem adds this linear inequality to the strictly convex fiscal-backing problem, and the unconstrained optimum violates it strictly, the inequality binds:

$$\sum_{t=0}^{\infty} \beta^t \left(\beta \hat{Q}_t + \hat{\pi}_t\right) = 0.$$

The central-bank present-value budget constraint then gives

$$(1-\beta) a_h \sum_{t=0}^{\infty} \beta^t \hat{H}_t^N = 0.$$

Since $\hat{H}_t^N \geq 0$ for every t, we obtain

$$\hat{H}_t^N = 0, \qquad t \geq 0.$$

The Treasury present-value budget constraint consequently requires

$$\sum_{t=0}^{\infty} \beta^t \hat{\tau}_t^N = 0.$$

Combining this restriction with the present-value Phillips curve yields

$$\hat{\pi}_0^N = \kappa \sum_{t=0}^{\infty} \beta^t \hat{y}_t^N.$$

The Euler equation and the binding central-bank budget provide the second restriction:

$$\hat{y}_0^N + \hat{\pi}_0^N - (1-\beta) \sum_{t=0}^{\infty} \beta^t \hat{y}_t^N = -\beta e_0.$$

Define

$$\mathscr{D} \equiv q_y \left[\kappa^2 + \beta(1-\beta)\right] + \beta q_\pi \kappa^2. \tag{105}$$

Solving the resulting quadratic problem gives

$$\begin{aligned}
\hat{\pi}_0^N &= -\frac{\beta e_0 q_y \kappa^2}{\mathscr{D}}, \\
\hat{\pi}_t^N &= 0, \qquad t \geq 1, \\
\hat{y}_0^N &= -\frac{\beta e_0 \left[q_y (1-\beta)(\beta+\kappa) + \beta q_\pi \kappa^2\right]}{\mathscr{D}}, \\
\hat{y}_t^N &= -\frac{\beta e_0 (1-\beta) \left[q_y (\kappa+\beta-1) - q_\pi \kappa^2\right]}{\mathscr{D}}, \qquad t \geq 1.
\end{aligned}$$

The corresponding asset-price responses are

$$\hat{Q}_0^N = \frac{e_0 q_y \kappa^2}{\mathscr{D}},$$
$$\hat{Q}_t^N = 0, \qquad t \geq 1,$$

and the tax path is recovered from the Phillips curve:

$$\hat{\tau}_t^N = \frac{\hat{\pi}_t^N - \kappa \hat{y}_t^N - \beta \hat{\pi}_{t+1}^N}{\gamma}.$$

The no-fiscal-backing welfare loss is

$$W^N = \frac{\beta^2 e_0^2 q_y \left[q_\pi \kappa^2 + q_y(1-\beta)\right]}{2\mathscr{D}}. \tag{106}$$

Thus, for every $\gamma > 0$, the no-fiscal-backing paths of output and inflation, and the associated welfare loss, are independent of γ; only the tax path required to implement the allocation changes. The case $\gamma = 0$ is distinct because tax-rate adjustments no longer enter the Phillips curve, and the Ramsey problem reduces to the baseline nondistortionary-tax case.

E.2 Proof of Proposition 16

Proof. Fix $0 < \beta < 1$ and $\kappa, q_y, q_\pi, a_h, b_h > 0$, and consider $\gamma > 0$. Appendix E.1 shows that, holding these parameters fixed, the no-fiscal-backing allocation and welfare loss are independent of γ. Define

$$\mathscr{D} \equiv q_y \left[\kappa^2 + \beta(1-\beta)\right] + \beta q_\pi \kappa^2.$$

Then

$$\mathscr{W}^N = \frac{\beta^2 e_0^2 q_y \left[q_\pi \kappa^2 + q_y(1-\beta)\right]}{2\mathscr{D}}.$$

Under fiscal backing, Appendix E.1 gives

$$\mathscr{W}^B(\gamma) = \frac{\beta^2 e_0^2}{2D_\gamma},$$

where

$$D_\gamma = \frac{\beta}{q_y} + \frac{(1+\chi_\gamma)^2}{q_\pi} + \frac{\kappa^2 \chi_\gamma^2}{(1-\beta)q_y},$$
$$\chi_\gamma = \frac{(1-\beta)a_h b_h}{(a_h + b_h)\gamma} > 0.$$

For $\gamma > 0$,

$$\frac{d\chi_\gamma}{d\gamma} = -\frac{\chi_\gamma}{\gamma},$$

and therefore

$$\frac{dD_\gamma}{d\gamma} = -\frac{2\chi_\gamma}{\gamma}\left[\frac{1+\chi_\gamma}{q_\pi} + \frac{\kappa^2\chi_\gamma}{(1-\beta)q_y}\right] < 0.$$

It follows that

$$\frac{d\mathscr{W}^B(\gamma)}{d\gamma} = \frac{\beta^2 e_0^2 \chi_\gamma}{\gamma D_\gamma^2}\left[\frac{1+\chi_\gamma}{q_\pi} + \frac{\kappa^2\chi_\gamma}{(1-\beta)q_y}\right] > 0.$$

Since $\mathscr{W}^N$ is independent of γ,

$$\frac{d}{d\gamma}\left[\mathscr{W}^N - \mathscr{W}^B(\gamma)\right] = -\frac{d\mathscr{W}^B(\gamma)}{d\gamma} < 0.$$

Hence, the welfare value of fiscal backing is strictly decreasing in the degree of tax distortion. Finally, as $\gamma \to \infty$, $\chi_\gamma \to 0$, so

$$D_\gamma \longrightarrow \frac{\beta}{q_y} + \frac{1}{q_\pi} = \frac{q_y + \beta q_\pi}{q_y q_\pi}.$$

Therefore,

$$\lim_{\gamma\to\infty} \mathscr{W}^B(\gamma) = \frac{\beta^2 e_0^2 q_y q_\pi}{2(q_y + \beta q_\pi)}.$$

Combining this limit with the expression for $\mathscr{W}^N$ gives

$$\lim_{\gamma\to\infty}\left[\mathscr{W}^N - \mathscr{W}^B(\gamma)\right] = \frac{\beta^2 e_0^2 q_y^3 (1-\beta)}{2(q_y + \beta q_\pi)\mathscr{D}} > 0.$$

The strict inequality follows from $0 < \beta < 1$ and $q_y, q_\pi, \kappa > 0$. Thus, greater tax distortions strictly reduce, but do not eliminate, the welfare value of fiscal backing. To clarify the mechanism behind the positive limiting welfare gap, the present-value Phillips curve implies, under fiscal backing,

$$\gamma\sum_{t=0}^{\infty}\beta^t\hat{\tau}_t^B = \hat{\pi}_0^B - \kappa\sum_{t=0}^{\infty}\beta^t\hat{y}_t^B.$$

Using the fiscal-backing solution in Appendix E.1,

$$\gamma\sum_{t=0}^{\infty}\beta^t\hat{\tau}_t^B = -\frac{\beta e_0}{D_\gamma}\left[\frac{1+\chi_\gamma}{q_\pi} + \frac{\kappa^2\chi_\gamma}{(1-\beta)q_y}\right].$$

Since $e_0 < 0$ and $\chi_\gamma \to 0$,

$$\lim_{\gamma\to\infty} \gamma \sum_{t=0}^{\infty} \beta^t \hat{\tau}_t^B = -\frac{\beta e_0 q_y}{q_y + \beta q_\pi} > 0.$$

Thus,

$$\sum_{t=0}^{\infty} \beta^t \hat{\tau}_t^B \longrightarrow 0,$$

even though the effect of tax adjustments on firms' pricing decisions remains nonvanishing. Moreover, the present-value Treasury and central-bank budget constraints imply

$$\sum_{t=0}^{\infty} \beta^t \hat{H}_t^B = -\frac{b_h}{a_h + b_h} \sum_{t=0}^{\infty} \beta^t \hat{\tau}_t^B \longrightarrow 0.$$

By contrast, under no fiscal backing,

$$\sum_{t=0}^{\infty} \beta^t \hat{\tau}_t^N = 0$$

for every $\gamma > 0$. Hence, even though both the present discounted value of tax-rate adjustments and fiscal support converge to zero under fiscal backing, the distortionary effect $\gamma\hat{\tau}_t$ does not vanish. The fiscal-backing allocation therefore does not converge fully to the no-fiscal-backing allocation, leaving a strictly positive limiting welfare gain. □

E.3 Derivation of the Phillips curve under the tax-rebate scheme

This appendix derives the New Keynesian Phillips curve under the tax-rebate scheme in Section 4.3. The lump-sum rebate does not enter firms' pricing decisions. As in the main text, firms receive a constant employment subsidy ς, chosen to offset the steady-state monopolistic distortion and sales-tax wedge:

$$1 - \varsigma = \frac{(\theta - 1)(1 - \bar{\tau})}{\theta}.$$

Firm i therefore solves

$$\max_{\{p_t(i)\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t \Lambda_t \left[(1-\tau_t) p_t(i) y_t(i) - (1-\varsigma) w_t N_t(i) - \frac{\varphi}{2} \left(\frac{p_t(i)}{p_{t-1}(i)} - 1 \right)^2 P_t Y_t \right],$$

subject to

$$y_t(i) = Y_t \left(\frac{p_t(i)}{P_t} \right)^{-\theta}, \qquad N_t(i) = y_t(i).$$

The demand curve implies

$$\frac{\partial y_t(i)}{\partial p_t(i)} = -\theta \frac{y_t(i)}{p_t(i)}.$$

Let $\pi_t(i) \equiv \frac{p_t(i)}{p_{t-1}(i)}$. Differentiating the firm's objective with respect to $p_t(i)$ gives

$$\begin{aligned} 0 = \Lambda_t \left[(1-\tau_t)(1-\theta) y_t(i) + \theta(1-\varsigma) w_t \frac{y_t(i)}{p_t(i)} - \varphi\big(\pi_t(i)-1\big)\pi_t(i)\frac{P_t Y_t}{p_t(i)} \right] \\ + \beta E_t \left[\Lambda_{t+1} \varphi\big(\pi_{t+1}(i)-1\big)\pi_{t+1}(i)\frac{P_{t+1}Y_{t+1}}{p_t(i)} \right]. \end{aligned}$$

In a symmetric equilibrium,

$$p_t(i) = P_t, \qquad y_t(i) = Y_t = N_t, \qquad \pi_t(i) = \pi_t.$$

Using $\Lambda_t = C_t^{-1}/P_t$, the household labor-supply condition

$$\frac{w_t}{P_t} = N_t^{\nu} C_t,$$

and the subsidy rule above, the symmetric pricing condition becomes

$$N_t C_t^{-1} \left[\varphi(\pi_t - 1)\pi_t + (\theta - 1)\left\{ (1-\tau_t) - (1-\bar{\tau}) N_t^{\nu} C_t \right\} \right] = \beta E_t \left[N_{t+1} C_{t+1}^{-1} \varphi(\pi_{t+1} - 1)\pi_{t+1} \right].$$

We linearize around a deterministic zero-inflation steady state, $\bar{\pi} = 1$, and define

$$\hat{\pi}_t \equiv \log \pi_t, \qquad \hat{\tau}_t \equiv \tau_t - \bar{\tau}, \qquad \hat{n}_t \equiv \log \frac{N_t}{\bar{N}}, \qquad \hat{c}_t \equiv \log \frac{C_t}{\bar{C}}.$$

The steady-state pricing condition implies

$$\bar{N}^{\nu} \bar{C} = 1.$$

Because the expression in square brackets in the symmetric pricing condition is zero in steady state, fluctuations in the multiplicative term $N_t C_t^{-1}$ do not enter at first order. Likewise, the inflation-adjustment term on the right-hand side is zero in steady state. The first-order approximation is therefore

$$\varphi \hat{\pi}_t - (\theta - 1)\hat{\tau}_t - (\theta - 1)(1 - \bar{\tau})(\nu \hat{n}_t + \hat{c}_t) = \beta \varphi E_t \hat{\pi}_{t+1}.$$

Rearranging,

$$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \frac{(\theta - 1)(1 - \bar{\tau})}{\varphi}(\nu \hat{n}_t + \hat{c}_t) + \frac{\theta - 1}{\varphi}\hat{\tau}_t.$$

Linear production implies $\hat{n}_t = \hat{y}_t$. Moreover, price-adjustment costs are second order around $\bar{\pi} = 1$, so the first-order resource constraint implies $\hat{c}_t = \hat{y}_t$. Hence,

$$\hat{\pi}_t = \kappa \hat{y}_t + \gamma \hat{\tau}_t + \beta E_t \hat{\pi}_{t+1},$$

where

$$\kappa \equiv \frac{(\theta - 1)(1 - \bar{\tau})(1 + \nu)}{\varphi}, \qquad \gamma \equiv \frac{\theta - 1}{\varphi}.$$

Thus, the lump-sum rebate removes cyclical fluctuations in the tax base from the Treasury's net revenue while leaving the marginal sales-tax wedge faced by firms unchanged. The sales-tax deviation therefore enters the Phillips curve directly through $\gamma \hat{\tau}_t$.

E.4 Treasury budget with distortionary tax

Under the tax-rebate scheme in Section 4.3, gross sales-tax revenue is $\tau_t P_t Y_t$, and the lump-sum rebate is

$$R_t = \tau_t P_t (Y_t - \bar{Y}).$$

Net tax revenue is therefore

$$\tau_t P_t Y_t - R_t = \tau_t P_t \bar{Y}.$$

With zero government purchases, $G_t \equiv 0$, the Treasury budget constraint becomes

$$Q_t^T B_t^T + \tau_t P_t \bar{Y} + P_t H_t = B_{t-1}^T.$$

Define

$$b_t^T \equiv \frac{B_t^T}{P_t}, \qquad \widehat{\tau}_t \equiv \tau_t - \bar{\tau}, \qquad \widehat{H}_t \equiv \frac{H_t - \bar{H}}{\bar{Y}}.$$

Dividing by P_t gives

$$Q_t^T b_t^T + \tau_t \bar{Y} + H_t = \frac{b_{t-1}^T}{\Pi_t}.$$

Linearizing around the deterministic zero-inflation steady state, where $\bar{Q}^T = \beta$, yields

$$\begin{aligned}
Q_t^T b_t^T &\simeq \beta \bar{b}^T (1 + \widehat{Q}_t^T + \widehat{b}_t^T), \\
\tau_t \bar{Y} &= \bar{\tau} \bar{Y} + \bar{Y} \widehat{\tau}_t, \\
H_t &= \bar{H} + \bar{Y} \widehat{H}_t, \\
\frac{b_{t-1}^T}{\Pi_t} &\simeq \bar{b}^T (1 + \widehat{b}_{t-1}^T - \widehat{\pi}_t).
\end{aligned}$$

The steady-state Treasury budget constraint is

$$\beta \bar{b}^T + \bar{\tau}\bar{Y} + \bar{H} = \bar{b}^T.$$

Subtracting this relation from the linearized budget constraint and dividing by $\bar{b}^T$ gives

$$\widehat{b}^T_{t-1} - \widehat{\pi}_t - \frac{\bar{Y}}{\bar{b}^T}(\widehat{\tau}_t + \widehat{H}_t) = \beta(\widehat{b}^T_t + \widehat{Q}^T_t).$$

Using

$$\frac{\bar{Y}}{\bar{b}^T} = (1-\beta)b_h,$$

we obtain

$$\widehat{b}^T_{t-1} - \widehat{\pi}_t - (1-\beta)b_h(\widehat{H}_t + \widehat{\tau}_t) = \beta(\widehat{b}^T_t + \widehat{Q}^T_t),$$

which is the linearized Treasury budget constraint in Section 4.3.

E.5 General case: Output-dependent tax revenue

We extend Section 4.3 by removing the lump-sum rebate of cyclical sales-tax revenue, so the Treasury collects $\tau_t P_t Y_t$ rather than $\tau_t P_t \bar{Y}$. All other assumptions of the linear-quadratic model remain unchanged.

Proposition 17. *Suppose that the Treasury collects sales-tax revenue $\tau_t P_t Y_t$, government purchases are zero, and $\bar{\tau} > 0$. Assume $\hat{b}^T_{-1} = \hat{b}^C_{-1} = H^* = 0$ and a one-time expansionary demand shock, $e_0 < 0$ and $e_t = 0$ for $t \geq 1$. Let $\mathscr{V}(\gamma) \equiv \mathscr{W}^N(\gamma) - \mathscr{W}^B(\gamma)$ denote the welfare value of fiscal backing, and define*

$$\gamma_{\max} \equiv \begin{cases} \infty, & q_y \geq \kappa q_\pi, \\ \dfrac{(1-\beta)q_y + \kappa^2 q_\pi}{\bar{\tau}(\kappa q_\pi - q_y)}, & q_y < \kappa q_\pi. \end{cases}$$

Then $\mathscr{V}(\gamma) > 0$ for every $\gamma > 0$, and $d\mathscr{V}(\gamma)/d\gamma < 0$ for $\kappa/\bar{\tau} \leq \gamma \leq \gamma_{\max}$. Moreover,

$$\lim_{\gamma \to \infty} \mathscr{V}(\gamma) = \frac{\beta^2 e_0^2 (1-\beta) q_y q_\pi^2}{2(q_y + \beta q_\pi)(q_y + q_\pi)} > 0.$$

Proof. Define $E \equiv -\beta e_0 > 0$, $c \equiv (1-\beta)a_h b_h/(a_h + b_h) > 0$, and $\chi_\gamma \equiv c/\gamma$. The steady-state public budgets imply $\bar{\tau}\bar{Y} = (1-\beta)(\bar{b}^T + \bar{b}^C)$. Using the definitions of a_h and b_h gives

$$c\bar{\tau} = 1 - \beta, \qquad \chi_\gamma = \frac{1-\beta}{\bar{\tau}\gamma}. \tag{107}$$

Intertemporal constraints. Without the rebate, the linearized Treasury budget constraint is

$$\hat{b}^T_{t-1} - \hat{\pi}_t - (1-\beta) b_h (\hat{H}_t + \hat{\tau}_t + \bar{\tau}\hat{y}_t) = \beta(\hat{b}^T_t + \hat{Q}_t).$$

Combining the Treasury and central-bank intertemporal budgets and eliminating remittances yields

$$\sum_{t=0}^{\infty} \beta^t (\beta \hat{Q}_t + \hat{\pi}_t) + c \sum_{t=0}^{\infty} \beta^t (\hat{\tau}_t + \bar{\tau}\hat{y}_t) = 0. \tag{108}$$

Let $S_y \equiv \sum_{t=1}^{\infty} \beta^t \hat{y}_t$. The Phillips curve and household Euler equation imply

$$\gamma \sum_{t=0}^{\infty} \beta^t \hat{\tau}_t = \hat{\pi}_0 - \kappa(\hat{y}_0 + S_y),$$

$$\sum_{t=0}^{\infty} \beta^t (\beta \hat{Q}_t + \hat{\pi}_t) = \hat{\pi}_0 + \beta \hat{y}_0 - (1-\beta) S_y - E.$$

Substituting into (108) and using (107) gives the consolidated government budget constraint

$$(1 - \kappa\chi_\gamma)\hat{y}_0 + (1+\chi_\gamma)\hat{\pi}_0 - \kappa\chi_\gamma S_y = E. \tag{109}$$

The central-bank budget and $\hat{H}_t \geq 0$ additionally require

$$\beta \hat{y}_0 + \hat{\pi}_0 - (1-\beta) S_y \geq E, \tag{110}$$

which is absent under fiscal backing.

Because tax-rate adjustments are unrestricted, the Phillips curve can be satisfied for any bounded output and inflation paths. Conditional on $(\hat{y}_0, \hat{\pi}_0, S_y)$, welfare loss is minimized by $\hat{\pi}_t = 0$ and $\hat{y}_t = (1-\beta)S_y/\beta$ for $t \geq 1$. The reduced objective is

$$\mathscr{W} = \frac{1}{2}\left[q_y \hat{y}_0^2 + q_\pi \hat{\pi}_0^2 + \frac{q_y(1-\beta)}{\beta} S_y^2 \right]. \tag{111}$$

Fiscal-backing equilibrium. Minimizing (111) subject to (109) gives

$$\mathscr{W}^B(\gamma) = \frac{E^2}{2D_\gamma}, \tag{112}$$

$$D_\gamma = \frac{(1-\kappa\chi_\gamma)^2}{q_y} + \frac{(1+\chi_\gamma)^2}{q_\pi} + \frac{\beta\kappa^2\chi_\gamma^2}{(1-\beta)q_y}$$

$$= D_0 + 2K\chi_\gamma + L\chi_\gamma^2, \tag{113}$$

where $D_0 \equiv 1/q_y + 1/q_\pi$, $K \equiv 1/q_\pi - \kappa/q_y$, and $L \equiv 1/q_\pi + \kappa^2/[(1-\beta)q_y] > 0$.

To show that the central-bank constraint binds under no fiscal backing, define

$$z \equiv (\hat{y}_0, \hat{\pi}_0, S_y)^\top, \qquad A \equiv \operatorname{diag}\left(q_y, q_\pi, \frac{q_y(1-\beta)}{\beta}\right),$$

$$f \equiv (\beta, 1, -(1-\beta))^\top, \qquad g_\gamma \equiv (1-\kappa\chi_\gamma, 1+\chi_\gamma, -\kappa\chi_\gamma)^\top.$$

The fiscal-backing optimum satisfies $z^B = (E/D_\gamma)A^{-1}g_\gamma$. Direct calculation gives

$$D_\gamma - f^\top A^{-1} g_\gamma = \frac{(1-\beta-\kappa\chi_\gamma)^2}{(1-\beta)q_y} + \frac{\chi_\gamma(1+\chi_\gamma)}{q_\pi} > 0.$$

Hence $f^\top z^B = E(f^\top A^{-1}g_\gamma)/D_\gamma < E$: the fiscal-backing optimum violates (110). Strict convexity implies that this constraint binds under no fiscal backing, and therefore $\mathscr{V}(\gamma) > 0$ for every $\gamma > 0$.

No-fiscal-backing equilibrium. When (110) binds, subtracting it from (109) and using (107) gives

$$\hat{\pi}_0 = (\kappa - \bar{\tau}\gamma)(\hat{y}_0 + S_y). \tag{114}$$

Define $k_\gamma \equiv \kappa - \bar{\tau}\gamma$, $R_\gamma \equiv k_\gamma^2/[(1-\beta)q_y]$, and $M \equiv \beta/q_y + 1/q_\pi$. The inverse-weighted Gram matrix associated with the two binding constraints is

$$G_\gamma \equiv \begin{pmatrix} M & 1/q_\pi \\ 1/q_\pi & 1/q_\pi + R_\gamma \end{pmatrix}, \qquad \det G_\gamma = \frac{\beta}{q_y q_\pi} + MR_\gamma > 0.$$

Solving the constrained quadratic problem yields

$$\mathscr{W}^N(\gamma) = \frac{E^2}{2}\frac{1/q_\pi + R_\gamma}{\beta/(q_y q_\pi) + MR_\gamma}. \tag{115}$$

Unlike the tax-rebate case, the no-fiscal-backing welfare loss depends on γ.

Monotonicity. Differentiating (112) and (115) gives

$$\frac{d\mathscr{W}^B(\gamma)}{d\gamma} = \frac{E^2\chi_\gamma(K + L\chi_\gamma)}{\gamma D_\gamma^2}, \tag{116}$$

$$\frac{d\mathscr{W}^N(\gamma)}{d\gamma} = \frac{E^2\bar{\tau}(\kappa - \bar{\tau}\gamma)}{(1-\beta)q_y q_\pi^2[\beta/(q_y q_\pi) + MR_\gamma]^2}. \tag{117}$$

Thus, $d\mathscr{W}^N/d\gamma \le 0$ whenever $\gamma \ge \kappa/\bar{\tau}$. Meanwhile, $d\mathscr{W}^B/d\gamma \ge 0$ if and only if $K + L\chi_\gamma \ge 0$.

If $q_y \ge \kappa q_\pi$, then $K \ge 0$, so this condition holds for all $\gamma > 0$. If $q_y < \kappa q_\pi$, then $K < 0$, and the condition is equivalent to $\gamma \le cL/(-K) = \gamma_{\max}$. Moreover,

$$\gamma_{\max} - \frac{\kappa}{\bar{\tau}} = \frac{(1-\beta+\kappa)q_y}{\bar{\tau}(\kappa q_\pi - q_y)} > 0,$$

so the stated interval is nonempty. Throughout $\kappa/\bar{\tau} \le \gamma \le \gamma_{\max}$, we therefore have $d\mathscr{W}^N/d\gamma \le 0$ and $d\mathscr{W}^B/d\gamma \ge 0$, with at least one inequality strict. Hence $d\mathscr{V}/d\gamma < 0$.

Limiting welfare value. As $\gamma \to \infty$, $\chi_\gamma \to 0$ and $R_\gamma \to \infty$. Equations (112) and (115) imply $\mathscr{W}^B \to E^2/(2D_0)$ and $\mathscr{W}^N \to E^2/(2M)$. Consequently,

$$\begin{aligned}\lim_{\gamma\to\infty} \mathscr{V}(\gamma) &= \frac{E^2}{2}\left(\frac{1}{M} - \frac{1}{D_0}\right) \\ &= \frac{\beta^2 e_0^2 (1-\beta) q_y q_\pi^2}{2(q_y + \beta q_\pi)(q_y + q_\pi)} > 0.\end{aligned}$$

This completes the proof. □

F Appendix for Section 4.2

F.1 Model and linearization

This subsection derives the equilibrium conditions for the reserve-in-utility extension, retaining the demand-shock specification in Appendix A.4. Reserve liquidity affects both the steady-state reserve price and the linearized central bank budget.

Household preferences and optimality conditions Let $b_t^C \equiv B_t^C/P_t$ denote real reserve holdings and $\Pi_t \equiv P_t/P_{t-1}$ gross inflation. Reserves are the only nominal asset held by households. Household preferences are

$$\max_{\{C_t, N_t, b_t^C\}_{t=0}^\infty} E_0 \sum_{t=0}^\infty \beta^t \delta_t \left[\frac{C_t^{1-\sigma}}{1-\sigma} + \frac{\omega}{1-\eta}(b_t^C)^{1-\eta} - \frac{N_t^{1+\nu}}{1+\nu}\right],$$

where $\beta \in (0,1)$, $\sigma, \eta, \omega > 0$, and $\nu \geq 0$. Logarithmic limits apply when $\sigma = 1$ or $\eta = 1$. The positive preference shifter δ_t multiplies all components of period utility.

The real household budget constraint is

$$C_t + Q_t^C b_t^C + T_t = \frac{w_t}{P_t} N_t + \Phi_t + \frac{b_{t-1}^C}{\Pi_t},$$

where T_t denotes lump-sum taxes net of employment-subsidy payments and Φ_t distributed profits net of the corresponding subsidy receipts. Netting the same subsidy out of both terms preserves the household budget constraint and is consistent with the Treasury budget below.

The first-order conditions, with λ_t denoting the current-value budget multiplier, imply

$$\begin{aligned}\lambda_t &= \delta_t C_t^{-\sigma}, \\ \frac{w_t}{P_t} &= N_t^\nu C_t^\sigma, \\ Q_t^C C_t^{-\sigma} &= \beta E_t\left[\frac{\delta_{t+1}}{\delta_t}\frac{C_{t+1}^{-\sigma}}{\Pi_{t+1}}\right] + \omega (b_t^C)^{-\eta}.\end{aligned}$$

Define the price of an otherwise identical one-period claim without liquidity services by

$$Q_t^f \equiv \beta E_t \left[\frac{\delta_{t+1}}{\delta_t} \left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \frac{1}{\Pi_{t+1}} \right].$$

This hypothetical claim need not be traded. The reserve Euler equation implies

$$\begin{aligned} Q_t^C - Q_t^f &= \omega C_t^{\sigma} (b_t^C)^{-\eta} > 0, \\ \frac{\partial \log(Q_t^C - Q_t^f)}{\partial \log b_t^C} &= -\eta. \end{aligned}$$

Thus, reserves command a higher price and lower pecuniary return than a nonliquid claim with the same payoff, while η governs the supply elasticity of their marginal liquidity value.

Public budgets and the steady state The central bank holds no assets, and the Treasury issues no debt. Their budget constraints and the remittance restriction are

$$\begin{aligned} Q_t^C b_t^C &= \frac{b_{t-1}^C}{\Pi_t} + H_t, \\ T_t + H_t &= 0, \qquad H_t \geq \underline{H}, \end{aligned}$$

where positive H_t denotes a transfer from the central bank to the Treasury.

Consider a deterministic steady state with $\bar{\delta} = \bar{\Pi} = 1$ and $\bar{b}^C > 0$. Household optimality gives

$$\begin{aligned} \bar{Q}^f &= \beta, \\ \bar{Q}^C &= \beta + \omega \bar{C}^{\sigma} (\bar{b}^C)^{-\eta}. \end{aligned}$$

Define

$$\bar{r} \equiv \frac{\beta}{\bar{Q}^C} \in (0,1).$$

Then

$$\bar{Q}^C = \frac{\beta}{\bar{r}}, \qquad \frac{\omega \bar{C}^{\sigma} (\bar{b}^C)^{-\eta}}{\bar{Q}^C} = 1 - \bar{r}.$$

Thus, $\bar{r}$ and $1 - \bar{r}$ are the steady-state shares of the pecuniary and liquidity components of the reserve price, respectively. The steady-state public budgets imply

$$\bar{H} = (\bar{Q}^C - 1)\bar{b}^C, \qquad \bar{T} = -\bar{H}.$$

Linearized asset-pricing and private-sector conditions Define log deviations by

$$\hat{C}_t \equiv \log(C_t/\bar{C}), \qquad \hat{y}_t \equiv \log(Y_t/\bar{Y}), \qquad \hat{b}_t^C \equiv \log(b_t^C/\bar{b}^C),$$
$$\hat{Q}_t^C \equiv \log(Q_t^C/\bar{Q}^C), \qquad \hat{Q}_t^f \equiv \log(Q_t^f/\bar{Q}^f), \qquad \hat{\pi}_t \equiv \log \Pi_t.$$

As in Appendix A.4, let

$$\hat{\delta}_t \equiv \log \delta_t, \qquad e_t \equiv E_t \hat{\delta}_{t+1} - \hat{\delta}_t.$$

Linearizing the nonliquid claim price and the reserve-price decomposition yields

$$\hat{Q}_t^f = \sigma \hat{C}_t - \sigma E_t \hat{C}_{t+1} - E_t \hat{\pi}_{t+1} + e_t,$$
$$\hat{Q}_t^C = \bar{r} \hat{Q}_t^f + (1-\bar{r})(\sigma \hat{C}_t - \eta \hat{b}_t^C).$$

The demand shock enters the reserve price through its pecuniary component, with coefficient $\bar{r}$.

The resource constraint remains

$$Y_t = N_t = C_t + \frac{\varphi}{2}(\Pi_t - 1)^2 Y_t.$$

Because price-adjustment costs are second order around zero inflation, $\hat{C}_t = \hat{y}_t$. Using reserve-market clearing, the linearized reserve-price equation becomes

$$\hat{Q}_t^C = \sigma \hat{y}_t - \bar{r}\sigma E_t \hat{y}_{t+1} - \bar{r} E_t \hat{\pi}_{t+1} - (1-\bar{r})\eta \hat{b}_t^C + \bar{r} e_t.$$

The baseline production and price-setting structure, including the employment subsidy that offsets the steady-state monopolistic distortion, gives

$$\hat{\pi}_t = \kappa \hat{y}_t + \beta E_t \hat{\pi}_{t+1}, \qquad \kappa \equiv \frac{(\theta - 1)(\sigma + \nu)}{\varphi}.$$

Thus, the Phillips curve retains its form, with a slope reflecting generalized consumption curvature.

Setting $\omega = 0$ gives $\bar{r} = 1$, recovering the nonliquid benchmark, and $\sigma = 1$ further recovers the logarithmic baseline. With liquidity services and $\sigma = \eta = 1$, the reserve-price equation simplifies to

$$\hat{Q}_t^C = \hat{y}_t - \bar{r} E_t \hat{y}_{t+1} - \bar{r} E_t \hat{\pi}_{t+1} - (1-\bar{r})\hat{b}_t^C + \bar{r} e_t.$$

Linearized public budgets Define normalized level deviations and the remittance bound by

$$\hat{H}_t \equiv \frac{H_t - \bar{H}}{\bar{Y}}, \qquad \hat{T}_t \equiv \frac{T_t - \bar{T}}{\bar{Y}}, \qquad H^* \equiv \frac{\underline{H} - \bar{H}}{\bar{Y}}.$$

Here $H^* = 0$ means that the remittance bound equals steady-state remittances. Linearizing the public budgets around the steady state gives

$$\frac{\beta}{\bar{r}}\left(\hat{Q}_t^C + \hat{b}_t^C\right) = \hat{b}_{t-1}^C - \hat{\pi}_t + \frac{\bar{Y}}{\bar{b}^C}\hat{H}_t,$$
$$\hat{T}_t + \hat{H}_t = 0, \qquad \hat{H}_t \geq H^*.$$

Importantly, the coefficient on reserve issuance is the steady-state reserve price $\bar{Q}^C = \beta/\bar{r}$.

Substituting the reserve-price equation yields

$$\hat{b}_{t-1}^C = \hat{\pi}_t + \frac{\beta}{\bar{r}}\sigma\hat{y}_t - \beta\sigma E_t\hat{y}_{t+1} - \beta E_t\hat{\pi}_{t+1}$$
$$+ \beta e_t + d\hat{b}_t^C - \frac{\bar{Y}}{\bar{b}^C}\hat{H}_t,$$

where

$$d \equiv \bar{Q}^C[1-(1-\bar{r})\eta] = \frac{\beta}{\bar{r}}[1-(1-\bar{r})\eta],$$
$$d - \beta = \frac{\beta(1-\bar{r})(1-\eta)}{\bar{r}}.$$

Thus, d is smaller than, equal to, or larger than β according as $\eta > 1$, $\eta = 1$, or $0 < \eta < 1$. In particular, d need not be smaller than one. The coefficients on expected future output, expected inflation, and the demand shock are $-\beta\sigma$, $-\beta$, and β, respectively.

Transversality and intertemporal valuation We impose the household transversality condition

$$\lim_{T\to\infty} E_0\left[\beta^{T+1}\lambda_{T+1}\frac{b_T^C}{\Pi_{T+1}}\right] = 0, \qquad \lambda_t = \delta_t C_t^{-\sigma},$$

which values terminal reserve repayments using household marginal utility.

Combining the reserve Euler equation with the central bank budget gives the exact finite-horizon identity

$$\lambda_0\frac{b_{-1}^C}{\Pi_0} = E_0\sum_{t=0}^{T}\beta^t\left[\delta_t\omega(b_t^C)^{1-\eta} - \lambda_t H_t\right]$$
$$+ E_0\left[\beta^{T+1}\lambda_{T+1}\frac{b_T^C}{\Pi_{T+1}}\right].$$

Under transversality, letting $T \to \infty$ yields

$$\lambda_0\frac{b_{-1}^C}{\Pi_0} = E_0\sum_{t=0}^{\infty}\beta^t\left[\delta_t\omega(b_t^C)^{1-\eta} - \lambda_t H_t\right].$$

Outstanding reserves are therefore valued through both net fiscal transfers and marginal liquidity services.

Under perfect foresight, forward iteration of the linearized central bank recursion instead yields the terminal term $d^{T+1}\hat{b}^C_T$. Eliminating it requires

$$\lim_{T\to\infty} d^{T+1}\hat{b}^C_T = 0,$$

which is not generally implied by household transversality. If $0 < d < 1$, bounded reserve deviations suffice; if $d > 1$, even convergence of reserve deviations to zero is insufficient without a stronger restriction on their rate of convergence.

Stabilization objective As in Section 4.2, policy analysis retains the quadratic stabilization objective in (13), with given weights $q_y, q_\pi > 0$. This objective measures output and inflation fluctuations rather than the second-order approximation to full household utility, which also includes reserve liquidity.

F.2 Proof of Proposition 14

Proof. We consider a one-time expansionary demand shock, $e_0 < 0$ and $e_t = 0$ for $t \geq 1$, with $\hat{b}^C_{-1} = H^* = 0$ and $\Delta = -\beta e_0 > 0$. Throughout, $q_y, q_\pi, \kappa > 0$, $\beta \in (0,1)$, and the Phillips-curve slope and stabilization weights are held fixed when varying σ or η. Define

$$\bar{Q}^C \equiv \frac{\beta}{\bar{r}}, \qquad\qquad d \equiv \bar{Q}^C[1-(1-\bar{r})\eta].$$

The assumption in the proposition implies $0 < d \leq \beta < 1$.

Intertemporal funding constraints. Under perfect foresight, the Phillips curve and the central-bank budget equation derived in Appendix F.1 are

$$\hat{\pi}_t = \kappa\hat{y}_t + \beta\hat{\pi}_{t+1},$$
$$\hat{b}^C_{t-1} = \hat{\pi}_t + \bar{Q}^C\sigma\hat{y}_t - \beta\sigma\hat{y}_{t+1} - \beta\hat{\pi}_{t+1} + \beta e_t + d\hat{b}^C_t - \frac{\bar{Y}}{\bar{b}^C}\hat{H}_t.$$

Because reserve deviations are bounded and $d < 1$, forward iteration eliminates the terminal term $d^{T+1}\hat{b}^C_T$ and gives

$$\mathscr{B}^R(\hat{y},\hat{\pi}) \equiv \sum_{t=0}^{\infty} d^t\left[\hat{\pi}_t + \bar{Q}^C\sigma\hat{y}_t - \beta\sigma\hat{y}_{t+1} - \beta\hat{\pi}_{t+1}\right] = \Delta + \frac{\bar{Y}}{\bar{b}^C}\sum_{t=0}^{\infty} d^t\hat{H}_t \geq \Delta. \tag{118}$$

The no-liquidity benchmark has $\bar{Q}^C = d = \beta$ and, at the same σ, κ, and welfare weights,

$$\mathscr{B}^{NL}(\hat{y},\hat{\pi}) \equiv \sum_{t=0}^{\infty} \beta^t\left[\hat{\pi}_t + \beta\sigma\hat{y}_t - \beta\sigma\hat{y}_{t+1} - \beta\hat{\pi}_{t+1}\right] \geq \Delta. \tag{119}$$

Since both funding functionals are linear and the stabilization loss is quadratic, any feasible no-liquidity allocation with funding capacity strictly above Δ can be scaled down to obtain a smaller loss. Thus the no-liquidity optimum satisfies $\mathscr{B}^{NL} = \Delta$, implemented with $\hat{H}_t = 0$ for all t.

No-liquidity optimum. Consider minimizing

$$\frac{1}{2}\sum_{t=0}^{\infty} \beta^t \left(q_y \hat{y}_t^2 + q_\pi \hat{\pi}_t^2\right)$$

subject to the Phillips curve and $\mathscr{B}^{NL} = \Delta$. Let μ be the funding multiplier. At $\mu = 1$, denote the optimal paths by $(\widetilde{y}_t, \widetilde{\pi}_t)$ and set $\widetilde{p}_t = \sum_{j=0}^{t} \widetilde{\pi}_j$, with $\widetilde{p}_{-1} = 0$. The first-order conditions give

$$\begin{aligned}
\lambda_t &= 1 - q_\pi \widetilde{p}_t, \\
q_y \widetilde{y}_0 &= \kappa + \beta\sigma - \kappa q_\pi \widetilde{p}_0, \\
q_y \widetilde{y}_t &= \kappa - (1-\beta)\sigma - \kappa q_\pi \widetilde{p}_t, \qquad t \geq 1,
\end{aligned}$$

where λ_t is the Phillips-curve multiplier. Define

$$M \equiv 1 - \beta + \frac{\kappa^2 q_\pi}{q_y}, \qquad\qquad z \equiv \sigma M - \kappa.$$

The unique bounded solution of these conditions and the Phillips curve is

$$\widetilde{p}_t = \frac{\kappa - (1-\beta)\sigma + z r^{t+1}}{\kappa q_\pi}, \qquad t \geq 0,$$

where $r \in (0,1)$ is the stable root

$$r \equiv \frac{2}{1 + \beta + \kappa^2 q_\pi / q_y + \sqrt{(1 + \beta + \kappa^2 q_\pi / q_y)^2 - 4\beta}}, \qquad (1-r)(1-\beta r) = \frac{\kappa^2 q_\pi}{q_y} r. \tag{120}$$

Consequently,

$$\begin{aligned}
\widetilde{y}_0 &= \frac{\sigma - rz}{q_y}, & \widetilde{\pi}_0 &= \frac{\kappa - (1-\beta)\sigma + rz}{\kappa q_\pi}, \\
\widetilde{y}_t &= -\frac{z}{q_y} r^{t+1}, & \widetilde{\pi}_t &= -\frac{(1-r)z}{\kappa q_\pi} r^t, \qquad t \geq 1.
\end{aligned}$$

Multiplying the unit-multiplier first-order conditions by the corresponding allocations, summing over time, and using the Phillips curve yields

$$\Omega_{NL} \equiv \mathscr{B}^{NL}(\widetilde{y}, \widetilde{\pi}) = \sum_{t=0}^{\infty} \beta^t \left(q_y \widetilde{y}_t^2 + q_\pi \widetilde{\pi}_t^2\right) > 0. \tag{121}$$

By linearity and strict convexity, the no-liquidity optimum is

$$(\hat{y}_t^{NL}, \hat{\pi}_t^{NL}) = \frac{\Delta}{\Omega_{NL}} (\widetilde{y}_t, \widetilde{\pi}_t), \qquad\qquad \mathscr{W}^{NL} = \frac{\Delta^2}{2\Omega_{NL}}. \tag{122}$$

Funding-capacity comparison and threshold. Evaluate the reserve-in-utility funding functional at the no-liquidity unit-multiplier allocation, and write $B_R \equiv \mathscr{B}^R(\widetilde{y},\widetilde{\pi})$. Summing geometric series gives

$$B_R = \frac{1}{q_y}\left[\sigma(\kappa+\bar{Q}^C\sigma) - \frac{rz[\kappa+\sigma(\bar{Q}^C-\beta r)]}{1-dr}\right],$$
$$\Omega_{NL} = \frac{1}{q_y}\left[\sigma(\kappa+\beta\sigma) - \frac{rz[\kappa+\beta(1-r)\sigma]}{1-\beta r}\right].$$

Define

$$\ell \equiv \bar{Q}^C - \beta = \frac{\beta(1-\bar{r})}{\bar{r}} > 0, \qquad h \equiv \eta - 1 \geq 0,$$
$$D \equiv 1-\beta r > 0, \qquad u \equiv \beta(1-r) > 0.$$

Then $d = \beta - \ell h$. Using the root identity in (120), direct simplification gives

$$B_R - \Omega_{NL} = \frac{\ell r}{q_y D(1-dr)}\left(K_2\sigma^2 + K_1\sigma - K_0\right), \tag{123}$$

where

$$K_2 \equiv Du + h(\ell D + rMu) > 0,$$
$$K_1 \equiv \kappa[D + hr(M-u)],$$
$$K_0 \equiv hr\kappa^2 \geq 0.$$

For $h > 0$, the quadratic has exactly one positive root, whereas for $h = 0$ it is positive at every $\sigma > 0$. Define

$$\sigma_c(\eta) \equiv \frac{\sqrt{K_1^2 + 4K_2K_0} - K_1}{2K_2} \geq 0. \tag{124}$$

It follows that

$$B_R > \Omega_{NL} \quad \Longleftrightarrow \quad \sigma > \sigma_c(\eta), \qquad \sigma_c(1) = 0.$$

To establish monotonicity in η, define, for $\sigma > 0$,

$$G(\sigma) \equiv \frac{r(\kappa - M\sigma)}{D\sigma} - \frac{\ell\sigma}{\kappa + u\sigma}.$$

The capacity difference can equivalently be factored as

$$B_R - \Omega_{NL} = \frac{\ell r\sigma(\kappa+u\sigma)}{q_y(1-dr)}\left[1-(\eta-1)G(\sigma)\right].$$

Moreover,

$$G'(\sigma) = -\frac{r\kappa}{D\sigma^2} - \frac{\ell\kappa}{(\kappa+u\sigma)^2} < 0,$$

$$\lim_{\sigma\downarrow 0} G(\sigma) = +\infty, \qquad \lim_{\sigma\to\infty} G(\sigma) = -\frac{rM}{D} - \frac{\ell}{u} < 0.$$

Thus, for $\eta > 1$, $\sigma_c(\eta)$ is the unique solution to $G(\sigma_c) = 1/(\eta-1)$. Implicit differentiation yields

$$\frac{d\sigma_c(\eta)}{d\eta} = -\frac{1}{(\eta-1)^2 G'(\sigma_c(\eta))} > 0.$$

Since $G(\sigma) \to +\infty$ as $\sigma \downarrow 0$, we also have $\lim_{\eta\downarrow 1} \sigma_c(\eta) = 0$. Together with $\sigma_c(1) = 0$, this proves the stated comparative statics, holding $\beta, \bar{r}, \kappa, q_y, q_\pi$ fixed.

Feasibility and welfare. Suppose $\sigma > \sigma_c(\eta)$ and set $m \equiv B_R/\Omega_{NL} > 1$. Consider

$$\hat{y}_t^F \equiv \frac{\hat{y}_t^{NL}}{m}, \qquad \hat{\pi}_t^F \equiv \frac{\hat{\pi}_t^{NL}}{m}, \qquad t \geq 0.$$

The Phillips curve is homogeneous, and linearity of the funding functional gives

$$\mathscr{B}^R(\hat{y}^F, \hat{\pi}^F) = \frac{\Delta}{m\Omega_{NL}} B_R = \Delta.$$

Set $\hat{H}_t^F = \hat{T}_t^F = 0$ and recover reserve holdings from the central-bank budget as

$$\hat{b}_t^{C,F} = \sum_{j=0}^{\infty} d^j \left[\hat{\pi}_{t+1+j}^F + \bar{Q}^C \sigma \hat{y}_{t+1+j}^F - \beta\sigma\hat{y}_{t+2+j}^F - \beta\hat{\pi}_{t+2+j}^F\right], \qquad t \geq 0.$$

The reserve-price equation in Appendix F.1 determines $\hat{Q}_t^{C,F}$. Since $d < 1$ and the candidate output and inflation paths are bounded and converge to zero, so do reserve holdings and reserve-price deviations. The construction satisfies the flow budget for $t \geq 1$; the date-zero budget holds because $\mathscr{B}^R(\hat{y}^F, \hat{\pi}^F) = \Delta = -\beta e_0$. Hence the scaled path is feasible.

Conversely, suppose such a scaled path is feasible for some $m > 1$. Equation (118) and $\hat{H}_t \geq 0$ require

$$\frac{\Delta}{m\Omega_{NL}} B_R = \mathscr{B}^R(\hat{y}^F, \hat{\pi}^F) \geq \Delta,$$

so $B_R \geq m\Omega_{NL} > \Omega_{NL}$ and therefore $\sigma > \sigma_c(\eta)$. This proves necessity and sufficiency for the existence of a uniformly scaled-down feasible allocation.

Finally, the constructed allocation has stabilization loss $\mathscr{W}^F = \mathscr{W}^{NL}/m^2 < \mathscr{W}^{NL}$. Since the RIU optimum minimizes the same quadratic objective over feasible allocations,

$$\mathscr{W}^R \leq \mathscr{W}^F < \mathscr{W}^{NL},$$

as required. □

F.3 Proof of Proposition 15

Proof. We maintain the one-time demand shock, initial conditions, and bounded admissible class stated in the proof of Proposition 14, with $\Delta > 0$. Let $q_y = q_\pi \equiv q > 0$, $\kappa > 0$, and define

$$s \equiv \frac{1-(1-\bar{r})\eta}{\bar{r}} \in (0,1), \qquad d = \beta s, \qquad \bar{Q}^C = \frac{\beta}{\bar{r}} = \beta + \ell.$$

All parameters other than σ are held fixed. We use the funding functionals $\mathscr{B}^R$ and $\mathscr{B}^{NL}$ and the stable root $r \in (0,1)$ derived in the proof of Proposition 14.

Quadratic welfare comparison. Let $\mathscr{H}_\beta$ denote the space of discounted-square-summable paths $X = \{(y_t, \pi_t)\}_{t\geq 0}$ satisfying $\pi_t = \kappa y_t + \beta \pi_{t+1}$, equipped with

$$\langle X, Z\rangle_\beta \equiv q \sum_{t=0}^{\infty} \beta^t (y_t z_{y,t} + \pi_t z_{\pi,t}), \qquad \|X\|_\beta^2 \equiv \langle X, X\rangle_\beta.$$

Because $0 < d < \beta < 1$, both funding functionals are bounded linear functionals on $\mathscr{H}_\beta$. Let $X_j(\sigma)$ be their respective Riesz representers:

$$\mathscr{B}^j(Z) = \langle X_j(\sigma), Z\rangle_\beta, \qquad j \in \{NL, R\}, \quad Z \in \mathscr{H}_\beta. \tag{125}$$

The representers are bounded paths: their first-order conditions reduce to a second-order recurrence with stable root r and forcing terms proportional to s^t (or to 1 under NL). At $s = r$, the bounded solution contains tr^t. Thus, the corresponding optima belong to the admissible class. Writing $\Omega_j(\sigma) \equiv \|X_j(\sigma)\|_\beta^2 > 0$, the projection formula for the quadratic minimization problem gives

$$\mathscr{W}^j(\sigma) = \frac{\Delta^2}{2\Omega_j(\sigma)}, \qquad \frac{\mathscr{W}^R(\sigma)}{\mathscr{W}^{NL}(\sigma)} = \frac{\Omega_{NL}(\sigma)}{\Omega_R(\sigma)}. \tag{126}$$

Since the funding functionals are affine in σ, their representers can be written as $X_j(\sigma) = U_j + \sigma V_j$. Define

$$F(\sigma) \equiv \Omega_R(\sigma) - \Omega_{NL}(\sigma) = A_2\sigma^2 + A_1\sigma + A_0, \tag{127}$$

where

$$\begin{aligned} A_0 &= \|U_R\|_\beta^2 - \|U_{NL}\|_\beta^2, \\ A_1 &= 2\big(\langle U_R, V_R\rangle_\beta - \langle U_{NL}, V_{NL}\rangle_\beta\big), \\ A_2 &= \|V_R\|_\beta^2 - \|V_{NL}\|_\beta^2. \end{aligned}$$

Applying (125) to X_{NL} gives

$$F(\sigma) = 2\left[\mathscr{B}^R(X_{NL}) - \Omega_{NL}\right] + \|X_R - X_{NL}\|_\beta^2.$$

The funding-capacity calculation (123) in the proof of Proposition 14 therefore yields

$$A_2 = \frac{2\ell r K_2}{qD(1-dr)} + \|V_R - V_{NL}\|_\beta^2 > 0, \tag{128}$$

where $D = 1-\beta r > 0$ and $K_2 > 0$ are defined there.

The constant term. To evaluate A_0, extend the unit-multiplier solutions algebraically to $\sigma = 0$. This is used only to compute coefficients; the proposition concerns $\sigma > 0$. Under R, the date-t first-order conditions at $\sigma = 0$ imply

$$\lambda_t = s^t - q\widetilde{p}_t, \qquad q\widetilde{y}_t = \kappa\lambda_t, \qquad \widetilde{\pi}_t = \widetilde{p}_t - \widetilde{p}_{t-1}, \quad \widetilde{p}_{-1} = 0.$$

Combining these equations with the Phillips curve, the unique bounded price-level path is

$$\widetilde{p}_t^R(0) = \frac{\kappa^2 r}{q(1-\beta rs)} \sum_{j=0}^{t} r^{t-j} s^j, \qquad t \geq 0. \tag{129}$$

This expression also applies when $s = r$. Under NL, setting $s = 1$ gives $\widetilde{p}_t^{NL}(0) = (1-r^{t+1})/q$, using $(1-r)(1-\beta r) = \kappa^2 r$. The generating function of inflation under R is therefore

$$G_s(z) \equiv \sum_{t=0}^{\infty} \widetilde{\pi}_t^R(0) z^t = \frac{\kappa^2 r(1-z)}{q(1-\beta rs)(1-rz)(1-sz)}.$$

At $\sigma = 0$, the funding functional is $\sum_{t\geq 0}(\beta s)^t(\pi_t - \beta\pi_{t+1})$. Thus

$$\Omega_R(0) = \frac{G_s(0) - (1-s)G_s(\beta s)}{s}, \qquad \Omega_{NL}(0) = \frac{\kappa^2 r}{q(1-\beta r)}.$$

Using the stable-root identity, simplification yields

$$A_0 = -\frac{\beta\kappa^2 r(1-s)}{q(1-\beta r)(1-\beta s^2)(1-\beta rs)^2} P(r,s), \tag{130}$$

where

$$P(r,s) \equiv r(1-\kappa^2) + [1-\beta r(1+r)]s - \beta r(2-\beta r)s^2 + \beta^2 r^2 s^3. \tag{131}$$

We first show $P(r,s) > 0$ whenever $0 < \kappa \leq 1$. Let r_1 denote the stable root for $\kappa = 1$, so that $r_1 \leq r < 1$. When $r = r_1$, direct use of the root equation gives

$$P(r_1,s) = \frac{s(1-\beta r_1)}{r_1}\left\{2r_1^2(1-\beta) + (1-r_1)\beta r_1(1-s)[2-\beta r_1(2+s)]\right\} > 0,$$

because $0 < r_1 < 1/2$. Also, $P(1,s) = (1+s)(1-\beta s)^2 > 0$, where κ^2 in P is eliminated using the root identity. For fixed (β, s), this elimination makes P a concave quadratic in r, with coefficient $\beta(1+s)(\beta s^2 - 1) < 0$. Hence $P(r,s) > 0$ for every $r \in [r_1, 1)$ and thus $A_0 < 0$ whenever $0 < \kappa \leq 1$.

The linear term when $\kappa > 1$. It remains to exclude two positive roots when $A_0 \geq 0$. This case requires $\kappa > 1$, hence $0 < r < r_1 < 1/2$. We show that then $A_1 > 0$.

Let $\mathscr{B}^{j,(1)}$ be the coefficient of σ in $\mathscr{B}^j$. By the representer identity,

$$\frac{A_1}{2} = \mathscr{B}^{R,(1)}(U_R) - \mathscr{B}^{NL,(1)}(U_{NL}).$$

From (129), $\widetilde{y}_t^R(0) = \frac{\kappa}{q}(s^t - q\widetilde{p}_t^R(0))$. Hence

$$\widetilde{y}_0^R(0) = \frac{\kappa}{q}\left(1 - \frac{\kappa^2 r}{1-\beta rs}\right),$$
$$\sum_{t=0}^{\infty}(\beta s)^t \widetilde{y}_t^R(0) = \frac{\kappa}{q}\frac{1-\kappa^2 r/(1-\beta rs)^2}{1-\beta s^2}.$$

Define

$$C(s,Q) \equiv \left(Q - \frac{1}{s}\right)\frac{1-\kappa^2 r/(1-\beta rs)^2}{1-\beta s^2} + \frac{1-\kappa^2 r/(1-\beta rs)}{s}.$$

Since $\mathscr{B}^{R,(1)}(U_R) = \frac{\kappa}{q}C(s,\beta+\ell)$ and $\mathscr{B}^{NL,(1)}(U_{NL}) = \frac{\kappa}{q}C(1,\beta)$,

$$\frac{qA_1}{2\kappa} = \ell\frac{1-\kappa^2 r/(1-\beta rs)^2}{1-\beta s^2} + C(s,\beta) - C(1,\beta). \tag{132}$$

The first term is strictly positive: $\kappa^2 r = (1-r)(1-\beta r) < (1-\beta rs)^2$ for $s \leq 1$. Direct algebra gives

$$C(s,\beta) - C(1,\beta) = \frac{\beta r(1-s)H(\beta,r,s)}{(1-\beta r)(1-\beta s^2)(1-\beta rs)^2}, \tag{133}$$

where

$$\begin{aligned} H(\beta,r,s) = {} & 1 + \beta(1-2s-4r+2r^2) \\ & + \beta^2 r[-r^2(1+s) + r(2+s-2s^2) + 3s^2 + 2s - 1] \\ & + \beta^3 r^2 s^2[r(1+s) - (2+s)]. \end{aligned}$$

For $0 < r \leq 1/2$ and $0 < \beta, s < 1$, this polynomial is strictly positive. To verify this, differentiation gives

$$\frac{\partial^2 H}{\partial r^2} = 2\beta(\beta s^2 - 1)[3\beta r(1+s) - \beta(2+s) - 2] > 0,$$
$$-\frac{\partial H}{\partial r}\bigg|_{r=1/2} = \frac{\beta}{4}[8 - \beta(1+9s+4s^2) + \beta^2(s^3+5s^2)] > 0.$$

The last bracket is decreasing in $\beta \in [0,1]$ and at $\beta = 1$ equals $(1-s)(7-2s-s^2) > 0$. Thus H is decreasing in r on $(0,1/2]$, and $H(\beta,r,s) \geq H(\beta,1/2,s)$, where

$$H(\beta,1/2,s) = (1-\beta)^3 + \frac{5-4s}{2}\beta(1-\beta)^2 + \frac{(1-s)(15-8s)}{8}\beta^2(1-\beta) + \frac{(1-s)^2(3-s)}{8}\beta^3 > 0.$$

Equations (132)–(133) therefore imply $A_1 > 0$ when $\kappa > 1$.

Existence and uniqueness of the threshold. By (128), $A_2 > 0$ for every admissible parameter configuration. If $A_0 < 0$, the quadratic $F(\sigma)$ has exactly one positive root,

$$\sigma_w \equiv \frac{\sqrt{A_1^2 - 4A_2A_0} - A_1}{2A_2} > 0,$$

and $F(\sigma) < 0$ if and only if $0 < \sigma < \sigma_w$. If $A_0 \geq 0$, then $\kappa > 1$ and $A_1 > 0$, so $F(\sigma) > 0$ for every $\sigma > 0$; define $\sigma_w = 0$ in this case. Consequently, for all admissible parameters,

$$\mathscr{W}^R > \mathscr{W}^{NL} \iff F(\sigma) < 0 \iff 0 < \sigma < \sigma_w.$$

Finally, $A_0 < 0$ whenever $\kappa \leq 1$, so $\sigma_w > 0$ in that region, whereas $\sigma_w = 0$ is possible when $\kappa > 1$. This proves the proposition. □

G Appendix for Section 5

G.1 Nonlinear Ramsey Problem and Equilibrium Conditions

This appendix specifies the exact nonlinear Ramsey problem underlying the quantitative analysis. Under perfect foresight, the planner chooses the entire transition path without linearizing equilibrium conditions or approximating household welfare.

Planner's objective Period utility and household welfare are

$$U(C_t,N_t,d_t) = \frac{C_t^{1-\sigma}}{1-\sigma} + \frac{\omega_D}{1-\eta_D}d_t^{1-\eta_D} - \chi_N\frac{N_t^{1+\nu}}{1+\nu}, \tag{134}$$

$$W_0 = \sum_{t=0}^{\infty}\beta^t\delta_t U(C_t,N_t,d_t), \tag{135}$$

with logarithmic limits when $\sigma = 1$ or $\eta_D = 1$. The preference shifter δ_t multiplies all components of utility.

Let $\bar{z}$ and $\bar{\lambda}$ denote the no-shock Ramsey steady-state allocation and its multipliers on the equilibrium restrictions $g(z_{t-1}, z_t, z_{t+1}; \delta_t, \delta_{t+1}) = 0$. To preserve commitments inherited from this steady state, the date-zero planner maximizes

$$J_0 = W_0 - \mathscr{B}_0, \tag{136}$$

$$\mathscr{B}_0 \equiv \frac{1}{\beta}\bar{\lambda}' \left[g(\bar{z}, \bar{z}, z_0; 1, \delta_0) - g(\bar{z}, \bar{z}, \bar{z}; 1, 1) \right]. \tag{137}$$

The inherited commitment term affects the date-zero Ramsey first-order conditions but is excluded from household welfare.

Preference shock The deterministic preference shock satisfies

$$\log \delta_t = \rho_e^t \log \delta_0, \qquad 0 < \rho_e < 1, \tag{138}$$

$$e_t^n \equiv \log \delta_{t+1} - \log \delta_t = -(1 - \rho_e) \log \delta_t. \tag{139}$$

The no-shock steady state has $\bar{\delta} = 1$, while $\delta_0 > 1$ implies an expansionary demand shock, $e_0^n < 0$.

Households Let $d_t \equiv D_t / P_t$ denote real deposits, and let Q_t^S be the price of a one-period claim providing neither liquidity nor collateral services. Household optimality implies

$$Q_t^S = \beta \frac{\delta_{t+1}}{\delta_t} \left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \frac{1}{\pi_{t+1}}, \tag{140}$$

$$Q_t^D = Q_t^S + \omega_D C_t^{\sigma} d_t^{-\eta_D}, \tag{141}$$

$$Q_t^T = Q_t^S (1 + \rho Q_{t+1}^T), \tag{142}$$

$$\frac{w_t}{P_t} = \chi_N N_t^{\nu} C_t^{\sigma}. \tag{143}$$

Banks Let ξ_t denote the shadow value of bank collateral. Bank optimality requires

$$Q_t^C (1 - \ell \xi_t) = Q_t^S, \tag{144}$$

$$Q_t^A (1 - \ell \rho_A \xi_t) = Q_t^S, \tag{145}$$

$$Q_t^D (1 - \xi_t) = Q_t^S. \tag{146}$$

Since deposit liquidity implies $Q_t^D > Q_t^S$, we have $\xi_t > 0$, so the collateral constraint binds:

$$Q_t^D d_t = \ell \left(Q_t^C b_t^C + \rho_A Q_t^A \bar{a} \right), \tag{147}$$

where the real supply of other bank assets is fixed at $a_t = \bar{a}$.

Firms With linear production, a constant employment subsidy ς, and Rotemberg price adjustment costs, the symmetric pricing condition is

$$N_t C_t^{-\sigma}\left[\varphi(\pi_t-1)\pi_t+(\theta-1)(1-\tau_t)-\theta(1-\varsigma)\chi_N N_t^{\nu}C_t^{\sigma}\right]=\beta\frac{\delta_{t+1}}{\delta_t}N_{t+1}C_{t+1}^{-\sigma}\varphi(\pi_{t+1}-1)\pi_{t+1}. \tag{148}$$

Treasury and central bank Let $b_t^T \equiv B_t^T/P_t$ and $b_t^C \equiv B_t^C/P_t$ denote real Treasury debt and reserves. Treasury debt is held by households and the central bank, $b_t^T = b_t^{T,H} + b_t^{T,C}$, with fixed central-bank holdings $b_t^{T,C} = \bar{b}^{T,C}$. Their respective budget constraints are

$$Q_t^T b_t^T + \tau_t Y_t + H_t = \frac{1+\rho Q_t^T}{\pi_t} b_{t-1}^T + G, \tag{149}$$

$$Q_t^C b_t^C + \frac{1+\rho Q_t^T}{\pi_t}\bar{b}^{T,C} = \frac{b_{t-1}^C}{\pi_t} + Q_t^T \bar{b}^{T,C} + H_t. \tag{150}$$

A positive H_t denotes a remittance from the central bank to the Treasury. Under no fiscal backing,

$$H_t \geq \underline{H}, \tag{151}$$

with $\underline{H} = 0$ in the baseline exercises. Under fiscal backing, H_t is unrestricted.

Resource constraint Goods-market clearing and linear production require

$$C_t + G + \frac{\varphi}{2}(\pi_t-1)^2 Y_t = Y_t = N_t. \tag{152}$$

Initial conditions and numerical solution Both fiscal regimes inherit the same no-shock Ramsey steady state, including $b_{-1}^C = \bar{b}^C$, $b_{-1}^T = \bar{b}^T$, and the steady-state commitment multipliers. Given the deterministic shock path, the planner jointly determines the nonlinear transition subject to the preceding equilibrium conditions and the regime-specific remittance restriction.

Numerically, the infinite-horizon problem is approximated by a long finite transition followed by an endogenous stationary continuation, whose discounted welfare contribution is evaluated exactly. The terminal date N is increased until equilibrium paths and welfare comparisons are invariant to further extensions.

We solve the nonlinear problem by continuation in the shock size. Accepted solutions must satisfy the original equilibrium conditions, Ramsey first-order conditions, complementary slackness under no fiscal backing, and a local second-order condition. The procedure does not linearize the equilibrium conditions but does not establish global optimality.

G.2 Calibration Details

Labor disutility The parameter χ_N governs labor disutility through the household intratemporal condition $w_t/P_t = \chi_N N_t^{\nu} C_t^{\sigma}$. At the reference allocation used to scale the numerical problem, $Y^{\text{ref}} = N^{\text{ref}} = 1$ and $C^{\text{ref}} = 0.8$. Normalizing the reference real wage to one implies $\chi_N = 1/[(N^{\text{ref}})^{\nu}(C^{\text{ref}})^{\sigma}] = 1.25$. This fixes the scale of labor disutility without imposing an additional empirical target on the final Ramsey steady state.

Price adjustment cost The Rotemberg parameter φ governs nominal price rigidity, with larger values flattening the New Keynesian Phillips curve. Linearizing the firm's pricing condition around the calibrated zero-inflation steady state yields $\hat{\pi}_t = \beta \hat{\pi}_{t+1} + \kappa \hat{y}_t + \gamma \widetilde{\tau}_t$, where

$$\kappa = \frac{(\theta - 1)(1 - \bar{\tau})}{\varphi} \left(\nu + \sigma \frac{\bar{Y}}{\bar{C}} \right). \tag{153}$$

We set $\varphi = 60$, implying $\kappa \simeq 0.203$. The quantitative analysis uses the exact nonlinear pricing condition rather than this approximation.

Deposit liquidity The parameter η_D governs the curvature of deposit liquidity services. The household deposit Euler equation implies

$$Q_t^D - Q_t^S = \omega_D C_t^{\sigma} d_t^{-\eta_D}, \qquad d_t = \left(\frac{\omega_D C_t^{\sigma}}{Q_t^D - Q_t^S} \right)^{1/\eta_D}, \tag{154}$$

where Q_t^S is the price of an otherwise identical one-period claim without liquidity services. Holding consumption and Q_t^S fixed, a larger η_D reduces the sensitivity of deposit demand to the liquidity premium. We set $\eta_D = 30$, implying that a one-percentage-point decline in the annual deposit rate reduces deposit holdings by approximately 3.3% locally.

The parameter $\omega_D = 3.36 \times 10^6$ targets an annual deposit liquidity premium of approximately one percentage point. Its large value reflects the CRRA normalization and high curvature: with $\eta_D = 30$, the term $d_t^{-\eta_D}$ is very small at the reference allocation, requiring a large ω_D to generate a modest marginal liquidity value $\omega_D C_t^{\sigma} d_t^{-\eta_D}$.

Collateral value of other bank assets The parameter ρ_A governs the collateral value of other bank assets relative to reserves. The bank collateral constraint is

$$Q_t^D d_t \leq \ell \left(Q_t^C b_t^C + \rho_A Q_t^A \bar{a} \right). \tag{155}$$

Reserves carry unit collateral weight, while other assets receive weight $\rho_A = 0.842$, providing 84.2% as much collateral capacity per dollar. This parameter determines relative collateral efficiency, whereas $\bar{a}$ determines the quantity of other bank assets.

Duration of Treasury debt The parameter ρ governs Treasury bond duration through a coupon stream $1, \rho, \rho^2, \ldots$, with larger ρ implying longer duration. At the zero-inflation steady state, Macaulay duration in years is $\mathscr{D} = 1/[4(1-\beta\rho)]$. With $\beta = 0.995$, we set $\rho = 0.9675$, giving a duration of approximately 6.7 years for Treasury securities held by households and the central bank.

Central-bank Treasury holdings We choose the real quantity of central-bank Treasury holdings, $\bar{b}^{T,C}$, jointly with other steady-state parameters to match a market-value asset-to-liability ratio of one, $\bar{Q}^T \bar{b}^{T,C} / (\bar{Q}^C \bar{b}^C) = 1$. The resulting $\bar{b}^{T,C} = 0.0343$ is small relative to output because it measures the quantity of long-duration bonds rather than their market value, which incorporates the present value of future coupons.

Other bank assets Banks hold a fixed supply $\bar{a}$ of other one-period assets, which contribute to collateral capacity as specified above. We calibrate $\bar{a}$ jointly with other steady-state parameters, targeting an annual net deposit rate of 1%. The supply of these assets affects collateral capacity, deposit issuance, the liquidity premium, and hence the equilibrium deposit return. The calibration yields $\bar{a} = 1.544$.

Government spending Constant government spending enters the resource and Treasury budget constraints. We choose G jointly with other steady-state parameters to match a government debt market value equal to annual GDP, $\bar{Q}^T \bar{b}^T / (4\bar{Y}) = 1$. Higher spending reduces the primary surplus and increases the required debt stock. The calibration yields $G = 0.167$. Because taxes and remittances are determined simultaneously, all five steady-state targets are matched jointly rather than calibrating G independently.

G.3 Welfare Measurement and Decomposition

Household welfare We measure the welfare cost of no fiscal backing by comparing household utility under the two Ramsey allocations. Let $j \in \{\text{FB}, \text{NFB}\}$ denote fiscal backing and no fiscal backing. Both regimes face the same preference shock $\{\delta_t\}_{t=0}^{\infty}$ and start from the same no-shock Ramsey steady state, including inherited state variables and commitment multipliers.

Period utility consists of consumption utility, labor disutility, and liquidity services from deposits:

$$U(C,N,d) = u_C(C) - u_N(N) + u_D(d),$$
$$u_C(C) = \frac{C^{1-\sigma}}{1-\sigma}, \qquad u_N(N) = \chi_N \frac{N^{1+\nu}}{1+\nu}, \qquad u_D(d) = \omega_D \frac{d^{1-\eta_D}}{1-\eta_D},$$

where $u_C(C) = \log C$ when $\sigma = 1$ and $u_D(d) = \omega_D \log d$ when $\eta_D = 1$. Discounted household welfare and the welfare cost of no fiscal backing are

$$W_0^j = \sum_{t=0}^{\infty} \beta^t \delta_t U(C_t^j, N_t^j, d_t^j),$$
$$\mathscr{L} = W_0^{\text{FB}} - W_0^{\text{NFB}}.$$

A positive $\mathscr{L}$ indicates that households are worse off without fiscal backing. The preference shifter δ_t weights all three utility components. We exclude the inherited-commitment term from household welfare. Since the two regimes share the same no-shock allocation, the welfare difference is zero in the absence of shocks.

Decomposition of welfare losses We decompose the welfare difference into three components: (i) changes in real consumption and labor, (ii) resources lost through inflation-related price adjustment, and (iii) changes in the liquidity services provided by deposits. The decomposition separates these sources of utility without changing the equilibrium allocations. In particular, each component is evaluated using the actual Ramsey allocation under each regime, rather than by solving a hypothetical economy in which one mechanism is removed.

The starting point is the resource constraint:

$$C_t^j = Y_t^j - G - \frac{\varphi}{2}(\pi_t^j - 1)^2 Y_t^j.$$

Output is divided between private consumption, government spending, and resources used by firms to adjust prices. This allows us to separate the welfare effects of real production and labor from the direct resource cost of inflation.

Real allocation component. First, consider the resources available for private consumption before deducting price-adjustment costs:

$$\widetilde{C}_t^j \equiv Y_t^j - G.$$

This quantity measures how much households could consume at the realized level of output if no goods were used for price adjustment. Importantly, we keep output and labor at their actual equilibrium levels. Thus, $\widetilde{C}_t^j$ is an accounting measure, not consumption in a separately solved zero-inflation equilibrium.

We define the real allocation component as

$$\mathscr{L}_{\text{real}} = \sum_{t=0}^{\infty} \beta^t \delta_t \left[u_C(\widetilde{C}_t^{\text{FB}}) - u_C(\widetilde{C}_t^{\text{NFB}}) - u_N(N_t^{\text{FB}}) + u_N(N_t^{\text{NFB}}) \right].$$

This component compares the utility from goods available for consumption, net of the disutility of producing them. It is positive when the real allocation under no fiscal backing delivers lower

welfare, even before accounting for price-adjustment costs and deposit services. It includes the effects of monetary policy and inflation on actual output and labor.

Inflation component. Inflation creates an additional welfare cost because firms use real resources to adjust prices. These resources would otherwise be available for consumption. Define

$$A_t^j \equiv \frac{\varphi}{2}(\pi_t^j - 1)^2 Y_t^j,$$
$$\mathscr{R}_t^j \equiv u_C(\widetilde{C}_t^j) - u_C(\widetilde{C}_t^j - A_t^j) = u_C(\widetilde{C}_t^j) - u_C(C_t^j) \geq 0.$$

Here, $\mathscr{R}_t^j$ measures the loss of consumption utility caused by the resources absorbed by price adjustment. We compare this cost across the two regimes:

$$\mathscr{L}_\pi = \sum_{t=0}^{\infty} \beta^t \delta_t \left(\mathscr{R}_t^{\text{NFB}} - \mathscr{R}_t^{\text{FB}} \right).$$

A positive value indicates that price adjustment absorbs more resources, in utility terms, under no fiscal backing. This component captures only the direct resource cost of inflation. The indirect effects of inflation on output and labor are already included in the real allocation component, so they are not counted again here.

Deposit component. Finally, deposits provide liquidity services that enter household utility directly. The deposit component is

$$\mathscr{L}_d = \sum_{t=0}^{\infty} \beta^t \delta_t \left[u_D(d_t^{\text{FB}}) - u_D(d_t^{\text{NFB}}) \right].$$

This component is positive when households receive fewer liquidity services under no fiscal backing. Conversely, if no fiscal backing leads to greater deposit holdings, the component is negative because the additional liquidity services improve household welfare. A negative contribution therefore offsets some of the losses from real allocations and inflation.

To see why these three components add up, note that actual consumption satisfies $C_t^j = \widetilde{C}_t^j - A_t^j$. Period utility can therefore be written exactly as

$$U(C_t^j, N_t^j, d_t^j) = \underbrace{u_C(\widetilde{C}_t^j) - u_N(N_t^j)}_{\text{Real allocation}} - \underbrace{\mathscr{R}_t^j}_{\text{Inflation resource cost}} + \underbrace{u_D(d_t^j)}_{\text{Deposit services}}.$$

Taking the discounted welfare difference between the two regimes gives

$$\mathscr{L} = \mathscr{L}_{\text{real}} + \mathscr{L}_\pi + \mathscr{L}_d.$$

This identity is exact, with no approximation or residual. In particular, using actual consumption in the real component would already include price-adjustment costs and would therefore double count them when the inflation component is added.

Consumption-equivalent cost To express welfare losses in economically interpretable units, we calculate the permanent reduction in consumption under fiscal backing that would make households indifferent between the two regimes. Let $\lambda < 1$ denote this proportional reduction, holding labor and deposit services at their fiscal-backing paths:

$$\sum_{t=0}^{\infty} \beta^t \delta_t U\left((1-\lambda)C_t^{\mathrm{FB}}, N_t^{\mathrm{FB}}, d_t^{\mathrm{FB}}\right) = W_0^{\mathrm{NFB}}.$$

Define

$$S_{\mathrm{FB}} = \sum_{t=0}^{\infty} \beta^t \delta_t (C_t^{\mathrm{FB}})^{1-\sigma}.$$

The exact consumption-equivalent reduction is

$$\lambda = \begin{cases} 1 - \exp\left(-\dfrac{\mathscr{L}}{S_{\mathrm{FB}}}\right), & \sigma = 1, \\ 1 - \left[1 - \dfrac{(1-\sigma)\mathscr{L}}{S_{\mathrm{FB}}}\right]^{1/(1-\sigma)}, & \sigma \neq 1. \end{cases}$$

For $\sigma \neq 1$, the expression requires $1 - (1-\sigma)\mathscr{L}/S_{\mathrm{FB}} > 0$. Under the baseline logarithmic specification, $S_{\mathrm{FB}} = \sum_{t=0}^{\infty} \beta^t \delta_t$. We report

$$\mathrm{CE} = 10^4 \lambda,$$

so that one unit represents a welfare cost equivalent to 0.01%, or one basis point, of permanent consumption. A positive value indicates a welfare loss from no fiscal backing.

Contributions in consumption-equivalent units Figure 6 reports the three welfare components in consumption-equivalent units. Since the consumption-equivalent transformation is nonlinear, converting each utility component separately would generally produce numbers that do not sum to the total welfare cost.

To preserve the exact decomposition, we instead allocate the total consumption-equivalent cost proportionally to the three utility components. Define the common conversion factor

$$k_W = \begin{cases} \dfrac{\mathrm{CE}}{\mathscr{L}}, & \mathscr{L} \neq 0, \\ \dfrac{10^4}{S_{\mathrm{FB}}}, & \mathscr{L} = 0, \end{cases}$$

where the second expression is the continuous limit as $\mathscr{L} \to 0$. The reported contributions are

$$\mathrm{CE}_{\mathrm{real}} = k_W \mathscr{L}_{\mathrm{real}}, \qquad \mathrm{CE}_{\pi} = k_W \mathscr{L}_{\pi}, \qquad \mathrm{CE}_{d} = k_W \mathscr{L}_{d},$$

and satisfy

$$\mathrm{CE} = \mathrm{CE}_{\mathrm{real}} + \mathrm{CE}_{\pi} + \mathrm{CE}_{d}.$$

Thus, the three contributions show how much of the total welfare cost is accounted for by real allocations, inflation resource costs, and deposit services. Positive contributions increase the total welfare cost, while negative contributions offset it.

These contributions are an accounting decomposition, not the welfare effects of separately eliminating individual mechanisms. They preserve the signs and relative magnitudes of the underlying utility differences. Individual contributions can exceed the total or be negative when welfare gains in one component offset losses in another. At the common no-shock allocation, all three contributions are zero.